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DEVELOPMENT OF MULTIFUNCTIONAL POLYPROPYLENE WITH STRUCTURED SURFACE LAYER

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Independent work and indicating references

I, the undersigned, Csenge Vámos declare, that the current PhD thesis is my own work, and only the resources referred have been used in the document. Each part used literally or taken over paraphrased from other sources have been obviously indicated by the identification of its source.

Budapest, 10/02/2025

.....
Csenge Vámos

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TABLE OF SYMBOLS AND ABBREVIATIONS

Latin characters

Symbol	Designation, note, value	Unit
A	Surface area of the sample used for BET measurement	cm ²
CA	Contact angle	°
CA _{adv}	Advancing contact angle	°
CA _{rec}	Receding contact angle	°
CA _{CB}	Cassie-Baxter contact angle	°
CA _w	Wenzel contact angle	°
CAH	Contact angle hysteresis	°
C _{scat}	Scattering cross-section	m ²
E _{perf}	Perforation energy	J/mm
E _{total}	Total absorbed energy	J
f	Fractional areas	-
GF%	GF weight percent	%
I _{in}	Intensity of the incident light	W/m ²
I _{scat}	Intensity of the scattered light	W/m ²
I _{solar}	ASTM G173 Global solar intensity spectrum	W/m ²
l ₁	Thickness of the treated sample	cm
l ₂	Thickness of the sample's bulk phase	cm
l ₃	Thickness of the sample's porous layer	cm
m	Weight of the solvent-treated sample	g
N	Number of the single scatterer	-
n _r	Radial refractive index	-
n _t	Tangential refractive index	-
r	Roughness factor	-
R	Distance between the scattering center and the observer	m
R ₀	Reflectance of the surface	%
R(λ)	Reflectance of samples at the wavelength	%
r _f	Roughness factor of the wet surface	-
P _{scat}	Power of total scattered light	W
S _{BET}	Specific surface area	m ² /g
S _{BET} [*]	Calculated specific surface area	m ² /g
V	Volume of the solvent-treated sample	cm ³
V _{scat}	Volume of the single scatterer	m ³
V _{vol}	Volume of N number of single scatterers	m ³
T	Characteristic spreading time	s
t	Droplet spreading time	s
T _{cp}	Crystallization peak temperature	°C

T_m	Melting temperature	°C
$T_{\text{Tot surface}}$	Total surface area of sample used for BET measurement	cm ²
TIS	Total integrated scattering	%
TSR	Total solar reflectance	%
v	Thickness of the sample used for impact test	mm
X_c	Degree of crystallinity	%

Greek characters

Symbol	Designation, note, value	Unit
α_R	Light incident angle	°
γ_{SV}	Solid– vapor surface tension	N/m
γ_{SL}	Solid-liquid surface tension	N/m
γ_{LV}	Liquid-vapor surface tension	N/m
ΔH_m	Crystal melting enthalpy	J/g
ΔH_0	Crystal melting enthalpy of 100% crystalline polymer	J/g
Δn	Birefringence	-
DT	Temperature difference	°C
ε	Porosity	%
η	Packing fraction	-
θ_s	Stationary contact angle	°
θ_0	Fitted initial contact angle	°
λ	Wavelength of the incident light	nm
ρ_a	Apparent density	g/cm ³
ρ_t	True density	g/cm ³
ϕ	Weight fraction of PP	-
Ω	Spatial angle	°

Abbreviations

Abbreviation	Designation
α -iPP	Alpha nucleated isotactic polypropylene
AFM	Atomic force microscope
ASA	Acrylonitrile-styrene acrylate
BaSO ₄	Barium sulfate
CO ₂	Carbon dioxide
DSC	Differential scanning calorimetry
GF	Glass fiber
HNT	Halloysite nanotube
LDPE	Low-density polyethylene

MAPP	Maleated polypropylene
m-PP	Metallocene catalyst polypropylene
NA	Nucleating agent
NA-21 E	Hydroxy aluminum bis(2,4,8,10-tetra-trans-butyl-6-hydroxy-12H-dibenzo[d,g][1.3.2]dioxaphosphocin-6-oxide)
NIPS	Non-solvent induced phase separation
NIR	Near-infrared
PC	Polycarbonate
PDMS	Polydimethylsiloxane
PE	Polyethylene
PET	Polyethylene terephthalate
POM	Polarized optical microscope
PP	Polypropylene
PS	Polystyrene
PMMA	Polymethyl methacrylate
PVA	Polyvinyl alcohol
P(VdF-HFP)	Poly(vinylidene fluoride-co-hexafluoropropene)
RMS	Root mean square
SEM	Scanning electron microscope
SiO ₂	Silicon dioxide
TiO ₂	Titanium dioxide
TIPS	Thermally induced phase separation
TIS	Total integrated scattering
UV	Ultraviolet
UV-NIL	UV-based nanoimprint lithography
VIS	Visible band
XPS	X-ray photoelectron spectroscopy
ZN-PP	Ziegler-Natta catalyst polypropylene
ZnO	Zinc dioxide

1. INTRODUCTION

Nature serves as an inexhaustible source of inspiration for science and technology, offering numerous solutions to complex challenges. Scientists have paid great attention to replicating structures or behaviors observed in nature to develop innovative applications. This approach leverages nature's millions of years of evolution, where effective solutions have already been refined to address specific problems. It is fascinating how examples from the natural can offer transformative development opportunities across various fields. One such great example is the *Cyphochilus* beetle [1], renowned for its white and remarkable water-repellent scales which cover the insect's body. The beetle's scales achieve exceptional solar reflectivity through a unique nanostructured design that scatters light across the entire solar spectrum. Such multifunctional materials inspired by nature could be particularly valuable in several fields of life where different properties should meet in one product. The combination of solar reflectivity and water-repellency as we can see in the case of this beetle, offers dual benefits: reducing energy consumption for cooling and protecting surfaces from water damage or contamination.

Solar energy often leads to undesirable thermal energy absorption, particularly in conventional roofing materials. This contributes to increased indoor temperatures and elevated energy demands for cooling. High sunlight reflection has emerged as a critical factor in mitigating the effects of intense solar radiation. For instance, the EU's new energy efficiency directive, effective since October 2023, tightens energy efficiency targets and accelerates the development of insulation solutions. By 2030, energy consumption must be reduced by 11.7 % compared to 2020 forecasts [2].

Today, the development of water-repellent, superhydrophobic surfaces has become increasingly important due to their highly diverse applications. Superhydrophobic materials are widely utilized in areas where minimizing contact with water or achieving self-cleaning properties is required. For instance, the self-cleaning windows of skyscrapers, the prevention of icing on aircraft bodies, or the filtration of oil spilled into seas and oceans.

In the context of sustainable materials development, the circular economy has become a driving force behind the push for mono-material products with enhanced recyclability. The ability to engineer multiple functionalities into a single material — such as superhydrophobicity and high solar reflectivity — offers an advantage in both performance and sustainability. Therefore, the customizability of surface properties is crucial for the successful design of mono-material products intended for advanced applications.

These days, thermoplastic polymers for industrial and everyday life purposes are gaining more and more attention, especially for semicrystalline polymers such as polypropylene (PP). Simultaneously, the industry has ever-increasing demands for the

materials used, so never-ending development and research are crucial to fulfill these demands. Polymers, particularly PP, are characterized by low surface energy and chemical inertness, which often face challenges with their advanced applications. Therefore, to extend the application and performance of PP materials, a significant portion undergo surface treatment processes before the final usage to optimize their surface properties.

Numerous surface treatment methods are known that can modify the chemical and morphological properties of polymer surfaces with little or no impact on the bulk properties. With appropriately chosen surface treatments, the surface structure of PP can be modified to such an extent that it results in superhydrophobic surface. To enhance the applicability of PP, it may be beneficial to develop new properties that further expand its range of uses.

For several years, endless efforts have been carried out to develop techniques to prepare superhydrophobic surfaces for industrial scale which are economical, easy to produce and highly efficient but most of them are restricted to only laboratory research phase. There is still much work required to be done for such surface preparation on a commercial scale [3].

Based on these perspectives, in my dissertation I aimed to develop a new, method for preparing hierarchically (multi-scale) structured surfaces on PP samples at Furukawa Electric Institute of Technology. This development includes the description of the phenomena occurring during each process step and identifying correlations between the processing parameters and the morphology of the created surface structure.

Furthermore, my research aimed to enhance two key functional properties of PP surfaces, superhydrophobicity, which improve water repellency and self-cleaning characteristics and solar reflectivity, which is essential for reflective thermal insulation applications.

Another goal, especially from the applicability point of view, was to analyze the correlation between the resulting structure and the mechanical properties. In my work, I examined how the developed method can be applied on reinforced PP composites and whether they can retain the created beneficial properties.

Finally, considering scalability and industrial relevance, my dissertation seeks to contribute to the development of a high-value, adaptable surface treatment technology that enhances the multi-functional potential of mono-material PP for diverse applications.

2. LITERATURE REVIEW

In this dissertation, I introduce a solvent-treatment method that transforms smooth PP surfaces into roughened ones. This surface modification imparts new properties to PP samples, such as water-repellency and higher solar light reflectivity. The literature review supports this work through an examination of the key concepts and techniques related to surface wettability, roughened surface creation and surface scattering.

Chapter 2.1 discusses wetting fundamentals, which are crucial for understanding the wetting behavior of roughened surfaces. Since superhydrophobicity and water-repellency arise from surface roughness, **Chapter 2.2** explores wetting models that describe how liquids interact with textured surfaces.

In **Chapter 2.3**, I provide a general overview of current physical surface modification methods used to create superhydrophobic surfaces. **Chapter 2.4** discusses the light reflection properties of materials, followed by **Chapter 2.5**, which focuses on the optical properties of spherulitic structures. Finally, **Chapter 2.6** reviews techniques for enhancing solar reflectance and the thermal insulation of coatings.

2.1. Surface wettability

Wetting is a commonly occurring phenomenon when a liquid comes in contact with the solid surfaces, and it is closely related to the interactions at the solid-liquid interfaces. When a liquid droplet is placed on a solid surface, three types of wetting behavior can be distinguished: the liquid perfectly wets the surface if it completely spread over its surface, or contact wetting occurs if the droplet adheres to the solid surface at a certain angle and forms a spherical cup and non-wetting occurs if the droplet retains its original shape with minimal contact. The ability of a liquid droplet to spread on the surface or, on the contrary, to keep its original drop shape will be governed by the equilibrium relationship between the attractive, cohesive forces that arise in the droplet and the adhesive forces at the liquid-solid phase boundary. When the adhesive force between the liquid and solid surface is greater than the cohesive forces among the liquid molecules enables the droplet to spread out on the surface, or when the cohesive force exceeds the adhesive force, the liquid droplet can keep its spherical shape. The balance between adhesive and cohesive forces determines the wettability of solid surfaces [4].

The surface tension determined by the cohesive forces in the material plays a decisive role in the contact angle formation of a liquid drop with the solid surface. The wetting behavior of a solid material is quantified by measuring the contact angle (CA) of a drop of liquid when it is placed on a surface. The CA between the solid, liquid, and vapor interfaces can be expressed according to Young's equation [4].

$$\cos CA = \frac{\gamma_{SV} - \gamma_{SL}}{\gamma_{LV}} \quad (2.1)$$

where γ_{SV} , γ_{SL} and γ_{LV} correspond to surface tensions at solid–vapor, solid-liquid, and liquid-vapor interfaces, respectively.

The degree of wetting is characterized by the CA, which can theoretically range from 0° to 180°. If the CA reaches the 0° perfect wetting takes place meanwhile, the increasing value of the CA shows the decrease of the wettability of the surface against the investigated liquid [5].

According to the behavior of the examined surface against water, we can determine surfaces having hydrophobic ($CA > 90^\circ$) or hydrophilic ($CA < 90^\circ$) characteristics (**Figure 2.1**). The transition of hydrophobic-hydrophilic behavior is difficult to determine, so in general, the cutoff point is put at 90° . In extreme cases, when the CA is below 10° we can talk about superhydrophilicity or when the CA is more than 150° , we can talk about superhydrophobicity [5].

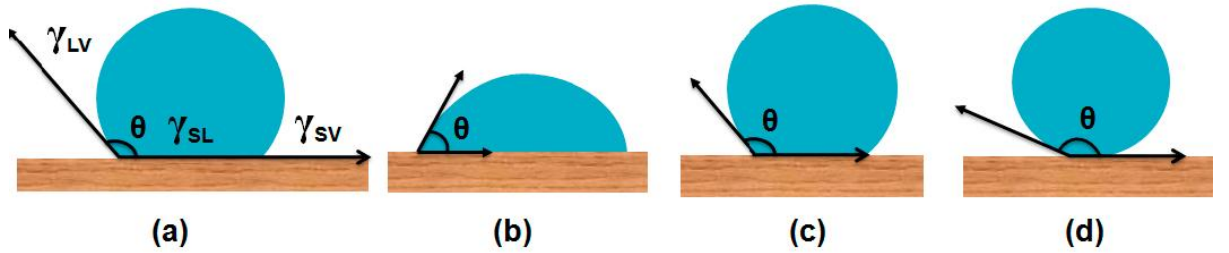


Figure 2.1 Schematic diagram of interfacial tension of all three phases (a) hydrophilicity (b), hydrophobicity (c) and superhydrophobicity (d) [6]

For the surface wetting characterization, two types of CA values are used: static CA and dynamic CA. For a flat surface, a static CA, which is nearly equal to Young's angle, can be determined by sessile drop measurements, where a drop of liquid is deposited on the surface, and the value is obtained by a goniometer [7].

The dynamic CA is non-equilibrium CA. In general, on a heterogeneous surface, different CA values can be measured depending on whether the solid-liquid contact is increased (the advancing contact angle (CA_{adv})) or decreased (the receding contact angle (CA_{rec})) as it is shown in **Figure 2.2**.

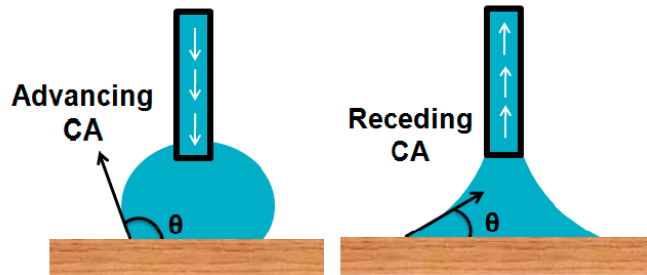


Figure 2.2. Schematic diagram of dynamic CA for measurement of CAH [6]

This difference between maximum CA_{adv} and minimum CA_{rec} is called contact angle hysteresis (CAH) (Equation 2.2). The more significant the difference (more considerable hysteresis), the more water drops will stick to the surface. The increasing tendency of CAH indicates increasing surface (chemical and/or geometric) heterogeneity. Generally, surfaces with a CAH of less than 10° are called water-repellent surfaces, which means water droplets will roll off extremely easily [7].

$$CAH = CA_{adv} - CA_{rec} \quad (2.2)$$

where CAH is the contact angle hysteresis, CA_{adv} and CA_{rec} corresponds to the advancing and receding contact angle.

2.2. Wetting models of roughened surfaces

The roughened surfaces created through the presented solvent-treatment method, exhibit specific wetting properties, including superhydrophobicity. Understanding the mechanisms that govern wetting behaviour is crucial, as interactions between liquids and textured surfaces can lead to a wide range of effects, influencing applications such as self-cleaning coatings. The following section explores the two primary wetting models for roughened surfaces. On superhydrophobic surfaces, CA exceeding 150° can be measured, this very high CA is referred to as the apparent CA, as it differs from the 'real' CA measured on smooth, flat surfaces. The superhydrophobic surfaces are usually hierarchically structured, and have micro- and nanoscale roughness, which contributes to their unique wetting behavior.

The above introduced Young's equation refers to an ideally smooth surface and can not be used to explain the exact mechanism of wetting in the case of surfaces having a certain degree of roughness. The wetting characteristic of rough surfaces can be described by two distinct homogeneous and heterogenous wetting models [8, 9].

The Wenzel model (homogeneous wetting model), explains how a liquid droplet behaves on a rough surface when it completely fills in the surface irregularities, leaving no trapped air (**Figure 2.3**). According to this theory, the Wenzel contact angle (CA_W) observed on a textured surface is linked to the equilibrium Young contact angle of a smooth surface by a roughness factor (r) (Equation 2.3), which represents the ratio of the real surface area to its flat projection [9].

$$\cos CA_W = r \cos CA \quad (2.3)$$

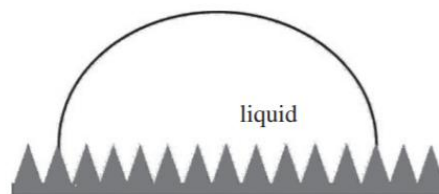


Figure 2.3 Wenzel wetting state [10]

Since roughness factor is always greater than one, roughness amplifies the material's natural tendency: if the surface is hydrophilic ($CA < 90^\circ$), the droplet spreads even more, whereas if it is hydrophobic ($CA > 90^\circ$), the droplet becomes even more spherical. Additionally, the model emphasizes that the enhanced interfacial area due to the surface's microstructures alters the balance of forces at the contact line, thereby modifying the droplet's equilibrium shape. However, while the Wenzel approach clarifies the relationship between roughness and wettability, it does not account for the phenomenon of CAH. A consequence of this kind of wetting state is that the droplet will stick to the surface quite well [10].

In the Cassie-Baxter model (heterogeneous wetting model), a liquid droplet rests primarily on the top of a rough surface, with air pockets trapped in the valleys beneath it (**Figure 2.4**). This results in a composite interface where only part of the drop contacts the solid material, while the remaining area is supported by air. This reduced contact area leads to lower adhesion, allowing droplets to roll or bounce off the surface with ease—a key factor in self-cleaning behavior. Unlike the Wenzel model, where the liquid fully wets the rough surface, the Cassie-Baxter mechanism is particularly useful for designing surfaces with hierarchical (double scale roughness) or porous textures that exhibit superior superhydrophobic properties [8].

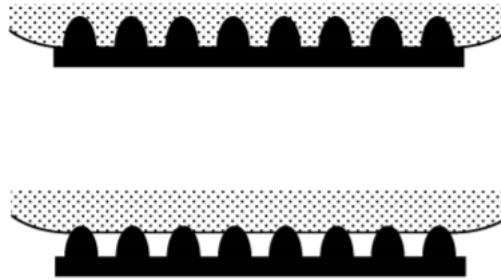


Figure 2.4. Illustration of the homogeneous wetting state (top) and the heterogeneous wetting state (bottom) [11]

According to Cassie-Baxter model [8], the resulting Cassie-Baxter contact angle (CA_{CB}) is described by Equation 2.4:

$$\cos CA_{CB} = r_f f \cos CA + f - 1 \quad (2.4)$$

where f is the relative fractions of the solid and air fractions underneath the drop and r_f is the roughness ratio of the substrate wetted by the liquid. It can be seen that if $f = 1$ and $r_f = r$, the equation is the same as the Wenzel equation. Thus, the Cassie-Baxter equation is useful to describe both types of wetting models.

It is important to note that even if a surface exhibits superhydrophobic characteristics, it does not necessarily mean it will also be water-repellent. The water-repellent property is best characterized by the CAH. When evaluating CAH, superhydrophobic surfaces are often categorized into the previously discussed

wetting states. For instance, if a surface is described by the Wenzel model, turning the surface upside down will cause the water droplet to remain adhered, hanging on the surface. A good example of the Wenzel wetting state is the rose petal. Despite having a roughened surface structure, a rose petal can hold a water droplet even when inverted. In contrast, the Cassie-Baxter model is well illustrated by the lotus leaf. Water droplets on a lotus leaf easily roll off due to its hierarchical surface structure [12, 13].

The primary difference between these behaviors lies in the hierarchical micro- and nanostructures of the surfaces. To illustrate these differences, two natural examples can be compared: the rose petal and the lotus leaf. **Figure 2.5** shows the SEM images of lotus leave and rose petal.

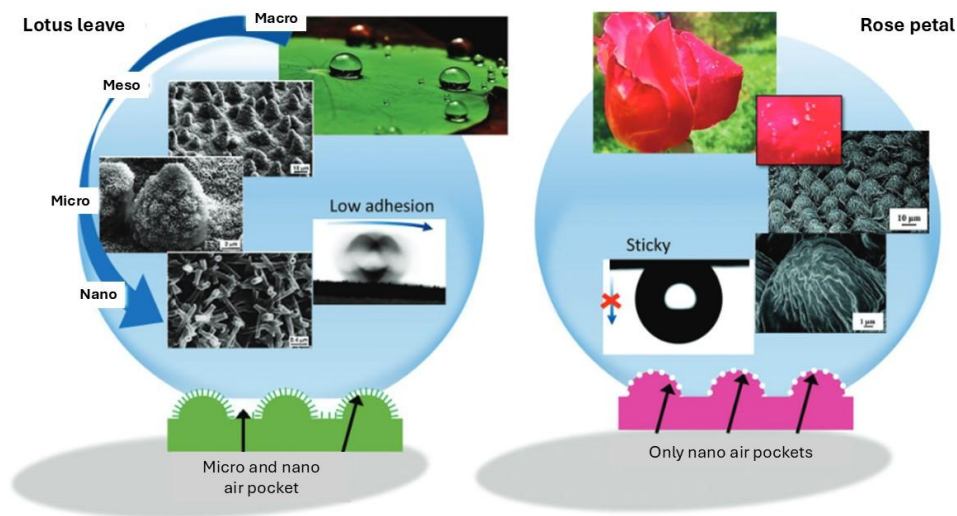


Figure 2.5 SEM images of lotus leave and rose petal [14]

Rose petals have larger structures on the surface compared to the lotus leaf, allowing liquid to impregnate the microstructure and partially penetrate the nanostructure. In contrast, the surface of a lotus leaf contains randomly distributed spikes and grooves that create roughness. These structures prevent water droplets from wetting the microstructural spaces between the spikes. Air becomes trapped beneath the water layer, creating a heterogeneous surface. This significantly reduces adhesion forces, enabling water to roll off effortlessly—a phenomenon known as the "self-cleaning" effect. This ability is utilized for self-cleaning purposes, where water droplets roll off the leaf completely, carrying dirt and debris [12, 13].

Since my research primarily focuses on the surfaces having hierarchical structure, I will provide examples in the following section on how surface roughness can be created on polymer surfaces.

2.3. Techniques for fabricating rough surface

The formation of superhydrophobic surfaces is essentially determined by two crucial parameters, appropriate surface roughness and low surface energy [15, 16]. The

first parameter is based on the preparation of micro- and nano- hierarchically structured surface, where the surface has an initially low surface energy. The second parameter is to fabricate a certain roughness and then modify it with low-surface-energy material [17].

In the following section, those methods are presented that primarily create superhydrophobic characteristics on polymer surfaces by forming well-ordered structures. These preparation methods are generally divided into two main categories: top-down and bottom-up methods. Top-down methods enable the creation of a highly controlled surface structure through material disintegration. In the case of bottom-up methods, the roughened surface is obtained by a material addition process or a self-assembly process [18-24].

2.3.1. Top-down methods

Etching method

Etching is a widely utilized technique for generating rough surface textures, prized for its affordability and straightforward implementation in superhydrophobic surface fabrication. The process is broadly categorized into chemical (wet) etching and dry (plasma) etching [25].

In chemical etching, liquid reagents are used to dissolve the upper layers of the substrate through reactions at the liquid–solid interface. This method, which does not require sophisticated equipment, is well-suited for industrial applications but typically results in less uniform and more irregular textures. On the other hand, dry etching employs reactive gases or plasma to precisely remove material, allowing for the controlled creation of three-dimensional structures at a faster rate due to the generation of highly reactive species. However, the need for complex, high-energy equipment makes this approach more costly and less accessible for large-scale manufacturing. Despite their effectiveness, both techniques inherently produce delicate surface features since they form nanoarchitectures by eroding the original material [25].

Recent advances in one-step dry etching have enabled the rapid fabrication of superhydrophobic poly(dimethylsiloxane) (PDMS) surfaces using plasma-based methods. In one approach, researchers developed a plasma etching technique that produces nanostructures with features smaller than 50 nm, resulting in highly water-repellent PDMS surfaces. This process not only makes surfaces water repellent but also allows for precise tuning of surface wetting properties, ranging from moderate to extreme water repellency. Such nanostructured surfaces are not only ideal for external water-repellent applications but also show promise in droplet-based microfluidic systems, where the increased surface area can significantly lower flow resistance [26].

Numerous studies introduces a scalable technique to create highly organized nanostructures on a variety of polymer surfaces. These method combines laser interference patterning with a plasma etching process to produce arrays of nanowires

or pillar-like features with high aspect ratios on substrates such as polyethylene terephthalate (PET), polyvinyl alcohol (PVA) and PP. For polymers that absorb UV light, the process relies on direct laser-induced ablation to generate periodic patterns, whereas UV-transparent materials are first coated with a thin photoresist layer before patterning. By adjusting the laser energy and etching dosage, the researchers can precisely control the hierarchical structure and morphology of the resulting substrates, as it is presented in **Figure 2.6** [27, 28].

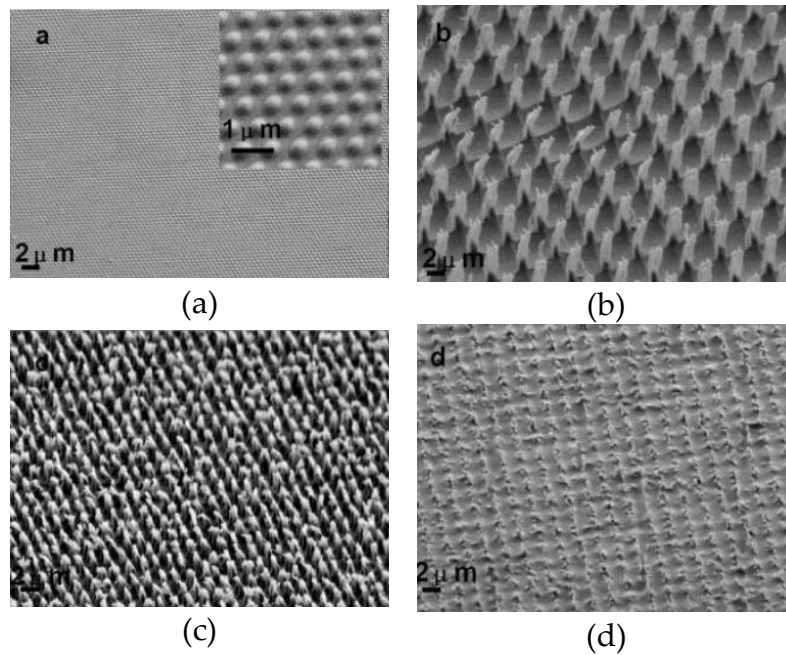


Figure 2.6 Scanning electron microscopy (SEM) images of high-aspect-ratio polymer-nanostructures of various polymer materials, (a) 1.0 μm period of PDMS film; (b) 2.5 μm period of PET nanowire array; (c) 2.5 μm period of PVA film nanowire array; (d) 2.5 μm period of PP nanowire array [27]

Lithography

For creating rough surfaces with regular structures, lithography approaches could be an effective method. Generally, lithography consists of three types of surface preparation steps: coat, expose and develop, where the aim is to create a patterned substrate by irradiating the substrate coated with a designed featured mask, as can be seen on **Figure 2.7**. During the exposing step, light travels through the patterned mask and starts to create the designed pattern of on the photosensitive layer. In the subsequent development step, which is usually etching, the formed pattern will be transferred into the substrate upper layer [4].

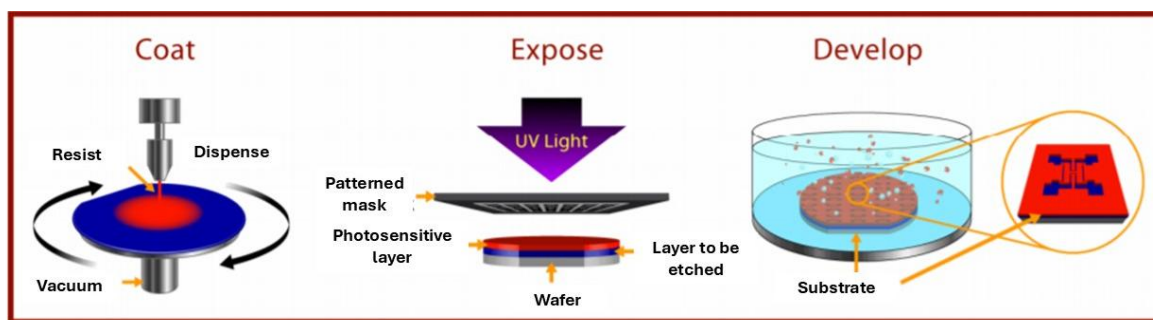


Figure 2.7 Main surface preparation steps of lithography [29]

Chen and coworkers [30] developed a method using nanosphere lithography to produce extensive, periodic arrays of nanospheres with tunable wettability. Initially, a monolayer of monodisperse polystyrene beads was formed via spin-coating, and the bead sizes were subsequently decreased from 440 nm to 190 nm through oxygen plasma etching. The arrays were then transformed into superhydrophobic surfaces by depositing a 20 nm thick gold film followed by modification with 1-octadecanethiol. This strategy effectively reduced the solid–liquid contact area, enabling the CA to be adjusted from 132° to approximately 170° (**Figure 2.8**). The study clearly demonstrates that the engineered nanostructure, rather than just the low-surface-energy coating, plays a decisive role in achieving superhydrophobicity.

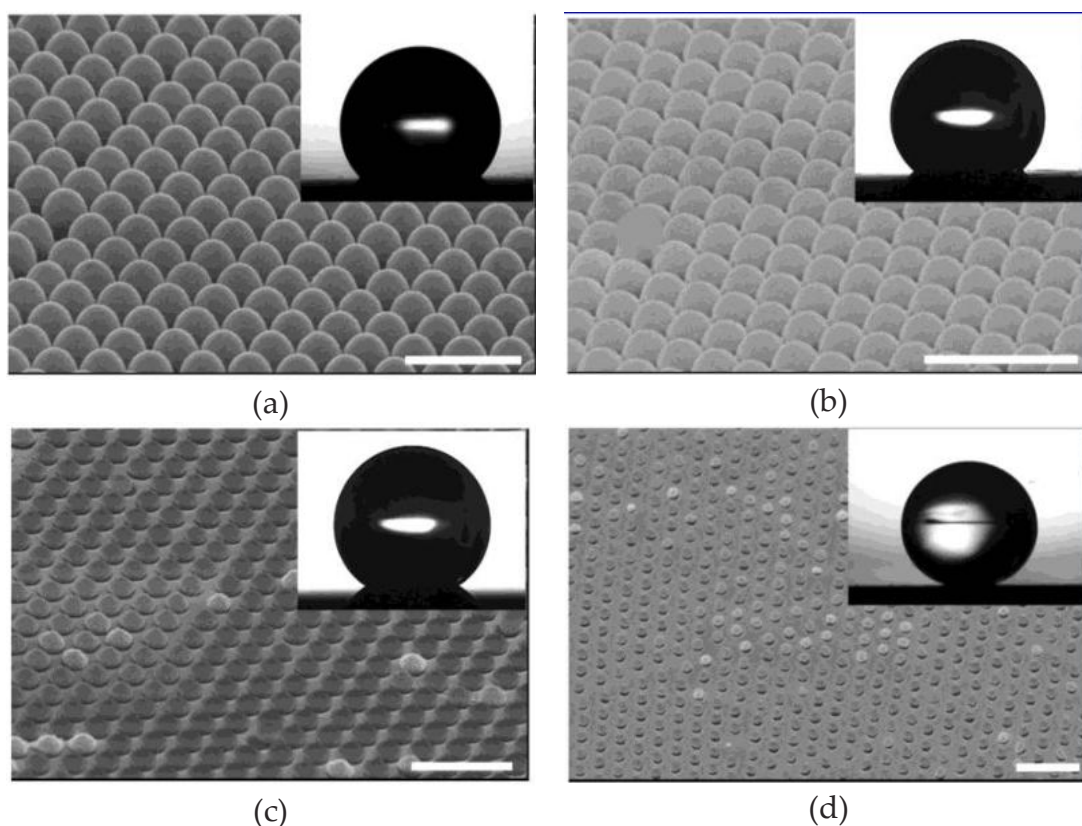


Figure 2.8 SEM images of the size-reduced PS beads and the CA measurement on the corresponding modified surfaces. The diameters of polystyrene beads and CAs on these surfaces were measured to be (a) 400 nm, 135°, (b) 360 nm, 144°, (c) 330 nm, 152°, and (d) 190 nm, 168° [30]

Martin et al. [31] employed electron beam lithography to fabricate nanopillars on silicon substrates and examined how the pillars' geometry influences water repellency after surface modification with octadecyltrichlorosilane. Keeping the spacing between pillars constant, they varied the height-to-diameter ratio and discovered that taller, slimmer structures enhance and stabilize superhydrophobicity compared to unmodified smooth surfaces.

Among recent lithographic techniques, nanoimprint lithography has come into the spotlight. Assumably due to its high precision, and flexibility in patterning surfaces. The joint application of nanoimprint lithography with diverse radiations such as UV, laser, or e-beam is a common practice [32, 33].

Kim et al. [34] presented a new approach to produce transparent, superhydrophobic PET films that combine high optical clarity with robust mechanical durability. The process begins by fabricating an elastomeric PDMS stamp via lithographic techniques, which is then patterned using UV imprinting to generate micro-pillar arrays. These patterned micro-cavities are subsequently filled with a UV-curable polymer through drop-casting, and additional UV exposure facilitates the transfer of the microstructures onto the PET substrate. Once the PDMS stamp is removed, the resulting hierarchically textured surface exhibits CAs close to 170° and very low CAH. Remarkably, the superhydrophobic properties are preserved even after 2000 bending cycles under stringent conditions, underscoring the film's exceptional flexibility and durability. This method holds significant promise for practical applications, including those in the glass industry, where both transparency and long-term water repellency are critical.

Lithography is a useful technique for producing superhydrophobic surfaces, but its processes, including the use of nanoscale roughness templates or vacuum-based surface coatings, are too costly for large-scale applications.

Templating

The templating technique is an economical, and reliable approach for generating patterned surfaces essential for superhydrophobicity. In this method, a master template—sourced either from natural formations [35] and synthetic surfaces [36] —is chosen to provide the desired design. A polymer resin, such as PDMS, is introduced into the master template's features and then applied to a substrate, resulting in a negative imprint of the original pattern. Once the master is removed by dissolution or peeling, the negative template can be reused to replicate the positive pattern by repeating the process. Both the inverse and replicated templates serve as valuable tools for fabricating surfaces with controlled micro- or nanoscale roughness, leading to superhydrophobic behavior (**Figure 2.9**) [37].

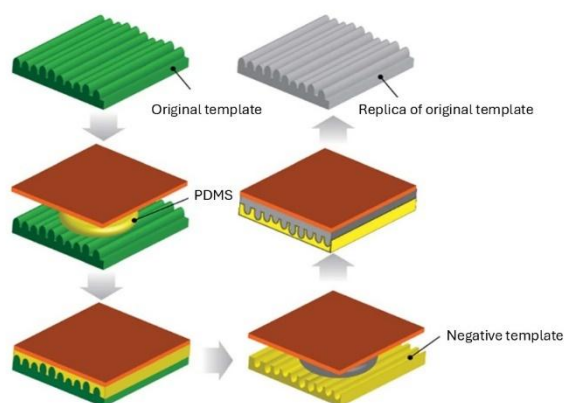


Figure 2.9 Schematic diagram of the templating technique to produce micro and nano-scale roughened surface [17]

Sun et al. [38] introduced an new strategy to replicate the superior water-repellent properties of a lotus leaf by using the leaf itself as a template and casting with PDMS (**Figure 2.10**). This technique successfully reproduces the hierarchical structure of the natural surface. The resulting positive replica demonstrates a CA of approximately 160° , closely copying that of the natural lotus leaf. This approach, utilizes nanoscale casting of a biological template, offers a compelling alternative to traditional microfabrication or chemical synthesis methods for creating complex textured surfaces.

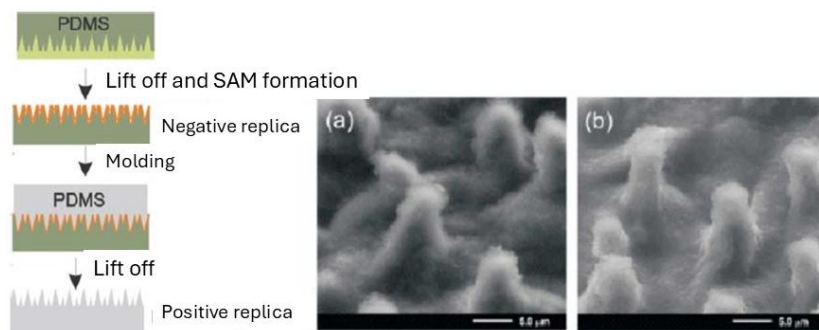


Figure 2.10 Schematic diagram of (left) the lotus leaf replication process and (right) SEM images of (a) the natural lotus leaf and (b) its positive replica [38]

Lee et al. [39] investigated the possibility of reproducing the textures found on plant leaves using UV-based nanoimprint lithography (UV-NIL). By transferring the fine micro- and nanostructures from leaves of species such as tulip tree, silver maple, bamboo, and lovegrass into a photoreactive polymer, researchers were able to create rough films that display significantly enhanced water repellency—demonstrated by an increase in CAs of about 90° compared to smooth surfaces. Although the exact duplication of the original leaf patterns was not fully achieved but indicating a high degree of structural accuracy. These findings confirm that UV-NIL, performed without additional chemical treatments, provides an efficient and straightforward method for fabricating hydrophobic films.

2.3.2. Bottom-up methods

Dip coating

Dip-coating is a versatile method widely used to achieve superhydrophobic surfaces. The process starts by thoroughly cleaning the substrate to eliminate any contaminants. Next, the substrate is immersed in a chemical bath that contains nanoparticles, which not only generate the necessary rough texture but also incorporate agents that lower the surface energy. Finally, the treated substrate is subjected to a drying or curing phase that stabilizes the coating, resulting in a durable, water-repellent surface (**Figure 2.11**) [17].

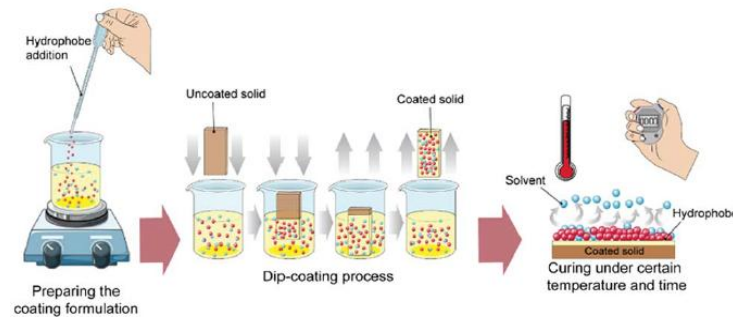


Figure 2.11 Schematic illustration of the dip-coating process [17]

Cui et al. [40] developed superhydrophobic coatings on epoxy substrates by integrating micro- and nanoscale features through a three-stage process (**Figure 2.12**). The fabrication began with sandblasting to create a rough micro-textured foundation on the substrate. Next, the surface was immersed in a solution containing silicon dioxide (SiO_2) particles combined with an epoxy adhesive, promoting the formation of nanoscale features on the pre-roughened surface. A final dip-coating step with a modified epoxy adhesive further reduced the surface energy, thereby enhancing water repellency. This method resulted in coatings exhibiting CAs close to 167.8° and CAHs as low as around 7.2° , while also demonstrating excellent durability against mechanical abrasion and exposure to various chemicals. The study highlights the importance of both hierarchical structuring and low surface energy in achieving superhydrophobicity and presents a scalable approach for fabricating such surfaces on a large scale.

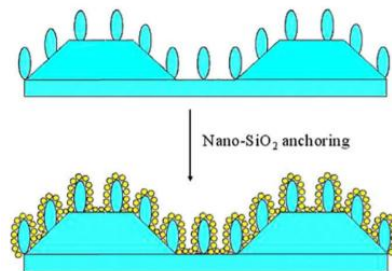


Figure 2.12 Schematic illustration of the multi-scale structures prepared on the sandblasted paint surface by nano-SiO₂ anchoring [40]

In one study, commercial silica particles were functionalized by introducing methyl groups through a reaction with methyltrichlorosilane. These modified particles were sequentially deposited in multiple layers using a dip-coating process, resulting in a coating that exhibits a CAs of about 153° and a CAHs close to 8° . The final surface not only demonstrates exceptional water repellency and low adhesion but also retains its superhydrophobic characteristics under the stress of water jet impact, making it a promising strategy for industrial applications requiring durable, low-maintenance, and self-cleaning surfaces [41].

In addition to the dip-coating method, several other techniques are known in the coating family that are commonly used for superhydrophobic coating formation such as spin coating, spray coating, flow coating. In the case of a spin coating, the substrate is rotated at high speed around an axis perpendicularly to the coating plane. Spin coating is a widely employed method for fabricating thin films by rapidly rotating a substrate to evenly distribute a small volume of coating solution. This technique, along with alternatives like spray and flow coating, is valued for its ability to produce uniform layers, which is particularly useful in creating superhydrophobic surfaces. Moreover, in the fabrication of electronic devices, spin coating plays a crucial role in determining film morphology and molecular organization. By adjusting the spinning duration, it is possible to control the microstructure evolution during polymer solidification. Additionally, the use of processing additives can further influence the drying behavior and final properties of the films, thereby affecting overall device performance [42].

Spray-coating is another frequently used technique for fabricating superhydrophobic surfaces, where a uniformly dispersed solution of functional compounds is atomized and directed onto a substrate. Following deposition, the evaporation of the solvent solidifies the coating onto the surface. This technique has been effectively utilized in addressing environmental challenges such as oil-water separation. For instance, the combination of superhydrophobic coating with an adhesive can produce a cellulose sponge uniformly decorated with TiO_2 nanoparticles. The resulting material exhibited CAs reaching 171° . Notably, the coating maintained its performance under chemically aggressive conditions and repeated mechanical abrasion, underscoring its potential for reliable industrial application [43].

Sol-gel process

The sol-gel method is a well-established technology for producing three-dimensional networks of nanoparticles that impart superhydrophobic properties to coated substrates. The sol-gel method is a versatile, solution-based process that creates a three-dimensional macromolecular network with superhydrophobic properties. Initially, a sol is prepared by dissolving a metal salt or metal alkoxide—typically in an alcoholic medium—and then inducing hydrolysis. This hydrolysis, particularly of metal alkoxide groups, produces silanol groups that undergo condensation (or polycondensation) reactions, forming a precursor. Under mild reaction conditions that allow for kinetic control, this precursor gradually gels to form a continuous network,

with various morphologies and structures achievable. Hydrophobic additives are often incorporated during this process to ensure the final coating exhibits superhydrophobicity [44, 45].

A novel optically clear and superhydrophobic silica coating on glass has been developed by combining controlled sol-gel synthesis with targeted surface modifications. This process involves precisely adjusting the hydrolysis and condensation reactions of various silica precursors to create nanometer-scale roughness using nanoclusters and nanoparticles. Such microstructuring not only preserves high visible light transmittance (above 90 %) and minimizes reflectivity (under 10 %) but also facilitates the trapping of air beneath water droplets, thereby enhancing water repellency. Subsequent surface functionalization with a self-assembled layer of water-repellent agents, such as fluoroalkylsilane, results in CA of approximately 165°. Overall, the integration of these chemical and structural strategies offers a promising route for further improving the hydrophobic performance of silica-based coatings [46].

Cotton fabrics were modified to achieve significant water and oil repellency through a dual-step surface treatment that creates hierarchical nanoscale structures. Initially, a silica sol with a controlled particle size was deposited onto the cotton via a conventional pad-dry-cure process. This was followed by the application of a synthesized PDMS, which chemically altered the fabric surface. The resulting dual-scale structure—arising from the dense silica nanoparticle layer—plays a pivotal role in imparting superhydrophobic characteristics. The treated fabrics (Figure 2.13) exhibited water and oil contact angles of 145° and 131°, respectively, underscoring the effectiveness of the combined sol-gel and surface modification strategy [47].

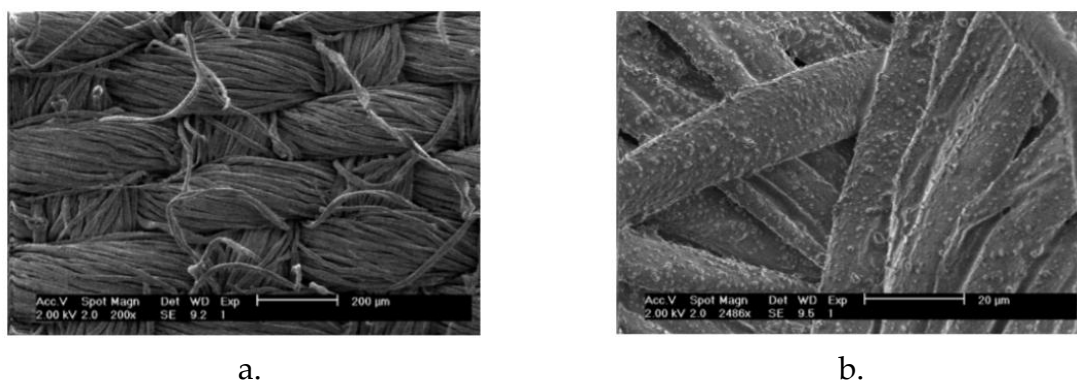


Figure 2.13 SEM image of the cotton textile (a) and (b) the particle covered cotton textile modified with PDMS [47]

Phase separation

In the phase separation method, a homogeneous solution is first prepared by dissolving a polymer in a high-boiling-point solvent at elevated temperatures. Phase separation is induced by external parameters, allowing these methods to be categorized based on how the process is initiated [48].

For thermally induced phase separation (TIPS), external factors such as temperature or pressure induce phase separation and solidification, and the removal of the solvent results in the formation of a microporous structure [49]. In contrast, non-solvent induced phase separation (NIPS) is initiated by introducing a non-solvent that is miscible with the solvent, thereby precipitating the solution into two distinct phases. As a result, a bicontinuous structure is formed, consisting of a polymer-rich phase that creates the patterned structure [50] and a polymer-poor phase that develops into a porous network [51, 52].

A novel approach was developed to fabricate superhydrophobic polycarbonate (PC) coatings by utilizing a TIPS mechanism initiated through solvent treatment. PC sheets were immersed in acetone, which induced swelling and partial melting of the polymer. As the solvent penetrated the material and subsequently evaporated, the polymer crystallized into a hierarchical architecture consisting of micron-scale spherulites decorated with nano-scale fibrils. This dual-scale texture not only yielded CA above 150° but also resulted in low CAH, efficient droplet roll-off, and high impact resistance. The precise morphology of the surface could be controlled by adjusting treatment parameters such as immersion time, acetone concentration, and drying conditions. This solvent-induced crystallization strategy shows promise for application to other solvent-polymer systems to produce similarly robust superhydrophobic surfaces [53].

Zhang et al. [50] developed a novel fabrication strategy to create superhydrophobic PC surfaces via NIPS. Initially, PC is dissolved in dichloromethane, and as the solvent gradually evaporates, the substrate is immersed in a range of non-solvents. This dipping and drying process—conducted under ambient conditions without requiring additional low-surface-energy agents or strict environmental controls—leverages the ability of carefully chosen non-solvents (selected for their strong swelling effect on the polymer, low compatibility, and high boiling points) to induce phase separation. The result is a hierarchical micro/nano-structured coating that mimics the lotus leaf effect, with its surface properties adjustable through variations in the non-solvent treatment, as shown in **Figure 2.14**. This versatile approach is especially promising for fabricating superhydrophobic coatings on substrates with intricate geometries, such as microfluidic channels or biomedical implants, and even opens the possibility for developing colored superhydrophobic surfaces.

Phase separation can be used with semi-crystalline polymers [54-56]. Since it is simple and also cost-effective in industrial-scale production [57, 58]. Methods for producing roughened structured surfaces on PP using phase separation are becoming more widespread [51, 59]. A large number of publications investigate the effects of the processing parameters of TIPS on surface structure formation. These processing parameters include solvent type and its temperature [54], solution concentration [49], and cooling conditions [60]. Many different textures can be produced by altering the concentration of the polymer solution. Typically, a dilute polymer solution produces

a surface with cellular morphology, while a highly concentrated solution or gels lead to sphere-like structures [61]. The main advantages of phase separation methods are that they are easy to reproduce and control, and the structure is less prone to defects.

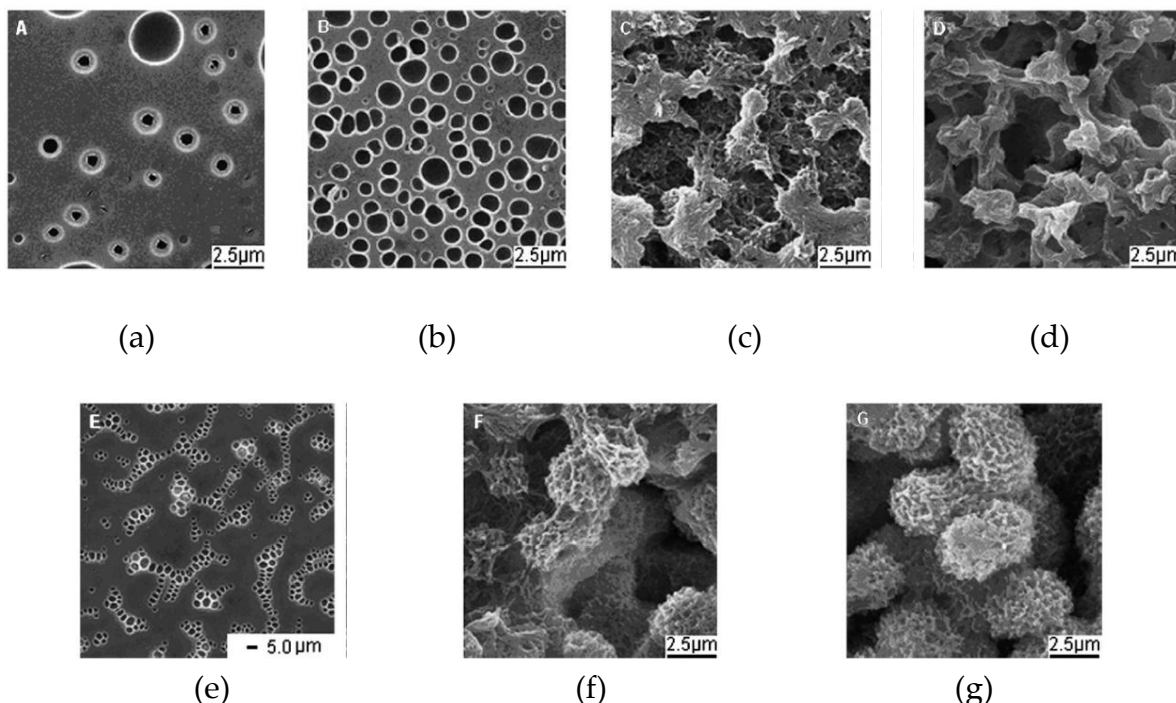


Figure 2.14 SEM images of the surface morphologies of PC coatings formed with different non-solvent. (a) before applying any non-solvent, (b) isopropyl alcohol, (c) acetone, (d) toluene, (e) benzene, (f) methyl ethyl ketone, and (g) 1, 4-dimethylbenzene non-solvents [50]

First, Erbil et al. [62] reported on superhydrophobic PP surfaces using the phase separation method with the appropriate selection of solvents, non-solvents, and cooling temperature (**Figure 2.15**).

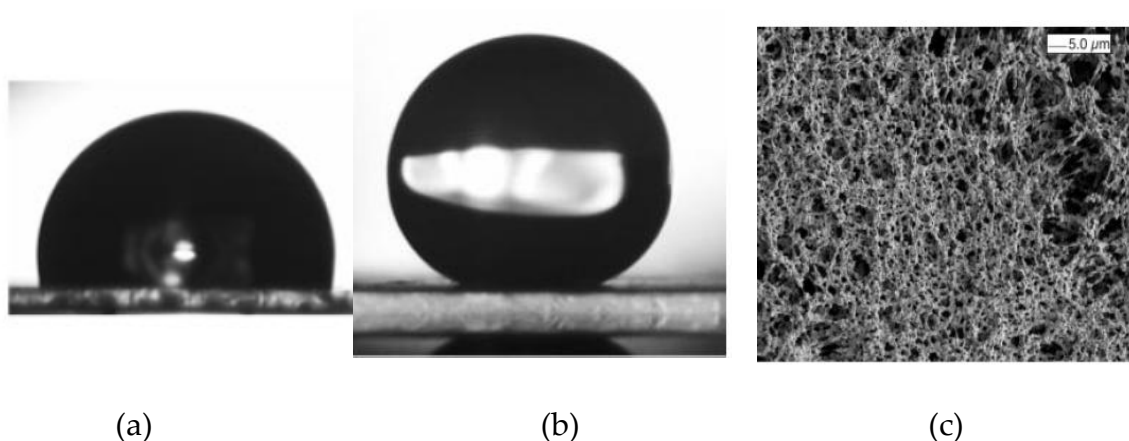


Figure 2.15 The CA of $104^{\circ} \pm 2^{\circ}$ on smooth PP surface (a) and of 160° on superhydrophobic PP coating (b) and SEM image of superhydrophobic PP coating (c) [62]

They used xylene as a good solvent to dissolve PP and methyl ethyl as a non-solvent to decrease crystallization time and increase nucleation rate, resulting in a

surface with smaller spherulite and larger pores, which contributes to enhanced surface hydrophobicity [63]. They could control the surface roughness by the concentration, temperature, and the amount of non-solvent [62]. Ji et al. [64] fabricated a stable superhydrophobic PP surface using acetone as the non-solvent. The roughened surface reached CA around $160^{\circ} \pm 1.9^{\circ}$ and a CAH less than $4^{\circ} \pm 1.9^{\circ}$.

Numerous researchers have studied the effect of solvents and the crystallization behavior of PP in surface structure formation during phase separation [65, 66].

In the case of non-nucleated PP grades, the course of crystallization depends on the regularity of the chain structure and the crystallization conditions [67]. Zhu et al. [68] systematically studied the effect of the tacticity of PP on surface wettability by changing the mass ratio of the composite coatings of atactic and isotactic PP. The created surface was water repellent and wetting behavior could be tuned from slippery to sticky with a change of the ratio of atactic and isotactic PP. The distinct wetting behaviors can be attributed to different morphologies, which resulted from the different crystallization behaviors of PP with different regularity.

We can also find an example of constructing superhydrophobic PP surfaces with the addition of a nucleating agent. The distribution and size of spherulites and the rate of phase separation can be influenced with nucleating agents, which can be inorganic and organic materials like salts, fillers, organic acids, polymers and apolar materials soluble in the molten polymer [69, 70]. Lui et al. [71] integrated halloysite nanotubes (HNTs) into a PP solution to modify the surface architecture and increase hydrophobicity, as can be seen in **Figure 2.16**.

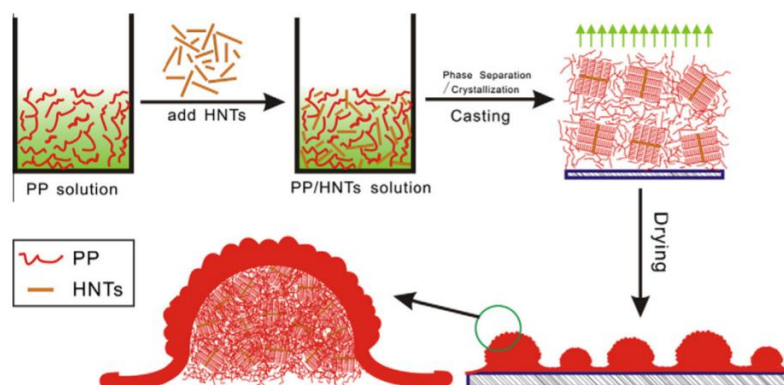


Figure 2.16 Schematic illustration of the mechanism of PP/HNTs hybrid spherulite superstructure formation [71]

The process began with blending to ensure the HNTs were uniformly dispersed in the PP solution, which was then cast onto substrates. As the solvent evaporated, phase separation occurred—forming distinct polymer-rich and polymer-poor regions—with the HNTs serving as nucleation sites that accelerated this separation. This led to the development of complex hybrid structures exhibiting both micro-scale and nano-scale roughness. The resulting surfaces demonstrated exceptional water-repelling properties, with CAs exceeding 160° and CAH below 10° . Furthermore, the

inclusion of HNTs improved the thermal stability of the PP, likely due to enhanced interfacial bonding.

The structure of superhydrophobic surfaces made from solutions is influenced by the polymerization catalyst. Zhang et al. investigated superhydrophobic surfaces made of metallocene catalyst PP (m-PP) and Ziegler–Natta catalyst PP (ZN-PP). Depending on the type of catalyst, the micro-sized particles formed on the surface differed. A micro-sized crystalline particle formed on the surface of the m-PP film, while the ZN-PP particle surface exhibited an amorphous state. They found that during film formation, m-PP had a higher crystallization rate and higher crystallinity. The results also revealed that the superhydrophobicity and thermal stability of the m-PP film were much better than those of the ZN-PP film [72].

2.4. Light scattering

The study of surface wettability and roughened surfaces provides valuable insight into how hierarchical structures influence liquid-solid interactions, enabling the development of superhydrophobic and water-repellent surfaces. However, beyond their wetting properties, micro and nanostructured surfaces also play a significant role in manipulating light through scattering phenomena. This ability to scatter light, both at the surface and within the volume, is essential for applications where high reflectivity is required. Therefore, understanding light scattering mechanisms serves as a critical bridge between surface morphology and the development of reflective and reflective heat insulating coatings, further expanding the multifunctionality of these engineered surfaces.

As the refracted light passes through the sample, it can lose energy by scattering as well as by light absorption. As far as scattering is concerned, both the bulk and surface properties of coatings should be considered. The scattering is analogous to surface reflection, but in this case, the phenomenon is caused by the fluctuation of the refractive index in the bulk phase. In the case of semi-crystalline polymers, scattering is primarily caused by supermolecular formations [73].

2.4.1. Volume scattering

Volume scattering occurs when light is scattered as it travels through the interior of a material, resulting from inhomogeneities in its optical properties, such as variations in refractive index, within the bulk. Volume scattering can be categorized into single-particle scattering, multiple scattering, and coherent scattering (**Figure 2.17**) [74]. As long as the number of scattering particles is not excessively large, volume scattering can be approximated as the sum of single-particle scattering events. This relationship is mathematically described using the packing fraction (η) described by the following Equation 2.5.

$$\eta = \frac{NV_{scat}}{V_{vol}} \quad (2.5)$$

where N is the number of the single scatterer, V_{scat} is the volume of the single scatterer and V_{vol} is the volume where the N single scatterer are located.

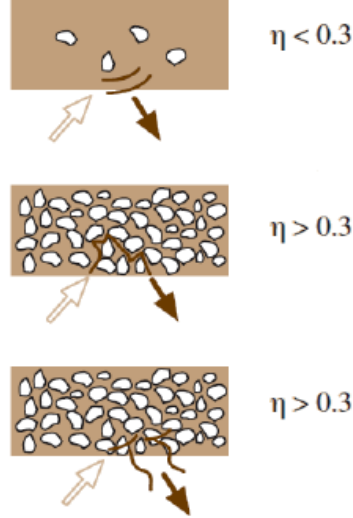


Figure 2.17 Dependence of single-particle, multiple and coherent scattering on packing fraction [74]

The power of total scattered light in all directions (P_{scat}) can be calculated as follows using Equation 2.6:

$$P_{scat} = C_{scat} \cdot I_{in} \quad (2.6)$$

where C_{scat} is the scattering cross-section and I_{in} is the intensity of the incident light. The power of the scattered light in a given direction can be characterized with Equation 2.7:

$$\frac{dC_{scat}}{d\Omega} = R^2 \frac{I_{scat}}{I_{in}} \quad (2.7)$$

where Ω is the spatial angle, R is the distance between the scattering center and the observer, and I_{scat} is the intensity of the scattered light.

The scattering cross-section can be calculated using the following Equation 2.8.

$$C_{scat} = \int_0^\pi \frac{dC_{scat}}{d\Omega} d\Omega \quad (2.8)$$

The total and differential scattering cross-sections can only be calculated for a few geometries, such as spheres.

In multiple scattering, light interacts with a large number of scattering particles, but no interference effects are observed between the scattered waves. Each scattering event is independent, and the resulting scattered intensity is simply the sum of the

intensities produced by individual particles. Multiple scattering typically occurs when the packing fraction (η), exceeds 0.3. Due to the complexity of interactions between the densely packed particles, multiple scattering phenomena are usually described through numerical simulations rather than analytical methods [74].

In contrast, coherent scattering involves interference effects between light waves scattered by different particles. Whether scattering is coherent depends on the relationship between the distances separating the scattering centers and the coherence length of the light. The coherence length is the distance over which the light waves remain phase-correlated. When the distances between scattering centers exceed the coherence length, the scattered waves are uncorrelated, and the scattering events can be considered independent. In such cases, the scattering is termed incoherent, and the total scattered intensity is the sum of the intensities from all scattering centers. However, when the distances between the scattering centers are smaller than the coherence length, interference effects arise. In this situation, the scattering is coherent, and the phases of the light waves scattered by different centers are correlated. The total scattered intensity is no longer a simple sum of individual contributions; instead, it is influenced by interference phenomena, which can amplify or diminish the intensity depending on the scattering vector. The scattering vector is the difference between the wave vectors of the incident and scattered light. In coherent scattering, the spatial arrangement and periodicity of the scattering centers play a critical role. For instance, periodic or quasi-periodic structures, enhance coherent scattering and result in directional or wavelength-dependent effects, such as structural colors [74].

When light interacts with matter, the nature of the scattering depends on various factors, including the particle size relative to the wavelength of light and whether the scattered light maintains its frequency. Elastic scattering occurs when the scattered light retains the same frequency as the incident light, whereas inelastic scattering involves a frequency shift. The latter is typically classified as Raman scattering or Brillouin scattering, depending on the mechanism of energy exchange between the photons and the material [75].

Elastic scattering can be further divided into Rayleigh scattering and Mie scattering, which describe how light interacts with particles of different sizes. Rayleigh scattering occurs when the particles inducing scattering are much smaller than the wavelength of the incident light. The scattered intensity is inversely proportional to the fourth power of the wavelength [76].

Mie Scattering occurs when the particle size is comparable to the wavelength of light. Unlike Rayleigh scattering, the scattered intensity in Mie scattering is less dependent on wavelength and more influenced by the particle's size and refractive index. Water droplets, ice particles, and smoke exhibit Mie scattering, which explains phenomena like the white appearance of clouds or the scattering of headlights in fog [77].

2.4.2. Surface scattering

A perfectly smooth surface reflects light exclusively in the specular direction, following the law of reflection, where the angle of incidence equals the angle of reflection. However, when a surface exhibits irregularities or roughness, light undergoes additional scattering, redistributing part of the incident light into directions other than the specular one. As illustrated in **Figure 2.18**, a rough surface not only produces specular reflection but also diffusely scattered light, which spreads over a wide angular range around the specular direction. The extent of this scattered light — often referred to as stray light — depends on the roughness of the surface relative to the wavelength of the incident light [74].

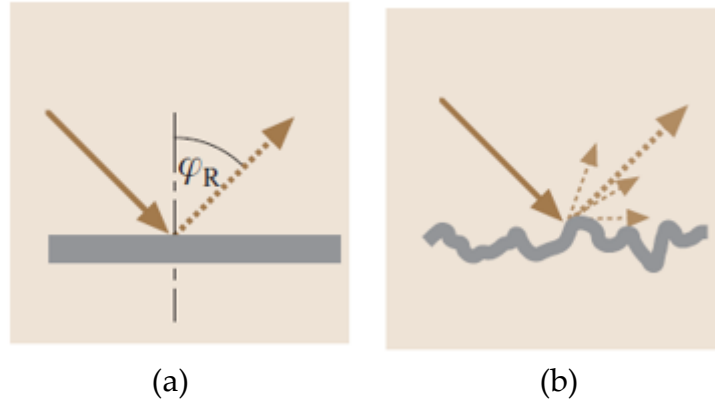


Figure 2.18 Schematic of (a) specular light reflection at a perfectly smooth surface and (b) light scattering at a rough surface, where solid line labels incident light and dotted line labels scattered (reflected) light [74]

Total Integrated Scattering (TIS) is a widely used optical technique developed to quantify surface roughness by measuring the ratio between the intensity of incident light and the intensity of scattered light [48, 78]. For the TIS model to be applicable, the following conditions must be satisfied: (1) the surface roughness must be significantly smaller than the incident wavelength and (2) the surface must be smooth and well-polished, ensuring minimal deviations from ideal specular reflection [79]. TIS value can be calculated by Equation 2.9

$$TIS = R_0 \left[1 - \exp \left[- \left(\frac{4\pi\sigma \cos \alpha_R}{\lambda} \right)^2 \right] \right] \quad (2.9)$$

where R_0 is the reflectance of the surface, σ is the root mean square roughness, α_R is the incident angle, λ is the wavelength of the incident light.

Based on the Equation 2.9, the following conclusions can be drawn [74]:

- Scattering is directly related to the root mean square surface roughness — smoother surfaces result in lower TIS values.

- Surface reflectance is inversely proportional to the amount of scattered light—as roughness increases, more light is scattered rather than being reflected in the specular direction.
- The scattering intensity increases as the wavelength decreases—shorter wavelengths interact more strongly with surface roughness, leading to higher scattering losses [79].

2.4.3. Optical properties of spherulites

Following the exploration of light scattering mechanisms, it is important to further examine the role of internal microstructures, such as spherulites, in influencing the optical properties of polymeric materials. Spherulites, common crystalline structures in semicrystalline polymers like PP significantly influence light propagation and scattering within the material. Understanding the optical behavior of spherulites not only enhances the design of reflective materials but also contributes to the development of surfaces with tailored optical performance for applications in reflective heat-insulating materials.

Spherulites, fundamental structures in semicrystalline polymers, form under quiescent crystallization conditions. They consist of radially growing lamellae or fibrils, which branch at small, non-crystallographic angles to achieve complete spatial filling (**Figure 2.19**). Two primary types of spherulites are recognized: radial, with straight fibrils, and ringed, characterized by twisted fibrils [80-82].



Figure 2.19 Schematic draw of central multidirectional growth spherulite [81]

Semicrystalline polymers are optically anisotropic birefringent materials. Birefringence (Δn) is an optical property of materials in which the refractive index depends on the polarization and propagation direction of light passing through the material. It serves as a measure of optical anisotropy. In spherulites, birefringence is determined by the difference between refractive indices in the radial and tangential directions (Equation 2.10) [83].

$$\Delta n = n_r - n_t \quad (2.10)$$

where n_r and n_t denote the radial and tangential refractive index of the spherulites, respectively [84, 85]. The refractive index of polymers is influenced by bond

polarizability, with photons encountering resistance from electrons during light propagation. In general, the refractive index is higher along polymer chains than perpendicular to them due to the density of bonds in the parallel direction. Within spherulites, lamellae are formed by folded polymer chains oriented nearly perpendicular to the spherulite radius, resulting in a tangential refractive index that typically exceeds the radial one. This alignment makes negative birefringence a common feature in polymers.

Positive birefringence arises when wide-angle fibril branching redirects chain alignment toward the radial direction, increasing the radial refractive index. The refractive index values reported in the literature for polymers typically represent average value, as they are influenced by factors such as chain orientation, conformation, and the presence of various phases [80, 82]. Most polymers exhibit refractive index values between 1.3 and 1.7, with of α -PP having a refractive index of approximately ~1.471 [86].

2.5. Fabrication of solar-reflective polymer materials

Lately, energy-saving, CO₂ emission, and environmental footprint reduction have become key expressions as climatic changes are undoubtedly related to human industrial activity. The industrial and residential building sectors consume significant amounts of electric energy. Moreover, increasing demands and improving living standards in developed and developing countries create additional energy consumption, partly because of the widespread use of air conditioning [87, 88]. Much of the energy used in buildings, approximately 40 %, is used for cooling or heating [89]. Therefore, it is important to create sustainable methods like reflective thermal insulation techniques to counteract the increase in the temperature of buildings. These methods help keep a desirable level of comfort while minimizing energy consumption. These methods have significant appeal due to their economic viability, making them suitable for buildings in developing countries. Heat and solar reflective techniques help to reduce most of the direct solar heat gain inside the building [90]. Also, the energy consumption of air conditioners could be reduced considerably if more incoming solar radiation could be reflected from the walls. As part of this endeavor, technologists work on developing materials with reflective and thermal insulation properties in the whole solar spectrum, which could effectively reduce energy demand [91].

The products having reflective thermal insulation surfaces can reflect solar energy rather than absorb it. About 5 % of the energy of solar radiation is in the ultraviolet band (UV) (200–400 nm), 43 % in the visible band (VIS) (400–800 nm) and 52 % in the near-infrared band (NIR) (800–2500 nm), therefore it is important to improve reflectivity in all these bands to increase the efficiency of solar reflectivity [92, 93]. Usually, the reflectivity of products can be evaluated with total solar reflectance (TSR). TSR shows the amount of energy of incident sunlight reflected from the surface.

Higher TSR means better reflectivity and higher thermal insulation in the range of 200 to 2500 nm [92].

In general, effective strategies can serve as a basis for producing solar-reflective polymer materials: adding optically active additives to the polymer matrix [94, 95] and creating porous polymers [96-98].

2.5.1. *Optically active additives*

Incorporating reflective additives, preferably fillers with high reflectivity in both the visible and near-infrared ranges, can effectively enhance the thermal insulation performance of materials [94, 98].

To achieve efficient solar reflectivity and, consequently, thermal insulation in engineered materials, high-refractive-index particles are typically embedded in low-refractive-index matrices to create a strong refractive index contrast. Reflective additives such as titanium dioxide (TiO_2), barium sulfate (BaSO_4), and zinc oxide (ZnO) are widely used in the fabrication of polymer composites with enhanced solar reflectivity [94, 98]. Numerous factors have been investigated, such as the size, amount and distribution of particles in the polymer matrix, the degree of adhesion between the particles and the matrix, and the thickness of the reflective coatings [99-101].

For example, the diameter of the particles greatly determines the intensity and directional distribution of scattered light. In accordance with Fresnel's rules, particle can effectively reflect light of a certain wavelength if the diameter of the particles is close to half the wavelength of the reflected light [102].

So far, TiO_2 is used in the largest amount as a reflective additive, since TiO_2 has a relatively high refractive index ($n \approx 2.6$) and little absorption in the visible spectrum. Commercially available TiO_2 pigments are one of the most effective additives to reflect visible light, as their particle size distribution is optimized within the range of 200 to 300 nm. However, their near-infrared reflectance remains quite moderate [103]. In order to increase the reflective ability of TiO_2 in the whole solar spectrum, multiple particle sizes are considered because the wider the size distribution of the particles, the wider the spectrum they can reflect, but at the same time, its implementation hides many challenges [104].

A uniform dispersion of high-refractive-index particles within a polymer matrix is essential for achieving optimal light scattering and, consequently, effective solar reflectivity. However, particle aggregation often occurs, particularly at higher additive concentrations, reducing the overall scattering efficiency. This issue has been addressed through various approaches, such as the use of dispersants, extenders, and polymer-encapsulated particles [94, 101]. Nevertheless, ensuring a homogeneous optical response by evenly distributing particles throughout the matrix remains a challenge.

However, TiO_2 particles have a strong tendency to aggregate, particularly at high loading levels. When aggregation occurs, the overlapping particles reduce the

available surface area for scattering, thereby diminishing their effectiveness in enhancing solar reflectivity [94].

One promising strategy to mitigate aggregation is surface modification of TiO_2 particles. For instance, Dong et al. [94] successfully coated TiO_2 particles with polystyrene (PS) and polymethyl methacrylate (PMMA) layers using emulsion polymerization. The polymer-coated TiO_2 particles exhibited improved dispersion within the polymer matrix, leading to enhanced multiple scattering of light. This resulted in a significant increase in NIR reflectance and an overall improvement in TSR (**Figure 2.20**) [94].

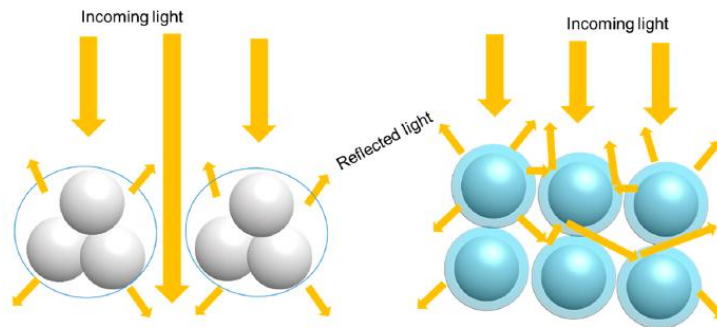


Figure 2.20 Schematic of the light scattering efficiency of surface coating produced using uncoated and coated TiO_2 particles [94]

The application of polymer coatings to TiO_2 particles has been shown to enhance their reflectance performance by 7-10 %, while the impact of the coating material itself on reflectance remains negligible [94]. This improvement arises from the increased spatial separation of TiO_2 particles, which prevents excessive clustering and maximizes light-scattering efficiency.

TiO_2 exists in multiple crystalline forms, including rutile, anatase, and brookite. Among these, rutile TiO_2 is the most widely used in reflective coatings due to its superior light-scattering efficiency in the visible spectrum [105].

A comparative study by Qi et al. [106] examined the scattering properties of rutile and anatase TiO_2 in acrylonitrile-styrene-acrylate (ASA)/ TiO_2 hybrid coatings. Their results showed that samples containing rutile TiO_2 exhibited higher solar reflectance than those with anatase TiO_2 , attributed to rutile's higher refractive index (**Figure 2.21**). However, the study also noted that TiO_2 absorbs ultraviolet (UV) radiation, which explains the lower reflectance in the UV range.

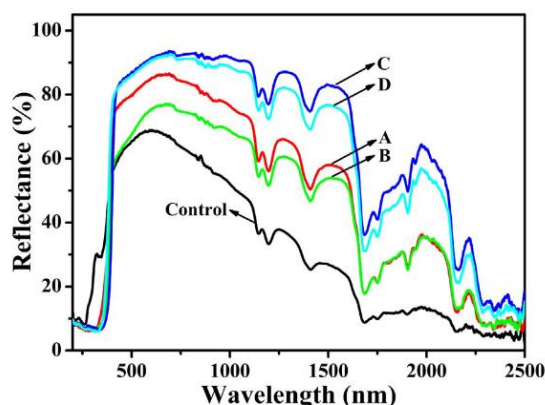


Figure 2.21 Reflection spectra of pristine ASA and ASA/TiO₂ hybrid materials (A) hydrophobic rutile nano-TiO₂, (B) hydrophilic anatase nano-TiO₂, (C) unmodified rutile TiO₂, (D) unmodified anatase TiO₂ [106]

2.5.2. Porous polymer layers

For years, conventional applications have been dominated by scattering systems utilizing inorganic nanoparticles. However, introducing porous structures is another strategy for producing solar-reflective polymer materials, contributing to reducing indoor heat and, as such, the energy consumption in buildings exposed to solar radiation. The porous layer formed on the surface of a products can significantly increase the reflectance value of the surface. By creating air voids in the porous structure, we can eliminate the need for reflective additives. The part of the incoming light that enters the porous layer undergoes multiple reflections on the surface of the pores, thereby reducing the amount of absorbed energy, and thus acts as a thermal insulation layer. In recent times, there has been a surge in research interest towards solar reflective porous polymer materials, owing to their promising structure, cost-effectiveness, ease of preparation, and wide availability [107].

Porous structures offer superior thermal insulation and solar reflective properties, due to the refractive index differences between the air voids and the polymer matrix [108, 109]. The micropores were found to strongly scatter visible and near-IR radiation, while nanopores enhanced scattering in the UV and visible regions [110].

Numerous simulations were conducted to explore the impact of microscopic parameters of the porous material on optical properties [111-113]. Typically, in these simulations, scattering units containing circular voids with varying diameters (0.1 μm to 10 μm) were modeled using Mie scattering theory. According to Mie scattering theory, solar reflectivity is maximized by scattering units that induce a significant refractive index mismatch within the hierarchical porous structure [114]. These results revealed a significant size-dependent effect on scattering efficiency and wavelength. Air voids with a diameter of 1 μm demonstrated the highest weighted average scattering efficiency due to their optimal scattering band coverage and higher efficiency in the solar spectrum. Although larger air voids (2 to 10 μm) exhibited slight

decreases in the weighted average scattering efficiency, they still provided substantial scattering across the entire solar range. Similarly, Mandal et al. [93] demonstrated that micropores ($\sim 5.0\ \mu\text{m}$) can efficiently scatter sunlight across the entire solar spectrum, while nanopores ($\sim 0.2\ \mu\text{m}$) enhance UV and visible scattering.

Increasing the porosity can reduce the dependence of reflectivity on thickness. Wang et al. [115] developed a multilayered PMMA film using a templating method that creates a dual-porosity structure: a well-organized, dense microporous surface combined with an underlying layer filled with randomly distributed nanopores. With a porosity of 60 % and a remarkably thin profile of just $160\ \mu\text{m}$, this film achieved a solar reflectance of 95 %. Yang et al. [94] developed a bioinspired bilayer design by combining a highly porous polyethylene (PE) aerogel with a PDMS film to create a flexible, superhydrophobic cooling coating. The PE aerogel, featuring about 97.9 % porosity and micropores around $3.8\ \mu\text{m}$, offers high solar reflectance (96 %)—when fabricated into a 2.7 mm thick layer. This structure not only effectively reflects sunlight but also provides thermal insulation from ambient heat. Moreover, the bilayer can be easily applied and removed from different substrates, making it a versatile solution for reflecting heat insulation material.

High porosity often comes along with poor mechanical strength and integrity. Therefore, developing porous polymer structures with balanced porosity and mechanical properties is still a challenge and of growing interest worldwide.

The morphology of porous polymers can be tailored through different fabrication methods, including microcellular foaming [116, 117], electrospinning [117], and phase separation [118, 119], etc.

2.6. Critical review, and the goals of the thesis

With the growing need for a circular economy and thus mono-material products with better recyclability, the diversity of requirements that must be met for the same material also increases. The surface property is one of the critical properties that must be customizable for successful mono-material products for advanced applications.

During the literature review, methods based on physical surface modification for creating superhydrophobic and reflective surfaces were primarily presented and compared. The literature review revealed that numerous surface modification techniques have been developed, enabling the simultaneous creation of various properties by altering the surface structure. These methods can be classified into two main categories. The "Top-down" approaches form the desired surface structure by disintegrating or degrading the surface, such as etching methods, lithography, and templating. In contrast, the "Bottom-up" methods create a new layer or coating on the substrate surface; examples include sol-gel techniques and phase separation methods. The solvent treatment described in my dissertation falls under the phase separation methods. In these techniques, a solution of a semicrystalline polymer is applied to the selected substrate, and by modifying environmental parameters, the evaporation of the solvent and the crystallization of the polymer is simultaneously initiated. The porous polymer layer that forms on the substrate is responsible for the superhydrophobic properties.

The literature shows that some surface modification methods, which enable the simultaneous development of multiple functional properties through engineered surface structures, are still in the laboratory research phase. While these methods hold promise, they face significant challenges. An easy-to-use and scalable approach is essential for designing products tailored to specific applications. However, very few studies focus on industrial applications due to several limitations. These include tedious multi-step processes, severe operating conditions, and the use of specialized instruments and chemicals. Additionally, such methods are often limited to small areas or specific materials. For roughened surfaces to be viable in industrial applications, low-cost materials, and single-step fabrication approaches should be prioritized. Furukawa Electric Institute of Technology has extensive experience in processing semicrystalline foams and studying the correlations between foam structure and mechanical and surface properties. Using this expertise, the focus has shifted toward developing methods to introduce new functionalities—such as water repellency and solar light reflectivity—to enhance the properties of products made from semicrystalline polymers. The goal is to generate practically relevant knowledge and solutions for surface morphology modification applicable to a wide range of semicrystalline polymers.

The literature review pointed out that phase separation-based technologies could be suitable for creating roughened coatings on the surface of selected semicrystalline polymers. These methods often use low-cost chemicals, making them potential candidates for large-scale applications. Most phase separation-induced methods use

two solvents: one that dissolves the polymer well and another that dissolves it poorly. While effective, these methods are less environmentally friendly due to the challenges in separating and regenerating the solvent mixture, and they often require post-treatment. Furthermore, an additional step may be necessary to increase interfacial adhesion between the substrate and the deposited layer to prevent exfoliation, adding complexity to the process.

To simplify the technology, creating a roughened surface directly from the substrate offers a promising alternative. This approach in mono-material products eliminates adhesion issues between the substrate and the roughened layer, enhancing usability and streamlining the production process.

Based on the literature review, I set the following goals for my dissertation:

- Development of a novel solvent-treatment method based on phase separation to create hierarchically structured surface layers on PP samples. This includes characterizing the resulting surface structure and establishing correlations between processing parameters and the developed morphology. Additionally, the effect of a commonly used nucleating agent on the surface morphology of PP samples will be investigated.
- Investigation of surface wettability of solvent-treated samples. This involves the examination of the surface wettability of the solvent-treated samples and the examination of wettability transitions of extruded PP sheets with hierarchical surface structures, ranging from superhydrophobic to superhydrophilic behavior, using air plasma treatment. The relationship between surface chemistry and wettability changes will be explored. Time-dependent hydrophobic recovery of hierarchically structured PP surfaces will also be studied by analyzing changes in CAs and surface composition over time.
- Evaluation of solar reflectance and thermal insulation properties of surface-treated samples. The investigation will analyze the combined effects of porous layer structures, the presence of optical additives, and nucleating agents on the structural and reflective properties of PP samples.
- Development of reinforced PP composite sheets with micro- and nano-structured porous outer layers to enhance the modulus and strength of the sheets. The solar reflectivity of the surface-treated samples will also be investigated.

3. MATERIALS, PROCESSING AND CHARACTERIZATION METHODS

This chapter presents the materials and processing methods that I used during the research process.

3.1. Materials

Tipplen H681F (MOL Petrochemicals, Hungary) PP homopolymer (melt flow rate = 1.7 g/10 min, 230 °C, 2.16 kg) was used in the experiments. This polymer grade was selected because it contains only essential additives like stabilizers and a slip agent, but does not contain any nucleating agent or any other special additives. Consequently, its structure can be manipulated within a wide range, which is favorable in my study. Also, PP has been extensively studied in the literature in the past decades, so the results obtained here can be compared to literature data.

An α -nucleating agent was added into the PP to change the crystallization process, especially nucleus density, and to modify the size of spherulites of the solvent-treated samples. The selected conventional heterogeneous nucleating agent (NA) was NA-21 E, produced by Adeka-Palmarole (Japan). 50 to 1000 ppm of NA was mixed into the polymer.

In order to modify the solar reflectance ability of the roughened samples, I prepared polymer–inorganic hybrid materials. TiO_2 and BaSO_4 are among the most widely used inorganic fillers and therefore are used for their well-proven high NIR reflective property. TiO_2 particles (KTR 600) with 98 % of the particles below 45 μm were supplied by Koltex Color s.r.o, Czech Republic. BaSO_4 particles with 1–4 μm particle size were supplied by Thermo Scientific, Belgium. Both reflective additives were added to the PP in amounts of 5, 10 and 20 wt%.

I investigated the mechanical behavior, particularly stiffness and strength, of the solvent-treated neat PP and glass fiber–reinforced PP composite samples, as the applied solvent treatment greatly influences mechanical properties. FGCS 3540 short GF was used, with an average length of 3 mm and an average diameter of 10 μm , obtained from Schwarzwälder Textil-Werke Heinrich Kautzmann GmbH (Germany). The amount of GF in the composites was 20, 30 and 40 wt%. To improve the interfacial adhesion between fibers and PP matrix, functionalized polymer, namely maleated polypropylene was used in glass fiber–reinforced PP composites. Exxelor PO 1020 maleated polypropylene (MAPP) coupling agent was obtained from ExxonMobil Chemical, Germany (density 0.9 g/cm³, melt flow rate 112 g/10 min (190 °C/1.2 kg)). MAPP was used at 10 wt% related to the amount of GF.

3.2. Processing methods

3.2.1. Sheet extrusion

For the production of masterbatches, PP was compounded with BaSO₄ and TiO₂ at 30 wt% and with a nucleating agent content of 1 wt%. The additive powders and PP pellets were pre-mixed and extruded with a co-rotating twin-screw extruder (Brabender Plasti-Corder, Duisburg, Germany, L/D ratio: 19, screw diameter: 20 mm) equipped with a rod die. The temperature profile from the hopper to the die was set to 190, 200, 200, 200, and 200 °C, with the die temperature at 200 °C. Screw speed was maintained at 20 rpm. The extruded filaments were pelletized with an S330 granulator (Rapid, Bredaryd, Sweden) and added to the PP pellets.

To produce 1 mm thick extruded sheets, a co-rotating twin-screw extruder (TSA, Cernobbio, Italy, L/D ratio: 40, D = 32 mm) was used, equipped with a 250 mm wide sheet die and a sheet line (Trocen QCAL-Quadro Calandra Linea R&G). For all compounds (PP/TiO₂, PP/BaSO₄, reinforced PP), the feed rate was 6 kg/h, and screw speed was 60 rpm. The extruder zone temperatures were set to 190, 195, 200, 205, and 210 °C, with the die temperature at 210 °C. The vertical cylinders were kept at 30 °C, and pulling speed was adjusted for a final sheet thickness of 1 mm. To investigate the influence of extruded sheet morphology on the phase separation method (**Chapter 4.1.3**), I prepared extruded sheets using two different (symmetric and asymmetric) tempered cylinder arrangements and two different sheet die lip gaps of 1.1 and 2.5 mm. Extruded sheets with symmetric tempered cylinder arrangement were prepared with vertical cylinders (Trocen QCAL-Quadro Calandra Linea R&G), the temperature was set to 30 and 80 °C, and pulling speeds were 1.0 and 1.3 m/min. Nucleated PP sheets were prepared with similar settings as neat PP, but with cylinder temperatures fixed at 30 °C and a die lip gap of 1.1 mm.

Additionally, neat PP sheets were produced with a single-screw extruder (LE 25-30/C) and an LCR300 flat-film line (Labtech Engineering Co. Ltd., Samutprakarn, Thailand) for the production of asymmetric tempered cylinder arrangement. The temperatures of the extruder zones were 190, 195, 200, 205 and 210 °C from hopper to die, and die temperature was 210 °C. The lips of the die were opened to 2 mm. Sheets were prepared with cylinder temperatures of 30 and 80 °C, and two different screw speeds. In one case, screw speed was 280 1/min with a pulling speed of 1.0 m/min and in the other case, screw speed was 300 1/min with a pulling speed of 1.4 m/min. The nominal thickness of the extruded sheet, regardless of the type of extruder, was 1 mm. The schematic diagram of extruders with different cylinder arrangements can be seen in **Figure 3.1**.

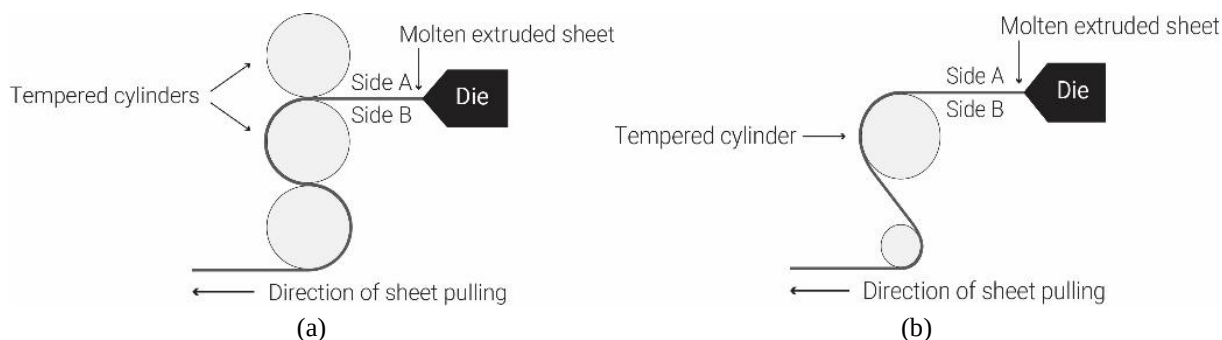


Figure 3.1 Schematic diagram of the sheet extrusion setup with symmetric (a) and asymmetric (b) tempered cylinder arrangements

3.2.2. Solvent-treatment and related conditions

The extruded samples were cut into rectangles of 150 mm × 200 mm. **Figure 3.2** shows the schematic diagram of the solvent treatment and the subsequent steps.

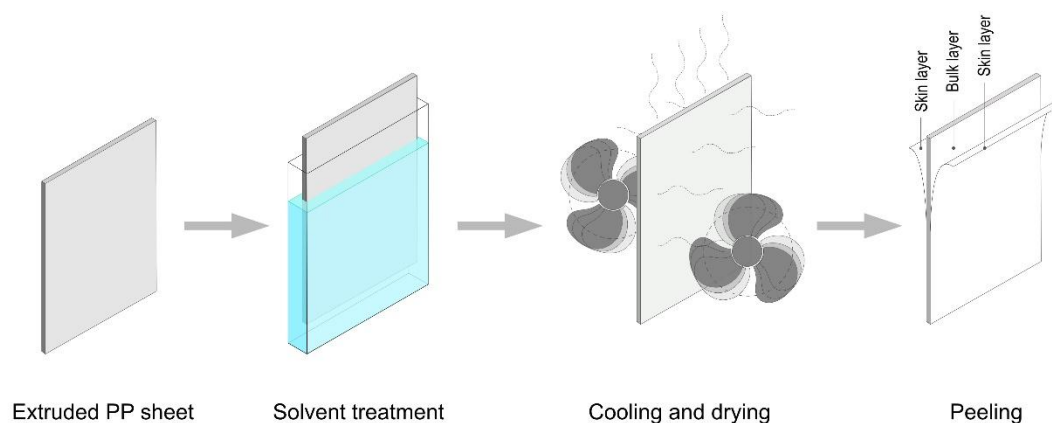


Figure 3.2 Schematic diagram of the solvent treatment

For the solvent treatment, I used xylene as a solvent because it is a good solvent of PP. For the investigation of the effects of the solvent treatment parameters, (**Chapter 4.1.1**) the PP samples were immersed in xylene (2 l). Solvent temperatures were 120, 125 and 130 °C and the duration of immersion was 30, 60, 90, and 120 s. Finally, the samples were dried at 30 °C for 24 h in an air-ventilated oven (UT6120, Thermo Fischer Scientific, USA), then for 24 h in a vacuum oven (FCD-3000, Faithful) at 30 °C. Finally, the skin layers were peeled off as a single layer (they were not broken during peeling off) from both sides of the sheet with duct tape (Professional 4613, Tesa, Hamburg). The remaining sheet and both sides of the peeled skin were investigated.

In the drying temperature investigation (**Chapter 4.1.2**), the samples were solvent-treated at 125 °C for 60 s and then dried at 30, 50, 70, 80, 100, 120 and 125 °C for 30 min. After drying, the samples were further dried at 30 °C for 24 h, then placed in a vacuum oven at 30 °C for 24 h in all cases. Finally, the skin layers were peeled off from both sides of the sheet.

For the investigation of the morphology of extruded sheet (**Chapter 4.1.3**), and for the investigation of plasma treatment (**Chapter 4.2**), the extruded sheets were immersed in xylene for 60 s at 125 °C. The following processing steps (drying and peeling) were similar to the investigation of solvent treatment parameters.

For the investigation of the effect of the nucleating agent (**Chapter 4.1.4**), samples were solvent-treated at 120, 125 and 130 °C for 60 s and 120 s. The following processing steps (drying and peeling) were similar to the investigation of solvent treatment parameters.

For the investigation of solar reflection ability (**Chapter 4.3**) and for the investigation of the reinforced PP composites (**Chapter 4.4**), the samples were solvent-treated at 125 °C 60, 120 and 180 s. The drying step was similar to that in the investigation of solvent treatment parameters. The samples in this experiments were not peeled

3.2.3. Preparation of plasma-treated samples

Air plasma treatment was carried out on extruded PP samples having smooth surface and also on samples having hierarchically structured surface (**Chapter 4.2**). The plasma surface treatment was conducted in an air atmosphere, using a cold plasma apparatus (Smartplasma 10, Plasma Technology GmbH, Herrenberg, Germany) operating in the radiofrequency range (13.56 MHz). The pressure inside the vacuum chamber (15 and 30 Pa), treatment time (60 to 600 s), and power (24 to 64 W) were varied during the plasma treatment, to explore its effect on the sample. Plasma-treated samples were stored at room temperature in a dust-free container afterward.

3.2.4. Heat treatment of plasma treated samples

To investigate the thermal stability of the introduced hydrophilic surface (**Chapter 4.2**), I carried out thermal aging tests on plasma treated samples (15 Pa, 40 W and 600 s) in a furnace at 70 °C or 150 °C for 60 min.

3.3. Characterization methods

This chapter details the characterization methods used in my research. I performed morphological analyses to investigate the surface layer structures formed on the solvent-treated samples. The physicochemical properties of the surfaces were characterized with surface wettability measurements and X-ray photoelectron spectroscopy (XPS). The solar reflectance and temperature-reducing capabilities of the samples were evaluated with optical methods and reflective insulation tests. Finally, the mechanical properties, including strength and stiffness, of the solvent-treated samples were assessed through mechanical testing.

3.3.1. Morphological analyses

Scanning electron microscopy

Surface and cross-section morphology were examined with a Jeol JSM-IT 200 scanning electron microscope (SEM). For the examination of surface and cross-sectional morphology, samples were cryo-fractured and made conductive by sputtering with a thin gold layer for 50 s with 10 mA in a vacuum before SEM imaging. The size of spherulites and the thickness of layers were analyzed with the use of surface and cross-sectional SEM images with the aid of the ImageJ software. Approximately 300 spherulite diameters were measured in the calculation of average spherulite size. For average layer thicknesses, measurements were taken at approximately 50 different locations across the cross-section of the samples.

Polarized optical microscopy

The morphology of the surface and the cross-sectional area of samples were studied with a Zeiss Axioscope polarized optical microscope (POM) equipped with a Leica DFC 320 digital camera. Micrographs were recorded with the Leica IM50 software.

Atomic force microscopy

The nanoscale morphology of the sample surfaces was analyzed with a PSIA XE-100 atomic force microscope (AFM, Park Systems Corp., Suwon, South Korea) in tapping mode. The data were collected in an area of 5 μm $\times$ 5 μm .

Porosity measurement

Porosity was determined from the true and apparent density according to Equation 3.1:

$$\varepsilon (\%) = \frac{\rho_t - \rho_a}{\rho_t} * 100 \quad (3.1)$$

where ε is the porosity, and ρ_t and ρ_a are the true and apparent density of the samples, respectively.

ρ_a was determined from 4 cm $\times$ 4 cm samples. I removed the skin layer from both sides of the treated samples and measured their weight. ρ_a is obtained by dividing the weight of the samples by their volume. ρ_t was measured on an Ultra pycnometer 1000 (Quantachrome, Boynton Beach, FL, USA) with helium at 20 $^{\circ}\text{C}$.

Specific surface area measurement

For the specific surface area measurement, samples were cut into 4 mm $\times$ 4 mm pieces and degassed for 24 h at 110 $^{\circ}\text{C}$. Low-temperature (-196.15 $^{\circ}\text{C}$) nitrogen adsorption measurements were performed on a NOVA 2000e (Quantachrome, Boynton Beach, FL, USA) automatic volumetric instrument. The apparent surface area S_{BET} was determined with the Brunauer–Emmett–Teller (BET) model [120]. From the

measured apparent surface area, I determined the calculated specific surface area (S_{BET}^*), which refers to the specific surface area of the porous layer. I used the thickness of the treated samples (l_1), the sample's bulk phase (l_2) and the sample's porous layer (l_3), determined from the SEM cross-section images, the apparent density (ρ_a) of the samples, and the true density (ρ_t) of the extruded sample. S_{BET}^* was calculated with the following equations:

$$V_1 = m_1 * \rho_a \quad (3.2)$$

$$V_1 = l_{(3)} * A + l_{(2)} * A \quad (3.3)$$

$$l_1 = l_2 + l_3 \quad (3.4)$$

$$A = V_1 / l_1 \quad (3.5)$$

$$V_2 = l_2 * A \quad (3.6)$$

$$m_2 = V_2 * \rho_t \quad (3.7)$$

$$m_3 = m_1 - m_2 \quad (3.8)$$

$$Tot_{surface} = S_{BET} * m_1 \quad (3.9)$$

$$S_{BET}^* = Tot_{surface} / m_3 \quad (3.10)$$

where V_1 , V_2 , V_3 are the volume of the solvent-treated sample, the porous layer and the bulk phase, respectively. m_1 , m_2 , m_3 are the weight of the solvent-treated sample, porous layer and bulk phase, respectively. A is the surface area of the sample used for BET measurement, $Tot_{surface}$ is the total surface area of the sample.

Thermal analysis

The melting and crystallization behavior of the samples (non-treated and solvent-treated) were studied by differential scanning calorimetry (DSC Q2000; TA Instruments, New Castle, USA) in a nitrogen atmosphere, according to ISO 11357-3:2018. Samples of approximately 15 mg were first heated from 30 °C to 220 °C at 10 °C/min. This temperature was maintained for 5 min to eliminate thermal history, then the samples were cooled to 30 °C at a rate of 10 °C/min. The degree of crystallinity (X_c) was calculated with Equation 3.11:

$$X_c = \frac{\Delta H_m}{\phi \Delta H_0} \times 100 \quad (3.11)$$

where ϕ is the weight fraction of PP, ΔH_m and ΔH_0 are the measured melting enthalpy of the first heating cycle and melting enthalpy of a 100 % crystalline sample (for PP it is 207 J/g [121]).

Determination of the fiber length

The average fiber length after the extrusion was measured, based on the fibers isolated from the ash of the composite materials pyrolyzed in a furnace at 550 °C for about 120 min. Extracted fibers were dispersed on a glass slide by water and evaluated with an optical microscope (Olympus BX51, Olympus, Germany). The average fiber lengths and fiber length distributions were determined based on the length of 500 fibers.

Determination of actual glass fiber content

The weight percent of the GF in the extruded composite samples was determined by calcination, according to the EN ISO 1172 standard. The specimen was placed in a furnace under an inert environment at 500 °C for 120 min, then dried in a desiccator. The weight percent of GF was calculated with Equation 3.12:

$$GF \% = \frac{m_2}{m_1} \times 100 \quad (3.12)$$

where m_1 is the initial weight of the composite sample and m_2 is the weight of the remaining GF after calcination.

3.3.2. Surface composition analysis

Changes in the chemical composition of surfaces were detected by XPS with a twin anode X-ray source (XR4, Thermo Fisher Scientific, USA) and a hemispherical energy analyzer with a 9-channel multi-channeltron detector (Phoibos 150 MCD, SPECS). The base pressure of the analysis chamber was approximately $2 \cdot 10^{-9}$ mbar. Samples were analyzed with a Mg K_{α} (1253.6 eV) anode without monochromatization. Peak fitting was carried out with the CasaXPS software. Wide-range scans and high-resolution narrow scans of the C1s and O1s photoelectron peaks were recorded.

3.3.3. Surface wettability

For checking the wettability of PP surfaces, CAs were measured in a drop shape analyzer (Krüss DSA 30, Hamburg, Germany) with the sessile drop method. To measure the CA_{adv} , 20 μ l droplets of MilliQ water were deposited on the surfaces by a drop-build up technique with the use of a microsyringe in a chamber with controlled temperature (25 ± 2 °C) and relative humidity (> 80 %). The CA_{rec} was measured while the volume of a sessile drop was slowly decreased by 10 μ l. CAH was calculated from the difference between the CA_{adv} and CA_{rec} . CA_{adv} of plasma-treated samples monitored as a function of time. The sessile drops were analyzed with the Advance image analysis software to calculate contact angles based on the tangent of the drop profile at the apparent intersection of the three phase contact line where the liquid–gas interface meets the solid–liquid interface.

3.3.4. Mechanical analysis

Tensile test

Tensile tests were performed with a Z010 universal tensile testing machine (Zwick/Roell GmbH, Germany) equipped with a 10 kN load cell according to MSZ EN ISO 527:2012. The tests were performed with a gauge length of 40 mm and a crosshead speed of 5 mm/min. Young's modulus was determined in the strain range of 0.05 % and 0.25 % by fitting a straight line to the stress–strain curve. The characteristic values of yield stress, tensile strength and elongation at break were also determined. The tensile tests were performed in parallel to the flow direction of extrusion. 5 dumbbell-shaped specimens were tested from each composition.

Impact test

Impact tests were conducted on 10 specimens, each measuring 80 mm x 80 mm, using a falling weight impact tester (Fractovis 9350, Ceast, Italy) equipped with a 20 mm diameter hemispherical dart, according to ISO 6603. Impact velocity was 4.4 m/s. The force-time diagrams were recorded during the tests, from which the perforation energy (E_{perf}) was calculated with Equation (3.13):

$$E_{perf} = \frac{E_{total}}{v} \quad (3.13)$$

where E_{total} is the total absorbed energy during the penetration and v is the thickness of the sample.

Peeling test

The peeling test was performed with a Z010 universal tensile testing machine (Zwick/Roell GmbH, Ulm, Germany) equipped with a 10 kN load cell, according to MSZ EN ISO:4578. 50 mm x 120 mm rectangular specimens were used for the tests. The skin layer on the samples was glued with adhesive tape (466, Tesa, Germany, Hamburg), the end of the adhesive tape was secured between the grips of the tensile machine, and the sample was placed in a standard roller holder used for the tear test. The test was performed at a speed of 1000 mm/min. The tear force was measured as a function of elongation. The test was carried out on 5 specimens in each case.

3.3.5. Reflectance investigation

The reflectance of solvent-treated PP composite samples was measured with a PerkinElmer 1050 spectrometer equipped with a 150 mm diameter Labsphere closed integrating sphere, according to ASTM 903-12, 2012. For the diffuse reflectance measurements, the specular mirror output of the PerkinElmer 150 mm integrating sphere was removed so only the diffuse part of the reflected light was measured. I also performed the total reflectance measurement with a closed integrating sphere to

register the total reflected light. In both cases, the measurement was carried out in the wavelength range of 200–2500 nanometers, with a resolution of 10 nanometers. Three different positions of each sample were randomly selected for measuring.

The area under the reflection spectra obtained during spectrophotometry is characterized with the total solar reflectance, TSR. This indicates the reflected energy percentage of solar radiation. For the TSR index, the spectral distribution of solar radiation must also be taken into account. To measure the amount of reflected solar radiation energy, we have to examine each range. In the examined wavelength range of 200–2500 nm, the total energy of solar radiation is 892.2873 W/m². The value of the TSR index can be determined with Equation 3.14 [92].

$$TSR = \frac{\int_{200}^{2500} R(\lambda) I_{solar}(\lambda) d\lambda}{\int_{200}^{2500} I_{solar}(\lambda) d\lambda} \quad (3.14)$$

where $R(\lambda)$ is the reflectance of samples at the wavelength from 200 to 2500 nm, and I_{solar} is the ASTM G173 Global solar intensity spectrum. TSR results are used to estimate the effectiveness of samples as solar reflective layers.

3.3.6. Temperature reduction ability investigations

Indoor test

Indoor near-infrared light measurements were conducted to characterize the reflective insulation property of samples with the help of an apparatus we designed (**Figure 3.4**). Two temperature sensors were set at the same height (40 mm) in both boxes. The reference sample, an extruded PP sheet, was always placed on one of the two boxes. I chose the duration of the tests in such a way that the temperature inside the boxes could stabilize in all cases. During evaluation, I plotted the temperature difference (ΔT) between the two boxes as a function of time. **Figure 3.4** shows the layout of the test.



Figure 3.4 The arrangement of laboratory-scaled "sun test"

In the tests, two chambers were fabricated with a wooden frame. The sides of the chambers were covered with a PS foam sheet. In addition, the outer sides of the PS foam sheet were sealed with aluminum (Al) foil to minimize heat convection and conduction. The top side of the chambers was covered with the selected samples. The inner temperature of the boxes was measured with K-type thermocouples. The boxes were irradiated from above with a 150 W light bulb, whose spectral distribution matches that of the solar spectrum well. The vertical distance between the lamp and the sample surface was 400 mm. A multi-channel temperature logger connected with two K-type thermocouples was used to detect the real-time temperatures inside the boxes. The test time was 1 h, and the internal temperature was recorded during the test.

Outdoor test

The reflective insulation performance of the samples was determined not only on an indoor test setup but under the actual application conditions (**Figure 3.5**) on the roof of a building, in natural sunlight: the tests were performed from 10:30 to 14:30 in Budapest on a sunny day on 18 July 2024.

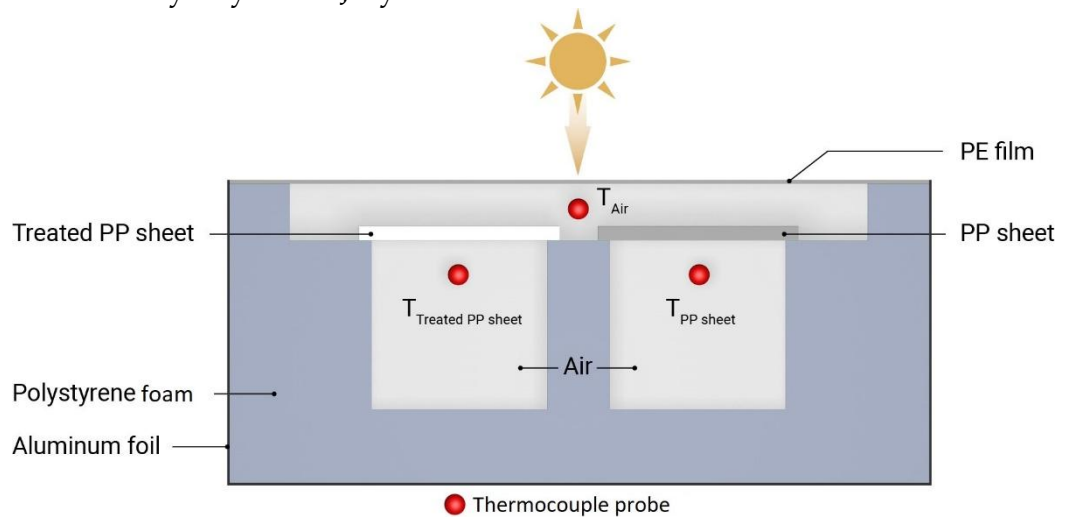


Figure 3.5 Schematic of the apparatus for testing reflective insulation performance

A highly infrared-transparent low-density polyethylene (LDPE) foil was used to suppress the influence of the outdoor environment (for example, wind) on the test. Ambient temperature was measured with a thermocouple placed under the LDPE foil, which was also covered with an Al foil to prevent the tip of the thermocouple from being heated by direct sunlight.

4. RESULTS AND DISCUSSION

This chapter presents and evaluates the experimental results obtained in this dissertation. The discussion begins with a concise introduction to the process steps of the solvent-based treatment I developed and an analysis of their effects on the morphology of the resulting surface layer. The details provided in **Chapters 4.1. Chapter 4.1** explore the surface wettability of the treated samples, specifically focusing on the correlation of surface wettability and the properties of the roughened surface. **Chapters 4.1** provides the basis for the following chapters.

Chapter 4.2 extends this investigation by demonstrating how wettability can transition between two extreme states: superhydrophobicity and superhydrophilicity.

Chapter 4.3 reveals the additional functional properties introduced by surface modification, specifically the enhanced solar light reflection and temperature-reduction capabilities.

Finally, in **Chapter 4.4**, I present the mechanical properties of the solvent-treated samples in the presence of reinforcing materials. While this chapter does not directly address the introduction of new surface functionalities, it highlights the potential drawbacks of the treatment method on mechanical performance. Strategies for compensating for the observed impairment of mechanical properties are also explored, providing a comprehensive perspective on the implications of the treatment process developed.

4.1. Development of the solvent-based surface treatment method

The development of solvent-based surface treatments is presented in five sections. First, the surface layer formed on the solvent-treated sample is characterized with various morphological analyses. In the second section, correlations between processing parameters and the resulting morphology are discussed. These include the effects of solvent temperature, immersion time and drying temperature on spherulite size and on the thickness of the skin and porous layers, and the removability of the skin layer. The third and fourth sections explore the impact of a nucleating agent for PP on surface morphology and the effect of the morphology of extruded sheets on the phase separation method and surface characteristics.

To introduce the technology and the resulting morphology, I used PP as a model material. PP is one of the most commonly used semicrystalline polymers. Its properties, such as crystallization behavior, are well-documented and extensively studied, making the morphological changes induced by the solvent treatment understandable and explainable based on our existing knowledge. At the Furukawa Electric Institute of Technology, the solvent based technology has also been successfully applied not only to PP but also to other semicrystalline polymers, such as PE, polylactic acid, and polyvinylidene fluoride. However, the technological parameters for these polymers are not discussed in this dissertation. It is worth noting

that for each semicrystalline polymer, the technological parameters and the solvent used for the treatment had to be re-optimized.

The solvent applied for the treatment was xylene, which is widely known as a good solvent for PP and is commonly used in phase separation methods. Although the potential health risks associated with xylene are a consideration, I chose this solvent because its adverse effects can be minimized through the closed-system application of the described method and by solvent regeneration.

The primary aim of this chapter is to provide a general overview of the developed method through the characterization of individual process steps. The method was optimized for xylene, and other solvents for PP in this dissertation are not discussed. However, at the Furukawa Electric Institute of Technology, we also investigated alternative solvents with lower environmental and health impacts, but the technological parameters for these are beyond the scope of this work.

In my dissertation, I worked with 1 mm thick extruded PP sheets. I investigated the solvent treatment parameters necessary for the successful treatment of 1 mm thick PP sheets. The examination of samples with different thicknesses, produced using different technologies, or with non-flat surfaces does not form part of this dissertation; I plan to analyze these in detail in future research.

4.1.1. Characterization of morphology

To determine the impact of solvent treatment on the morphology of samples, a surface and a cross-sectional morphology study was conducted with the use of SEM. To give a general overview of change in PP surface morphology, the morphology changes will be presented on a PP sample that was immersed in a 125 °C solvent for 60 s. **Figure 4.1** showcases the SEM micrographs of the surface of extruded PP and the surface of unpeeled, treated sample following the solvent treatment.

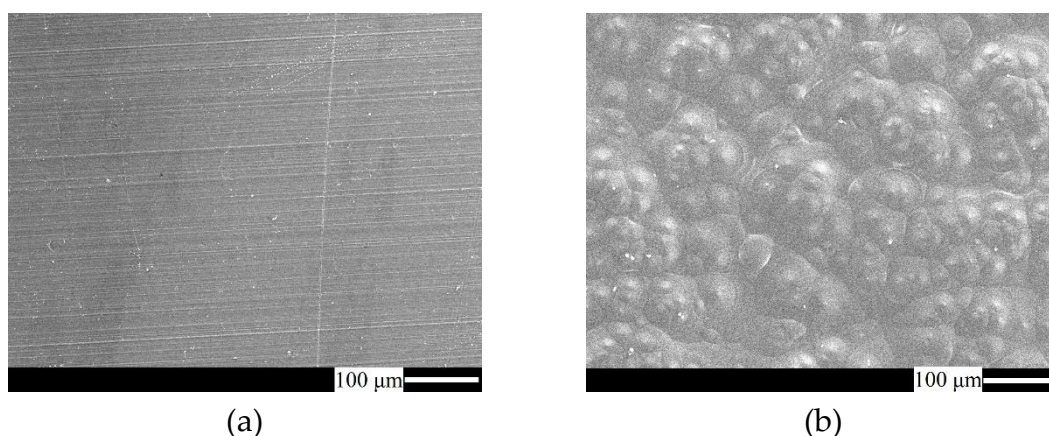


Figure 4.1 SEM images of (a) the surface of the extruded sheet and (b) the outer surface of the skin layer. The solvent temperature was 125 °C and immersion time was 60 s

The untreated sheet has a smooth surface (**Figure 4.1a**) due to extrusion, while after the treatment, the surface becomes rough (**Figure 4.1b**).

Figure 4.2 shows the surface morphology of the peeled sample and the inner side of the skin layer.

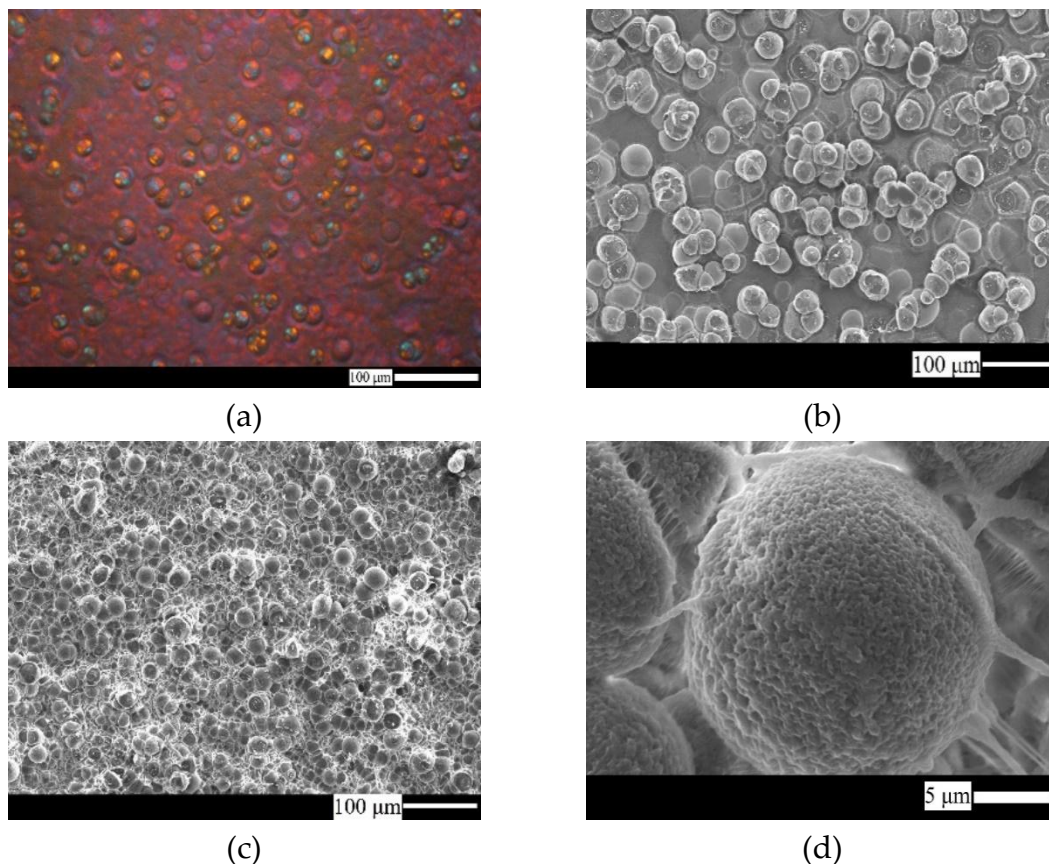


Figure 4.2 POM image (a) of the inner side of the skin layer, SEM images of (b) the inner surface of the skin layer, (c) the surface of the peeled sheet, (d) is a magnification of a particle in (c). Solvent temperature was 125 °C and immersion time was 60 s

The resulting outer layer is peeled off, and the surface of the inner side of the outer layer (**Figure 4.2a** and **Figure 4.2b**) and the peeled samples (**Figure 4.2c** and **Figure 4.2d**) exhibit sphere-like structures. The outer layer is hereinafter referred to as skin layer. **Figure 4.2a** shows the POM image of the inner side of the skin layer. The POM image reveals that the spherical structures exhibit a birefringent nature. The typical blue and yellow quarters indicative of birefringent spherulites can be seen [85].

The surface of the peeled samples consists of regular and well-separated spherulites, which suggests that they did not interfere with each other during their growth. Similar surface topographies can be found in the literature where the surface consists of spherulites with different sizes and voids between them, which confirm the porous nature of the surface [122, 123]. The enlarged SEM image (**Figure 4.2d**) reveals that PP crystallizes as typical spherulites with a micro- and nanoscale hierarchical structure from the PP gel. The surface of spherulites also has a roughness of several tens of nanometers. The nanoscale roughness of the spherulites is a result of the radially growing branched and banded chain-folded lamellae and the solvent treatment conditions [124].

Figure 4.3 shows the SEM images of the cross-section of the cryo-fractured samples before and after treatment. **Figure 4.3a** shows the cross-section of the extruded sheet, and **Figure 4.3b** shows the cross-section of the solvent-treated sheet before the peeling step. **Figures 4.3c** and **Figure 4.3d** show the cross-section of the developed porous structure shown in **Figure 4.3b** at a higher magnification.

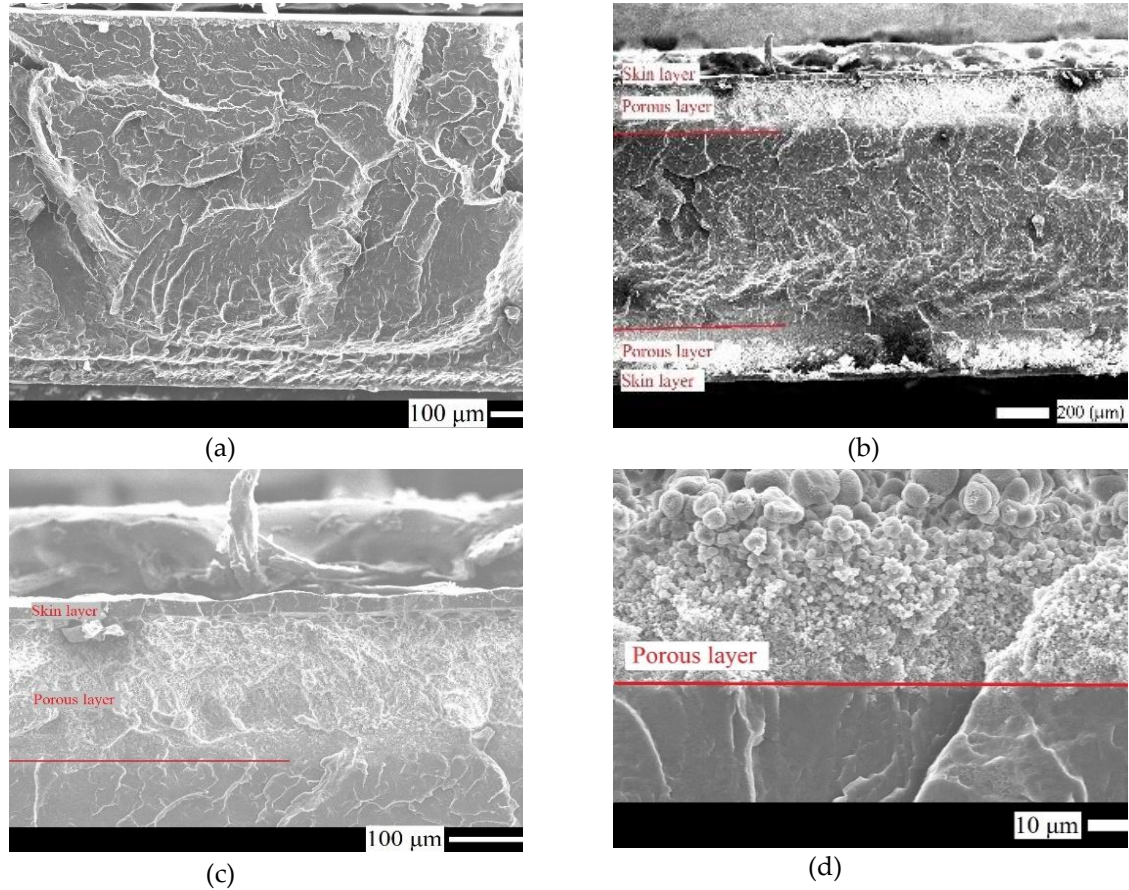


Figure 4.3 Cross-sectional SEM images of (a) the extruded PP sheet and (b) the sheet treated with a 125 °C solvent for 60 s, and the treated sheet at higher magnification (c) with and (d) without the skin layer

Under the skin layer, the cross-section has a porous layer consisting of spherulites. It is also noticeable that the structure of the porous layer is anisotropic. The cross-sectional view with larger magnification (**Figure 4.3d**) indicates that the spherulites are larger close to the surface than the spherulites in deeper layers of the porous structure. The size of the spherulites decreases in the direction of the bulk phase. There is a significant change in the cross-section of samples: below a certain depth, the porous structure turns into a dense material. The sharp change (marked with a red line in **Figure 4.3c** and **Figure 4.3d**) shows the depth of solvent penetration into the sample. To determine the thickness of the bulk phase and the porous layer, I used this sharp boundary to distinguish the two layers. **Figure 4.4** shows the layer thickness changes after the solvent treatment.

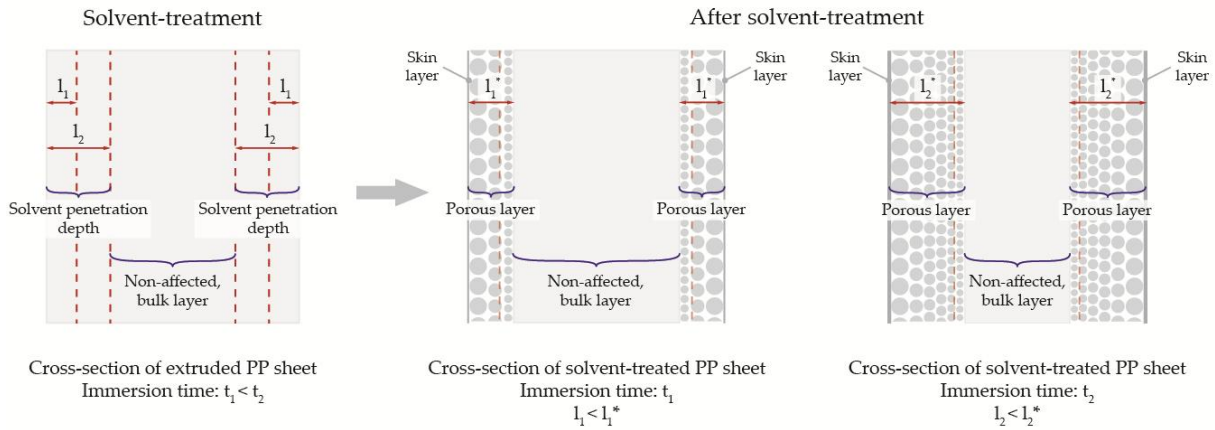


Figure 4.4 Schematic of layer thickness changes (bulk layer, solvent penetration depth, porous layer and skin layer) as a function of solvent treatment at different immersion times

Solvent penetration depth can be calculated based on the difference between the initial sheet thickness (1 mm) and the thickness of the remaining bulk phase after solvent treatment. The thicknesses of both the bulk phase and the porous layer were determined through analysis of cross-sectional SEM images.

Asymmetric membranes have a structure similar to the presented porous structure and the dense layer on top of it. An important type of asymmetric membranes is integrally skinned membranes, where the size of the pores gradually changes from very large to very fine and there is a skin layer on top of the membrane [125].

4.1.2. The effect of treatment parameters on the structure

Solvent treatment parameters (immersion time and solvent temperature)

Figure 4.5a shows how the thickness of the skin and the porous layer of the samples changes when immersion time is varied (from 30 s to 300 s) at a constant solvent temperature of 125 °C. In each case, the solvent treatment step was followed by a drying step at 30 °C.

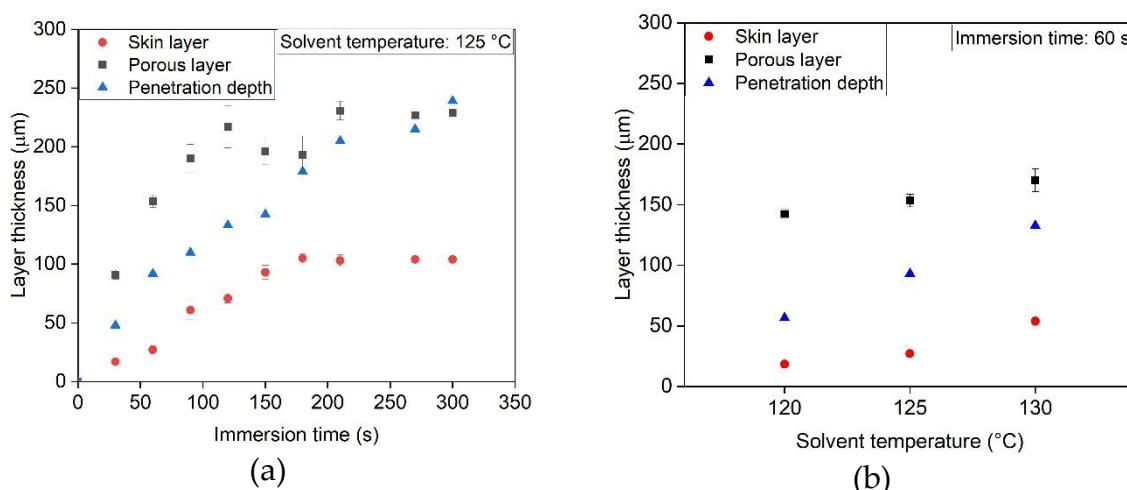


Figure 4.5 Changes in layer thicknesses as a function of treatment time at 125 °C (a) and as a function of solvent temperature with an immersion time of 60 s (b)

The treated samples have a symmetrical structure (**Figure 4.3b**), and the porous and skin layer are formed on both sides of the samples. The presented solvent penetration depth, skin and porous layer thicknesses in **Figures 4.5a** and **Figure 4.5b** belong to one side of the treated samples.

The thickness of the skin and porous layer increases with increasing immersion time. However, after 180 s, the thickness of the layers hardly increases. **Figure 4.5a** shows that a constant thickness of the layers was achieved at 125 °C.

Besides immersion time, solvent temperature also influences the formation of the structure. In **Figure 4.5b**, the thickness of the layers is shown as a function of solvent temperature with a constant immersion time. With increasing solvent temperature, the thickness of each layer increases.

The polymer–solvent interaction likely plays a key role in the formation of the structure created. The treatment starts with a solvent immersion step, when the hot solvent diffuses into the polymer sheet through the surface and toward the center of the samples like a sharp front [126]. The solvent swells the polymer, and as a consequence, a gel layer forms in the outer region of the sheet. The degree of swelling in the gel layer decreases towards the bulk phase. The thickness of the gel layer increases over time as the solvent diffuses deeper into the polymer. In the initial period, the swelling step dominates, which increases the thickness of the gel layer. When the degree of swelling reaches a critical level, the polymer molecule becomes independent of the adjacent molecules and leaves the surface and is dissolved in the solvent. Since the loosest molecules dissolve, the thickness of the gel layer decreases. After the rate of solvent diffusion into the bulk phase and polymer dissolution reaches a steady state at the given solvent temperature, the gel layer with constant thickness will gradually move toward the center of the sample [126].

During the drying step, the skin layer and the porous structure form [127, 128]. First, the solvent starts evaporating from the swelled surface and this leads to the formation of the skin layer. The solidified skin layer entraps the solvent molecules underneath, so their evaporation slows down.

The formation of the spherulitic porous structure in the cross-section is a consequence of the crystallization of PP and solvent evaporation during the drying step [127]. Various studies have demonstrated that a spherulitic structure forms from a semicrystalline polymer gel [129, 130]. Since the evaporation of the solvent is moderate due to the skin layer, the presence of the solvent during crystallization results in the formation of separate spherical spherulites and voids among them.

Table 4.1 presents the measured S_{BET} and S_{BET}^* and the ε of the samples. I took the porous layer forming on both sides of the samples into account to determine both porosity and specific surface area.

Table 4.1 Characteristics of solvent-treated PP sheets; the applied solvent temperature was 125 °C

Immersion time (s)	S_{BET} (m^2/g)	m_1 (g)	ρ_a (g/cm^3)	ρ_t (g/cm^3)	l_1 (cm)	l_2 (cm)	S_{BET}^* (m^2/g)	ε (%)
0				0.895				
30	0.8	5.16	0.79	0.905	0.108	0.090	12.6	12
60	1.9	4.50	0.73	0.912	0.112	0.081	16.3	19
90	1.6	4.94	0.68	0.911	0.116	0.078	12.5	24
150	2.7	4.07	0.69	0.911	0.110	0.071	16.3	24
210	2.5	3.91	0.62	0.914	0.105	0.059	12.4	31
280	1.5	4.58	0.56	0.911	0.103	0.057	12.7	40

The surface treated samples have a moderate S_{BET}^* of about 12–16 m^2/g . As **Table 4.1** shows, overall ε increases with immersion time because the ratio of the porous layer and the bulk phase increases with immersion time.

The experiments with different solvent temperatures and immersion times revealed that besides the thickness of the porous and skin layers, the size of the spherulites on the porous layer also changes as solvent treatment parameters are changed. The changes in average spherulite size and its standard deviation are summarized in **Table 4.2**.

Table 4.2 Spherulite sizes of the solvent-treated PP sheets

Immersion time (s)	Spherulite size (μm)		
	Solvent temperature (°C)		
	120	125	130
30	18.7 ± 4.1	23.9 ± 4.4	30.7 ± 7.0
60	16.7 ± 3.4	23.2 ± 5.5	29.6 ± 6.2
90	20.1 ± 5.6	24.2 ± 7.1	30.8 ± 9.2
120	16.0 ± 3.9	27.5 ± 9.5	30.2 ± 7.6

The results indicate that solvent temperature has a pronounced impact on average spherulite size. Increasing solvent temperature, results in the formation of larger spherulites from ~16 μm (120 °C) to ~ 30 μm (130 °C), regardless of immersion time, while the structure of the spherulites is constant. In comparison, spherulite sizes

do not change considerably with varying immersion times at a constant solvent temperature.

The resulting porous layer consists of spherulites that form via nucleation and growth steps [85]. The crystalline structure forming during extrusion is likely disintegrated in the swelling step and a semicrystalline gel is produced. The semicrystalline gel consists of a three-dimensional network structure, where the crosslinking points are crystallites [129, 131]. These crystallites could serve as self-nuclei during crystallization. The higher the degree of swelling, the more mobile the polymer molecules are, and the more pronounced the destruction of the crystalline structure is. The molecules under the preformed skin layer have the highest mobility and kinetic energy to crystallize into quiescent spherulites. With increasing solvent temperatures and degree of swelling, fewer nuclei are present, allowing the formation of larger spherulites [124].

Figure 4.6 presents the size distribution of the spherulites at different solvent temperatures with an immersion time of 60 s.

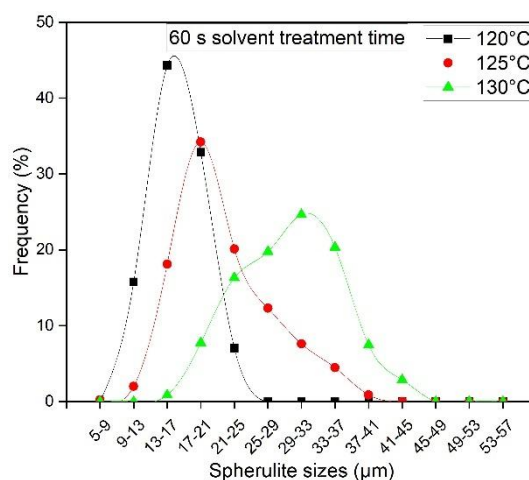


Figure 4.6 The effect of solvent temperature on the size distribution of spherulites with a constant immersion time of 60 s

With increasing temperatures, the spherulite size distribution curves shift towards higher size ranges and broaden, which meets the expectations of spherulitic crystallization under conventional crystallization conditions from polymer melt [132]. In our case, one can speculate that at higher solvent temperatures, the original structure of the sheets formed during the extrusion disintegrates more. Consequently, during the drying step of the sheets, fewer nuclei (self-seeds, self-nuclei) are available in the treated zone to initiate the growth of spherulites, leading to bigger spherulites with a broader distribution.

To investigate the separate effects of the temperature and time applied during solvent treatment, as well as the impact of the solvent (xylene) on the crystallinity of the samples, I treated the samples in silicone oil, which does not interact with PP, and in xylene under the same treatment conditions. **Table 4.3** shows the changes in the crystallinity of the annealed samples (immersed in silicone oil) and solvent-treated PP

sheets under varying solvent temperatures and immersion times. The reference sample is the extruded sample that was not solvent-treated.

Table 4.3 The degree of crystallinity of annealed and solvent-treated samples

Solvent temperature (°C)	Immersion time (s)	X _c (%)	
		Silicone oil	Xylene
Reference		39	
120	60	38	39
	30	37	38
	60	38	41
125	90	38	42
	120	38	43
	180	38	45
130	60	38	42

The changes in crystallinity of the annealed samples were negligible compared to the reference sample, irrespective of the applied temperatures and immersion times. All the samples solvent-treated with xylene displayed a monotonously growing X_c as a function of solvent temperature and immersion time. In the case of xylene-treated samples, the increased crystallinity can be attributed to the recrystallization in the solvent affected layer and not to the recrystallization of the bulk phase. As solvent treatment time is increased, the thickness ratio of the solvent-affected layer and the bulk phase increases as well, leading to the overall increment in X_c.

The thickness of the porous layer and the porosity of solvent-treated PP sheets increase with immersion time as well. These results show that longer immersion time results in the greater crystallinity, and thicker and more porous structure of the solvent-treated samples, which agrees well with the literature [61, 133].

In the initial stage of the research, I aimed to determine solvent treatment parameters that would allow for the formation of both the skin and porous layers. It was observed that when the solvent temperature was below 120 °C, the new structure did not form, regardless of the immersion time. Conversely, the highest investigated temperature was 130 °C because, from a technological perspective, the solvent treatment process became difficult to control above this temperature, even with the shortest immersion time (30 s). As expected, higher temperatures resulted in higher diffusion rates. In practice, this meant that the thickness of the bulk phase unaffected by the solvent decreased to such an extent that the sample deformed during the solvent treatment.

The created hierarchically structured surfaces change the wettability of the inherently smooth extruded surface. The untreated PP sheets showed a CA_{adv} and CAH of 102 and 7°, respectively. The surface wettability results of the solvent-treated PP samples, dried at 30 °C and peeled, are shown in **Table 4.4**.

Table 4.4 Contact angles measured on solvent-treated surfaces treated at 120, 125 and 130 °C for 30, 60, 90 and 120 s

Immersion time (s)	Solvent temperature (°C)					
	120		125		130	
	CA _{adv} (°)	CAH (°)	CA _{adv} (°)	CAH (°)	CA _{adv} (°)	CAH (°)
30	155.7 ± 4.6	2.9 ± 0.2	156.2 ± 0.4	0.9 ± 0.9	157.4 ± 0.6	1.7 ± 0.5
60	156.8 ± 2.3	2.6 ± 1.7	154.9 ± 1.2	1.1 ± 0.9	156.6 ± 1.1	2.3 ± 2.1
90	151.6 ± 2.3	1.8 ± 2.9	152.9 ± 1.2	1.9 ± 1.3	155.6 ± 1.1	1.4 ± 0.7
120	154.7 ± 1.1	2.5 ± 2.9	155.6 ± 1.2	3.6 ± 0.7	154.2 ± 2.4	0.9 ± 1.1

The increase in hydrophobicity (CA_{adv} is above 150°) and the low CAH, (less than 5°) can be attributed to the micro-, and nanoscale surface morphology (**Table 4.2**), since the treatment does not affect the chemical composition of the surface, only its morphology. Due to the formation of the hierarchically structured surface morphology, the surface has superhydrophobic and water-repellent characteristics, which agrees with other published studies [62, 134]. Changes in spherulite sizes did not cause a significant difference in the CA. Thus, changes in solvent treatment parameters (solvent-temperature and immersion time) do not affect the wetting behavior of the surface.

According to the heterogeneous wetting theories, including the Wenzel and Cassie-Baxter state, the water droplet on a surface of Cassie-Baxter state shows significantly lower CAH than a surface in the Wenzel state due to the presence of liquid–air and solid–air interfaces [135]. The results of the CA and surface morphology analysis may indicate the Cassie-Baxter’s wetting mechanism of the roughened surface created.

Drying temperature

The drying also plays a crucial role in controlling the formation of the structure, so I investigated the effect of drying temperature (between 30 °C and 125 °C), using constant solvent treatment conditions (temperature: 125 °C and immersion time: 60 s). **Figure 4.7** shows the thickness of the skin and the porous layer at different drying temperatures.

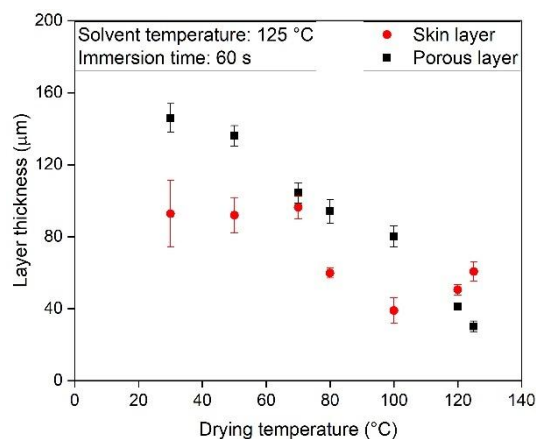


Figure 4.7 The effect of drying temperature on the structure of samples solvent-treated at 125 °C for 60 s

With increasing drying temperature, the thickness of the porous layer decreases; at 125 °C, the porous layer almost disappears. On the other hand, there is no clear relationship between drying temperature and the thickness of the skin layer. As the porous layer becomes thinner, it is more difficult to peel off the skin layer. The drying temperature determines the time of crystal growth, the rate of solvent evaporation and the rate at which the skin layer forms. At higher drying temperatures, the rate of solvent evaporation is higher. The observed declining trend in the thickness of the porous layer can be attributed to the higher drying temperature. At higher drying temperatures, the skin layer forms more slowly and solvent evaporation is faster, which lead to a thinner porous layer.

Figure 4.8 shows the relationship between drying temperature and the distribution of spherulite size on the surface of the sample after the peeling step.

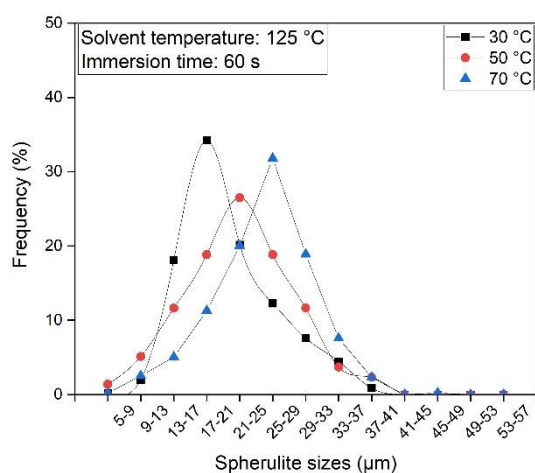


Figure 4.8 Spherulite size distribution on the samples cooled at different temperatures. The solvent treatment conditions were the same in all cases (solvent temperature is 125 °C and immersion time is 60 s)

The average spherulite sizes for samples dried at 30, 50 and 70 °C are 23.2 ± 5.5 , 19.4 ± 7 and 22.58 ± 5.9 μm, respectively. The calculation of the average size of spherulites is based on the size of 300 spherulites measured on the surface of the

sample. The average spherulite sizes obtained at different drying temperatures are not considerably different; however, the drying temperature has an effect on spherulite size distribution; spherulite sizes slightly shift towards higher size ranges at higher drying temperatures. It means that the drying temperature has an effect on the temporal distribution of the nuclei formed during crystallization. Symmetrical spherulite size distribution (**Figure 4.8**, 50 °C) indicates that the most of the spherulites (nuclei) formed at the same time during crystallization. An asymmetrical size distribution shifted to the higher spherulite sizes (**Figure 4.8**, 70 °C) means that more spherulites formed at the beginning of crystallization which were able to grow larger, while a shift to the lower temperature ranges (**Figure 4.8**, 30 °C) means that the majority of the spherulites formed close to the end of crystallization [132]. Spherulite sizes of samples dried above 70 °C are not presented because I could not peel the skin layer off the surface of these samples.

The effect of drying temperatures on surface wettability is presented in **Table 4.5**.

Table 4.5 Contact angles measured on surfaces solvent-treated at 125 °C for 60 s and dried at different temperatures

Drying temperature (°C)	CA _{adv} (°)	CAH (°)
30	154.9 ± 1.2	1.1 ± 0.9
50	142.8 ± 4.2	4.7 ± 3.7
70	137.4 ± 8.5	10 ± 6.3

Drying temperature correlates more significantly with surface wettability. The higher the drying temperature, the lower CA_{adv} is and the higher CAH is. Presumably, the decreasing trend in CA_{adv} and the increasing trend in CAH are caused by the damage to the porous structure during the peeling step. As the drying temperature increased, the samples became more difficult to peel. The surface of the sample dried at 70 °C is shown in **Figure 4.9**.

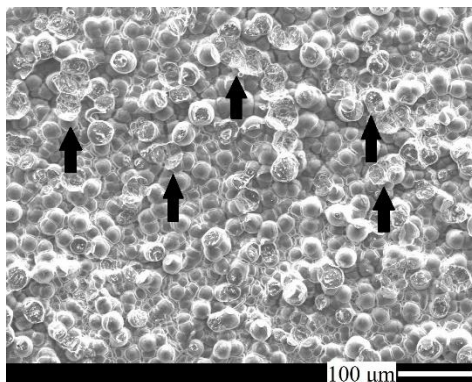


Figure 4.9 SEM image of superhydrophobic sample solvent-treated at 125 °C for 60 s and dried at 70 °C (the defects are shown with black arrows)

I was able to remove the skin layer up to 70 °C, but there are many defects on the surface as the skin layer was difficult to peel. These defect points are marked with black arrows in **Figure 4.9**.

4.1.3. *The effect of the morphology of extruded sheets on the porous structure*

The solvent treatment transforms the originally smooth extruded surface into a rough structure at both the micro- and nanoscale. The effect of this roughness on wetting behavior is evident in the significant increase in hydrophobicity compared to untreated PP.

To investigate the influence of the morphology of extruded sheets on the phase separation method and to further enhance hydrophobicity, extruded sheets with varying cross-sectional morphologies were prepared with two different extrusion techniques. **Table 4.6** contains the list of the prepared samples with the main settings. The S/PP/30/1 suffix refers to the type of extrusion, the material used, the temperature of the cooling cylinder, and the velocity of pulling.

Table 4.6 The extrusion settings and sample designations

Designation	Type of extrusion	Cylinder temperature (°C)	Pulling velocity (m/min)
S/PP/30/1	symmetric	30	1.0
S/PP/80/1		80	1.0
S/PP/30/1.3		30	1.3
S/PP/80/1.3		80	1.3
A/PP/30/1	asymmetric	30	1.0
A/PP/80/1		80	1.0
A/PP/30/1.4		30	1.4
A/PP/80/1.4		80	1.4

A POM study was carried out to determine the cross-sectional morphology of the samples. The morphological structure over the cross-section of the extruded samples with respect to the cooling temperature and the pulling velocity is shown in **Figure 4.10**. **Figure 4.10a-d** present the POM images of sheets cooled with symmetric cylinder arrangement and **Figure 4.10e-h** present POM images of sheets cooled with asymmetric cylinder arrangement.

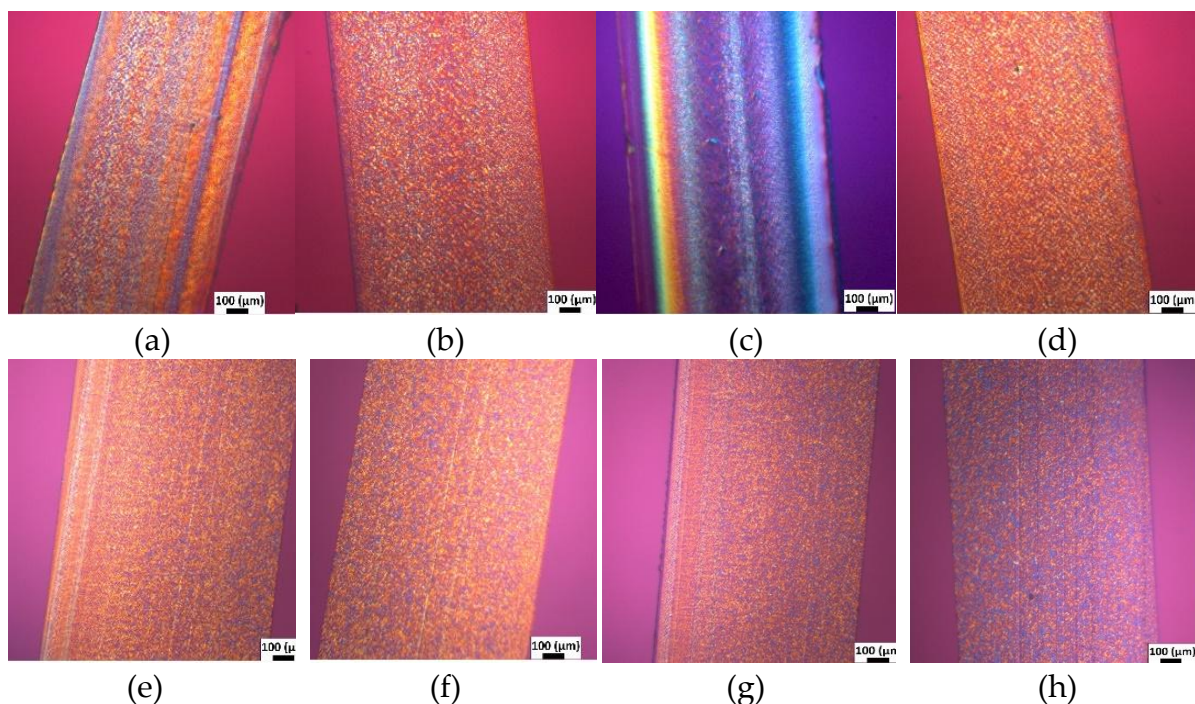


Figure 4.10 Cross-sectional POM images of extruded PP sheets, where (a) is S/PP/30/1, (b) is S/PP/80/1, (c) is S/PP/30/1.3, (d) is S/PP/80/1.3, (e) is A/PP/30/1, (f) is A/PP/80/1, (g) is A/PP/30/1.4 and (h) is A/PP/80/1.4

Based on the POM images, the different preparation methods result in samples with different cross-sectional morphology. On the one hand, I was able to produce samples (a-d, f and h) with a symmetrical cross-sectional structure. Samples a-d were contacted by tempered rollers on both sides, while samples f and h were contacted by a tempered roller (80 °C) on one side, with the opposite side being cooled by air. On the other hand, by applying the asymmetrical cooling technique with tempered roller (30 °C) and room temperature air, I can get samples (e, g) with asymmetrical shell-core morphology in the cross-section.

The thick core layer is composed of spherulites, which are formed under quiescent-like crystallization conditions. The high cooling rate of 30 °C near the shell layer leads to the formation of a highly oriented structure. Lowering the cooling rates by increasing the cooling temperature to 80 °C leads to a very thin shell layer since the oriented macromolecules have a longer time for relaxation.

On the non-contacting side, air cooling led to the formation of a shell layer. Presumably, the cooling rate of this side was slower, similar to the side cooled at 80 °C; the polymer molecules were able to crystallize in the form of spherulites. I compared the crystallinity of the extruded samples by DSC and established no significant differences (37–39 %) in the degree of crystallinity with different sample preparation methods.

To test the reliability of my experiments on the effect of extrusion parameters, I compared the cross-sectional images of the structure of extruded sheets with some publications studying the effect of cooling temperature and pulling velocity. Macauley et al. investigate the effect of various extrusion parameters on the morphological

properties of PP, such as conveying cylinder temperature, pulling velocity, sheet thickness and melt temperature. For sheet extrusion, they used a tempered cylinder arrangement similar to the arrangement I used in our sheet extrusion. They extruded a PP sheet of 1.4 mm and demonstrated that a pulling velocity between 7 and 9 m/min could increase crystallinity, while a lower pulling velocity in the range of 3 to 5 m/min does not result in a crystallinity difference. With a higher velocity, the molecules are more prone to the orientation of the machine direction. They increased sheet thickness from 0.4 to 1.4 mm. The thinner sheet showed a lower level of crystallinity because of the greater cooling rates used on the sheet. They also varied the temperature of the polymer melt and did not detect a significant influence on crystallinity. They detected an increase in overall crystallinity as the cooling temperature was increased from 30 to 80 °C. At the same time, they measured crystallinity separately inside the sheet and on both sides of the sheet that were in contact with the cooling cylinders. In this case, they found a significant difference in crystallinity along the cross-section of the sheet. On both sides of the sheet, crystallinity was lower than the bulk phase at all cooling temperatures. They concluded that the temperature of cooling cylinders plays a significant role in the morphological characteristics of an extruded PP sheet [136].

Similar to Macauley and his coworkers' results, I found that pulling velocity might have a minor effect on the morphology at the applied low velocity. By varying the temperature of cylinders, I also achieved morphology changes in the cross-section of the samples from core-shell to fully spherulite structure. The significance of the temperature of the cylinder can also be recognized in film extrusion, where only one side of the sheet is connected to the cylinder.

I examined the samples by SEM to compare the morphologies after the solvent immersion and drying process. **Figure 4.11** shows the results of the evaluation of cross-sectional SEM micrographs of the samples, the thickness of skin and the porous layer as a function of cooling temperature of the cylinders. Both sides of the samples were evaluated, and it is represented as side A and side B. The same color (blue or red) belongs to the same side of the sample.

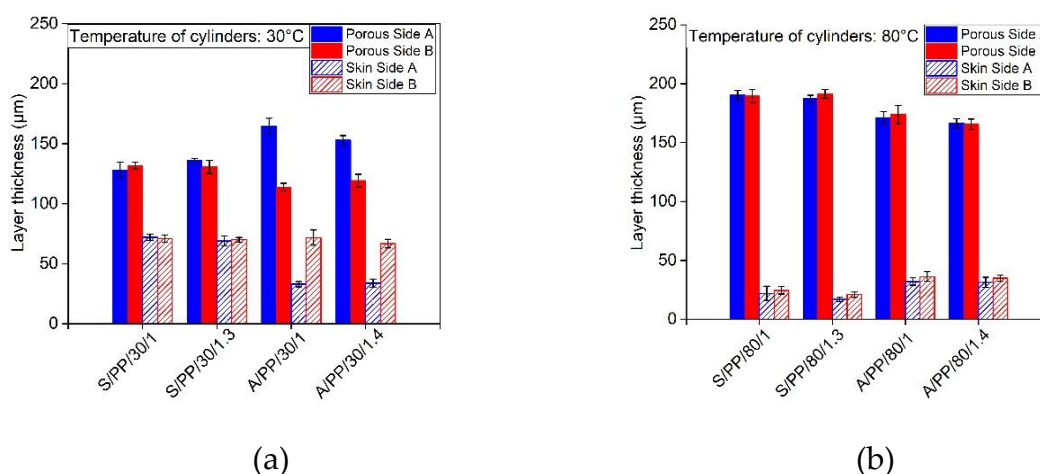


Figure 4.11 Thickness of the skin and porous layers of surface-treated samples cooled at (a) 30 °C and (b) 80 °C

Samples S/PP/30/1 and S/PP/30/1.3 have a symmetrical structure after extrusion. After solvent treatment, the thickness of the skin and porous layers on both sides of the samples is nearly the same. Samples A/PP/30/1 and A/PP/30/1.4 have asymmetric structures; the thickness of the skin and the porous layer are not equal on opposite sides of the samples. The A and B sides of samples S/PP/30/1 and S/PP/30/1.3 and the B side of samples A/PP/30/1 and A/PP/30/1.4 were tempered in a similar way (in contact with the cylinder at 30 °C), and the thicknesses of the skin and porous layer are nearly identical. The air-contact side A (A/PP/30/1 and A/PP/30/1.4) did not show a core-shell-like structure that would differ from the others. The thickness of the porous layer increased, and the thickness of the skin layer decreased on side A.

Figure 4.11b shows the layer thicknesses of the samples cooled at 80 °C. In the case of samples where the original materials have a symmetrical structure (samples S/PP/80/1 and S/PP/80/1.3), layer thickness values on the opposite sides of the samples are the same. Contrary to the results obtained with cooling to 30 °C, the samples cooled to 80 °C samples, even in the case of the asymmetric extruded structure (sample A/PP/80/1 and A/PP/80/1.4), did not show asymmetric morphology after the solvent treatment. The skin and porous layers had the same thickness.

Regarding drawing speeds, we can conclude that its effect can be neglected on the final morphology—the thickness of the layers was similar.

The skin layer of samples cooled to 30 °C was thicker and the porous layer was thinner than those of the samples cooled to 80 °C. No clear relationship can be established between the layer thicknesses of the samples and the drawing rate, regardless of the cooling rate in the investigated range.

After removing the skin layer, surface morphology was evaluated with spherulite size and water wetting behavior. The results are in **Table 4.7**.

Table 4.7 Surface wettability and spherulite sizes of the solvent-treated PP sheets

Samples	CA _{adv} (°)	CAH (°)	Spherulite size (μm)
S/PP/30/1	152.2 ± 1.7	1.1 ± 0.7	21.4 ± 7.3
S/PP/80/1	144.6 ± 9.3	8.0 ± 2.6	15.4 ± 4.3
S/PP/30/1.3	147.1 ± 3.9	3.2 ± 4.3	19.8 ± 6.2
S/PP/80/1.3	148.7 ± 1.1	1.3 ± 0.4	15.3 ± 3.1
A/PP/30/1	153.6 ± 3.2	2.2 ± 1.4	13.1 ± 3.2
A/PP/80/1	151.1 ± 4.0	2.7 ± 1.7	10.8 ± 3.5
A/PP/30/1.4	150.8 ± 1.5	4.0 ± 1.2	13.1 ± 3.2
A/PP/80/1.4	155.8 ± 3.0	1.8 ± 0.3	13.7 ± 4.2

The average size of the spherulites measured on the surface was in the range of 10 to 20 μm, regardless of the morphology structure of the extrudate which was subjected to the solvent treatment. The results indicate that most of the samples show superhydrophobic characteristics. In two cases (samples S/PP/80/1 and S/PP/80/1.3), some reduction can be observed. These lower CA results can be attributed to the mechanical degradation during the removal of the skin layer. In addition to their

superhydrophobicity, the samples also have water-repellent properties because the measured CAH is less than 10° .

Based on the investigation of extrusion parameters, the cooling rate at the cylinder has the primary effect on the thickness of the porous layer and the formation of the skin layer. On the other hand, I found that these processing parameters and methods have an almost negligible effect on the size of spherulites on the surface in the investigated range.

A key step in achieving the superhydrophobic surface is to remove the skin layer after the solvent treatment of the extruded sheets. Therefore, it is important to adjust the extrusion parameters so that the skin layer can be easily removed.

To quantify skin layer removability, I measured the force required to remove the skin layer from the surface by peeling tests. The average tear force results are shown in **Table 4.8**.

Table 4.8 Tear force of the skin layer of surface-treated sheets

Sample	Tear force (N)
S/PP/30/1	3.0 ± 1.0
S/PP/30/1.3	2.1 ± 0.7
S/PP/80/1	18.5 ± 4.6
S/PP/80/1.3	19.8 ± 4.1

The average tear force is a good indication that peeling the skin becomes increasingly difficult at higher cooling temperatures. A higher force was required to skin the samples cooled to 80°C . Here we have to note that in the case of samples cooled to 80°C , the average tear force does not reflect the detachment of the inhomogeneous skin layer from the surface. The separation of inhomogeneous skin layers is shown in **Figure 4.12**. The scatter of the results is well discernible on the blue straight and dash-dot curves.

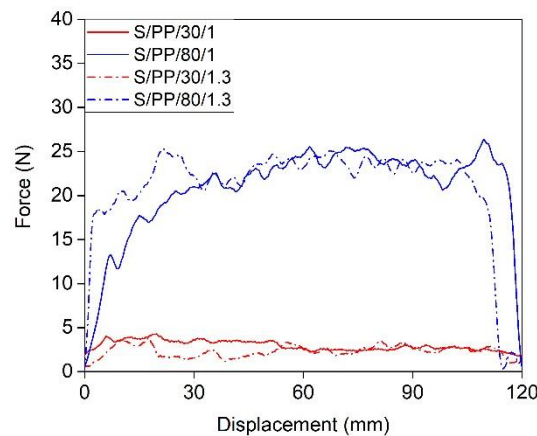


Figure 4.12 Typical peeling test curves

The experiments showed that the skin layer must reach a minimum thickness to avoid the failure of the structured surface underneath the skin layer during the peeling step. The results show that the surface of the extruded sheets with a core-shell structure has a thicker skin layer than the spherulitic structure in cross-section. Thus,

the structure of the extruded sheet cooled to different temperatures will affect the thickness of the skin layer. On the other hand, the cooling temperature of the cylinders does not change the size of the spherulites under the skin layer appreciably.

4.1.4. The effect of nucleating agent on the structure

The previously discussed results show that the surface of the extruded PP sheet can be roughened with the solvent treatment method I developed. However, by changing the parameters of the method or varying the cross-sectional morphology of the extruded sheets I cannot significantly alter surface wettability. Thus, I investigated the effect of a nucleating agent (ADK STAB NA-21E) on micro-, and nanoscale roughness. I used a solvent temperature of 125 °C and an immersion time of 60 s. To evaluate the changes in spherulite size, I compared the surfaces of the treated neat and treated nucleated samples by SEM images (**Figure 4.13**). The nanoscale roughness of the spherulites surfaces was determined with the use of AFM topographical images. **Table 4.9** summarizes the main surface geometry parameters and CA_{adv} of the nucleated and treated samples. Nanoscale roughness is presented with its root mean square (RMS).

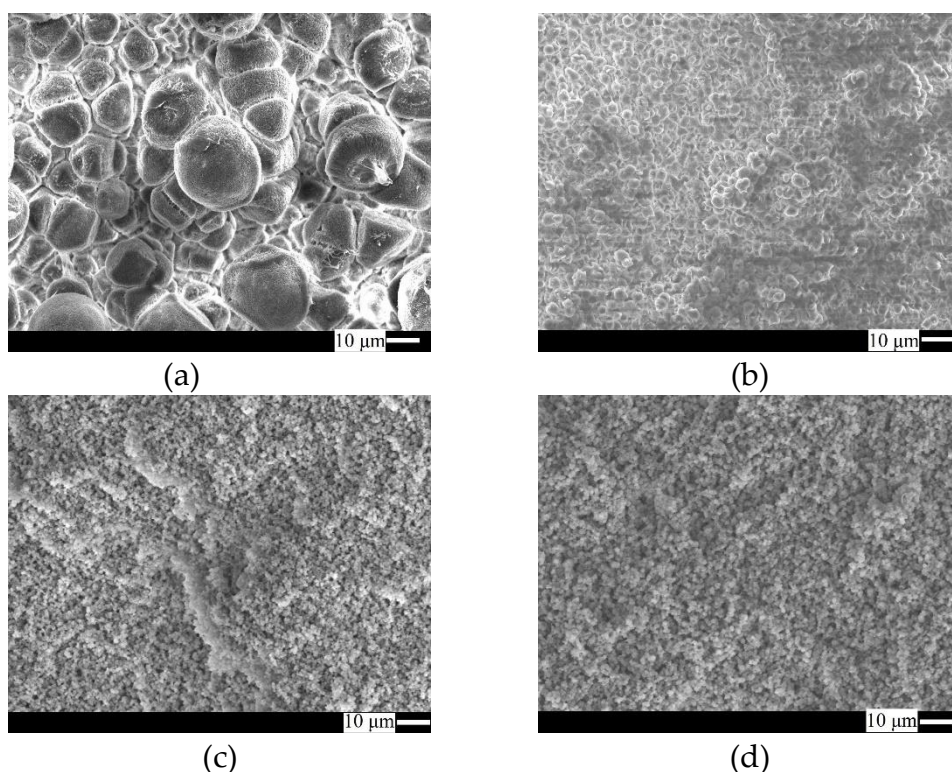


Figure 4.13 SEM images of the PP surface nucleated with ADK STAB NA-21E and treated at 125 °C for 60 s. (a) containing 0 ppm (neat PP), b) 50 ppm, c) 500 ppm, and d) 1000 ppm nucleating agent.

Table 4.9 Dependence of spherulite size, surface roughness and contact angle of nucleated PP on different concentrations of nucleating agent. The samples were treated at 125 °C for 60 s.

Nucleating agent concentration (ppm)	Spherulite size (μm)	RMS of nanoscale roughness (nm)	CA _{adv} (°)	CAH (°)
0	23.20 ± 5.50	52	154.90 ± 1.2	1.1 ± 0.9
50	3.37 ± 0.41	73	159.30 ± 4.2	2.3 ± 1.2
500	1.55 ± 0.21	67	163.15 ± 1.2	1.7 ± 0.7
1000	1.26 ± 0.15	-	163.00 ± 1.1	2.1 ± 1.2

Spherulite size decreased from $23.2 \pm 5.5 \mu\text{m}$ (neat PP) to $1.55 \pm 0.21 \mu\text{m}$ for the sample with 500 ppm nucleating agent and $1.26 \pm 0.15 \mu\text{m}$ for the sample with 1000 ppm nucleating agent. **Figure 4.13** shows that spherulite sizes changed only slightly when the amount of NA-21E is increased from 500 to 1000 ppm NA-21E since the efficiency of NA-21E reaches saturation around 500 ppm [137]. **Table 4.9** also shows that NA-21E at and above 500 ppm reduces average spherulite size by about one order of magnitude compared to non-nucleated PP. In the presence of a heterogeneous nucleating agent, the formation of a nucleus is more likely on the heterogeneous surface of the nucleating agent. Thus, the more crystal nuclei are created by the nucleating agent, the smaller the size of the forming spherulites is [124].

The results indicate that the nucleating agent reduced the size of spherulites significantly, but the nano-roughness analysis of the surfaces of the spherulites did not show remarkable differences between nucleated and non-nucleated samples (**Table 4.9**). The RMS of the nano-roughness of nucleated samples is quite similar to that of non-nucleated samples. The nano-roughness measured on the surface of the spherulites is a consequence of the chain-folded lamellae forming the spherulites and the solvent treatment conditions. Chain folding is affected by various parameters, such as crystallization temperature, chain regularity, molecular weight, and crystallization time [138]. The thickness of the chain-folded lamella depends inversely on the degree of supercooling. The similar nanoscale roughness results may be the consequence of the same treatment conditions. **Table 4.9** shows the effect of the nucleating agent on the CA_{adv} of the treated samples. The samples with a spherulite size below $3 \mu\text{m}$ had a CA_{adv} of more than 160° , indicating superhydrophobicity. The results suggested that increased hydrophobicity was mainly caused by the improved surface roughness as a result of the reduction in spherulite size.

Table 4.10 shows the effect of treatment conditions on the spherulite size of treated PP samples nucleated with 500 ppm nucleating agent.

Table 4.10 Spherulite sizes of the treated PP sheets nucleated with 500 ppm nucleating agent

Immersion time (s)	Spherulite size (μm)		
	Solvent temperature (°C)		
	120	125	130
60	1.47 ± 0.28	1.55 ± 0.21	1.50 ± 0.25
120	1.52 ± 0.31	1.50 ± 0.23	1.42 ± 0.22

The size of the spherulites on the surface of nucleated samples is almost identical regardless of solvent temperature and the immersion time, contrary to the results of the previously presented neat PP samples (**Table 4.2**). In nucleated samples, the nucleus density and thus the sizes of the spherulites are mainly determined by the concentration of the nucleating agent and not by the processing conditions. The crystalline parameters of the PP, NA-PP and solvent-treated NA-PP samples are shown in **Table 4.11**.

Table 4.11 Crystalline parameters of the PP, nucleated PP and solvent-treated nucleated PP. The samples were solvent-treated at 125 °C for 60, 120 and 180 s

Sample	T _m (°C)	T _c (°C)	ΔH _m (J/g)	X _c (%) [*]
PP	166.62	112.71	80	39
NA PP	164.91	128.18	85	41
NA PP_ 60 s	166.93	129.03	89	43
NA PP_ 120 s	167.19	128.88	91	44
NA PP_ 180 s	167.37	128.41	93	45

^{*} X_c is calculated from the first melting enthalpy

For the nucleated PP sample, the crystallization temperature (T_c) increased by 15.5 °C compared to the extruded PP sample. The T_c results suggest that the number of heterogeneous nuclei can be considered almost constant after the solvent treatment, regardless of the values of the treatment parameters, so crystal growth can start from almost the same number of nuclei. Increasing the solvent treatment time also led to a rise in X_c for the solvent-treated nucleated samples, from 41 % to 43 % (60 s), 44 % (120 s), and 45 % (180 s), which also show the phenomena of recrystallization after the solvent treatment.

4.2. Analysis of time-dependent hydrophobic recovery on plasma-treated superhydrophobic polypropylene

The hierarchically structured surfaces enable the creation of two extreme wettability states —superhydrophobicity and superhydrophilicity. This duality in surface properties opens pathways for diverse applications, where controlled wettability plays a crucial role.

A shift from superhydrophobic to superhydrophilic states can be achieved by altering the surface chemical composition. Among various methods, oxygen plasma treatment has been studied for several years as an eco-friendly technique for hydrophilic surface modification [139]. Plasma treatment can increase surface roughness [140] and/or modify surface energy [141, 142] depending on the treatment parameters. It can alter the outermost layers of the substrates and can reduce thermal degradation. Plasma treatment can also serve as an activation step, which is further combined with chemical functionalization [143].

Due to the phenomenon of aging in plasma-treated surfaces, their properties can be time-dependent, which will severely influence the practical applications of these

materials [144, 145]. Because of thermodynamic preference, the surface free energy of the plasma-treated polymer can decrease via reorientation of polar chemical groups from the top layer of the polymer into the bulk and the adsorption of airborne contaminations from the atmosphere on the chemically active sites of the surface [146-149]. Several studies suggested that the main aging mechanism is the reorientation of the hydrophilic groups formed by plasma treatment, however adsorption of airborne contaminations from the atmosphere could also contribute to the hydrophobic recovery [150-152].

Numerous studies have focused on the wettability recovery of plasma-treated polymer surfaces. According to these studies, the process of hydrophobic recovery can be influenced by many factors, such as the plasma parameters [153], relative humidity [154], the crystallinity of the polymer [155], and the temperature of the environment [156], among other things. Polymers with lower crystallinity have also been reported to be more reactive with plasma [157]. While studies have greatly revealed the parameters affecting hydrophobic recovery on plasma-treated PP surfaces and provided valuable applications for plasma treatments, comparing different surface topographies remains challenging as most research focuses on PP with a flat surface. Only a few studies have examined the aging behavior of plasma-treated, roughened, superhydrophilic PP surfaces [158-160]. However, the aging of hierarchically structured polymer surfaces could hinder practical applications. Additionally, the long-term stability of plasma-induced superhydrophilic, roughened PP surfaces under normal conditions has been rarely investigated. Since the applicability of these surfaces is determined by the rate of wettability recovery under ambient conditions, studying the stability of plasma-treated superhydrophilic PP surfaces is essential [161].

This chapter, unlike the surface treatment method based on physical surface modification presented in the previous section, investigates the feasibility of creating superhydrophilic surfaces through the chemical modification of structured surfaces using air plasma technology. As a model material, we used roughened PP surfaces, treated with a solvent at 125 °C for 60 seconds. This chapter provides a detailed analysis of the hydrophobic recovery of plasma-treated, roughened, superhydrophobic PP surfaces. The wettability transition of the hierarchically structured PP sheets, ranging from superhydrophobic to superhydrophilic behavior, is examined through air plasma treatment. Additionally, we explored the time-dependent hydrophobic recovery of these surfaces by studying changes in CAs and surface composition over time.

4.2.1. Air-plasma treatment induced superhydrophilicity on superhydrophobic surfaces

I investigated the effect of plasma treatment parameters on wettability with a focus on the influence of pressure, power and treatment time. The wettability results of the roughened PP samples show that a lower pressure (15 Pa compared to 30 Pa) is advantageous for the creation of a more hydrophilic surface (see Appendix, **Table 8.1**).

This finding is consistent with those of Yun et al., who found that lowering vacuum pressure is optimal for the plasma treatment of PP to achieve maximum hydrophilicity [157]. Subsequently, I focused on only 15 Pa and varied the other plasma treatment parameters.

Table 4.12 shows the combined effect of power and plasma treatment duration on the degree of hydrophilicity.

Table 4.12 The wettability of plasma-treated samples with the use of 15 Pa. The SHPP_LP_60 suffix refers to the surface morphology of PP (PP: samples with smooth surface, SHPP: samples with roughened surface), the applied power (low power (LP): 24 W; medium power (MP): 40 W; high power (HP): 64 W), and plasma treatment time (60 s, 300 s, 600 s)

Designation	Power (W)	Treatment time (s)	CA _{adv} (°)
PP	-	-	100 ± 1
PP_LP300	24	300	20 ± 2
PP_LP600	24	600	24 ± 2
PP_MP300	40	300	21 ± 1
PP_MP600	40	600	19 ± 1
PP_HP300	64	300	18 ± 1
PP_HP600	64	600	12 ± 2
SHPP	-	-	152 ± 1
SHPP_LP60	24	60	40 ± 3
SHPP_LP300	24	300	~0
SHPP_LP600	24	600	~0
SHPP_MP60	40	60	~0
SHPP_MP300	40	300	~0
SHPP_MP600	40	600	~0
SHPP_HP60	64	60	~0
SHPP_HP300	64	300	~0
SHPP_HP600	64	600	~0

CA_{adv} measurements were performed immediately after the plasma treatment. According to the surface analysis, one can conclude that in these cases the plasma treatment has little or no influence on the surface morphology of both PP and SHPP samples. Thus, changes in wettability are due to changes in the chemical composition of the surface due to the introduced functional groups. As expected, increasing power and treatment time leads to more hydrophilic surfaces. For samples with the original smooth PP surface, the treatment reduced the CA_{adv} from ~100 to ~20°, making the samples hydrophilic. At the same treatment conditions, the previously roughened sample showed a CA_{adv} of 152 and ~0° before and after plasma treatment, respectively. This impressive reduction in CA_{adv} is due to the morphology of the sample, which affects the apparent contact angle measured on the surface [162]. The sample with microscale and nanoscale roughness exhibits higher hydrophobicity before plasma treatment and also exhibits higher hydrophilicity after plasma treatment. The surface wettability—based on the high CA_{adv} and low CAH values (see **Table 4.4**) of the SHPP sample—can presumably be interpreted in terms of the Cassie-Baxter’s wetting model

[8]. I further note that extensively long, high-power plasma treatment (64 W for 600 s) led to the discoloration (yellowing) of the samples, due to degradation.

4.2.2. Wetting kinetics and aging of plasma-treated superhydrophilic surfaces

As mentioned before, plasma treatment can produce surfaces that are not in a state of thermodynamic equilibrium, giving way to aging kinetics and changes in wettability. Studying these changes is crucial from a practical point of view, because it determines how long the surfaces retain their properties after treatment. The thermodynamical value of the cosine of CA is defined by an equilibrium of three interfacial tension values, but in practical circumstances, measured CA is not equivalent to thermodynamical CA. Slight deviations (i.e. evaporation of the liquid) can result in the measured CA being time-dependent or undergo spreading due to surface roughness, heterogeneity or porous structure [163-165].

I observed that the water droplet wetting kinetics differed on various plasma-treated surfaces, all exhibiting superhydrophilic wetting behavior with a similar equilibrium CA_{adv} of nearly 0° . This phenomenon is exclusive to plasma-treated surfaces that were previously roughened. To find the cause of this unique droplet spreading behavior, I investigated the kinetics of wetting on differently plasma-treated samples, while also exploring aging effects on the wetting behavior of plasma-treated surfaces (**Table 4.13**).

Table 4.13 Wetting kinetics: spreading time (s) needed for water droplets to reach complete wetting ($\sim 0^\circ$) after droplet deposition measured on different aging days

	Spreading time for complete wetting (s)				
	Immediately after treatment	1 day after treatment	2 days after treatment	3 days after treatment	7 days after treatment
SHPP_LP60	not reached	not reached	not reached	not reached	not reached
SHPP_LP300	4	24	33	44	180
SHPP_LP600	0	4	5	9	12
SHPP_MP60	40	not reached	not reached	not reached	not reached
SHPP_MP300	1	4	20	20	20
SHPP_MP600	0	0	0	0	0
SHPP_HP60	10	150	180	not reached	not reached
SHPP_HP300	0	0	9	9	10
SHPP_HP600	0	0	no data	no data	10

In **Table 4.13**, the spreading times for each sample indicate the time required for complete wetting after different aging days. In **Table 4.13**, „not reached” refers to when the complete wetting could not be achieved within the frame of the investigation time, which was 60 minutes, for example, in the case of SHPP_MP60 and SHPP_HP60 samples. „No data” means that the spreading time was not measured, for example, in the case of SHPP_HP600. The importance of plasma treatment time is clearly shown by the fact that at a given pressure, as the plasma treatment time increases, film wetting develops faster. The samples plasma-treated for the shortest time (SHPP_LP60, SHPP_MP60, SHPP_HP60) either did not reach or required the longest time to develop film wetting.

Up to a certain point (SHPP_MP600: 15 Pa, 40 W, 600 s), the time needed to reach film wetting continuously decreases with higher power and longer treatment time. Due to the more intense treatment parameters, the concentration of hydrophilic groups forming on the surface increases, which allows faster spreading on the surface [166]. In the case of SHPP_MP600, the drop immediately reaches near-zero equilibrium CA even after seven days of aging. Applying more intense parameters than in the case of SHPP_MP600, the time needed for the drop to achieve film wetting increases again possibly due to degradation.

The effect of hydrophobic recovery can be observed in wetting kinetics data (droplet spreading) measured after various aging times. These data allow us to compare the hydrophilicity of samples on which water droplets exhibit identical, near-zero equilibrium CAs.

To determine the changes in surface functional groups as a function of aging time, I performed XPS studies for the plasma-treated roughened and smooth samples.

Table 4.14 shows the calculated atomic compositions of the plasma-treated roughened samples (SHPP_LP300) and smooth samples (PP_LP300) as determined by XPS at different aging times. SHPP_LP300 was selected for XPS analysis due to the pronounced change in its wetting kinetic.

Table 4.14 Surface composition of roughened (SHPP_LP300) and smooth (PP_LP300) samples measured by XPS in comparison with CA_{adv} measurements

		SHPP_LP300				PP_LP300			
		elemental composition (%)		O/C atomic ratio	CA_{adv} (°)	elemental composition (%)		O/C atomic ratio	CA_{adv} (°)
		C	O			C	O		
Aging time (day)	Before plasma treatment	97.8	2.1	2.1	152	97.9	2.0	2.0	100
	0	71.0	20.5	28.9	0	71.0	19.8	27.9	20
	1	73.6	18.8	25.5	0	73.8	17.6	23.8	29
	2	75.7	17.1	22.6	0	75.4	16.1	21.4	41
	3	77.7	16.1	20.7	0	77.3	14.6	18.9	47
	7	79.6	14.6	18.3	0	79.5	12.9	16.2	52

*O/C atomic ratio: the relative amount of oxygen atom compared to carbon atom on the surface

Table 4.14 shows that after the plasma treatment, the concentration of oxygen increased significantly due to the introduction of polar functional groups. The introduced oxygen functional groups resulted in samples with a hydrophilic surface. Immediately after the plasma treatment (day 0), the oxygen concentration on the surface of sample SHPP_LP300 was slightly higher compared to sample PP_LP300. During aging, both samples showed a similar decrease in oxygen concentration but the CA_{adv} measured on their surfaces significantly differed.

The roughened sample with higher crystallinity ($X_C \approx 41$ %) had a higher concentration of atomic oxygen compared to samples with a smooth surface and lower

crystallinity ($X_c \approx 39\%$). This result can be controversial since lower crystallinity was previously found to have greater chemical reactivity with plasma [155, 157]. However, the increase in the specific surface area of the roughened samples provides a larger surface area for the formation of functional groups, which could counterbalance its greater crystallinity. As shown, 7 days of aging resulted in a 28.8 % decrease in oxygen concentration with SHPP_LP300, and 34.8 % with PP_LP300 samples. All this clearly shows that the increase in crystallinity hinders hydrophobic recovery due to higher structural regularity in the polymer matrix. Despite the decreasing oxygen concentration, the hierarchical surface structure enables film wetting, even after multiple days of aging.

For the characterization of the functional groups formed on the samples and their changes as a function of aging time, the best-fitted high resolution C1s spectra were investigated. **Figure 4.14** and **Figure 4.15** show the comparison of C1s spectra of the SHPP_LP300, and PP_LP300.

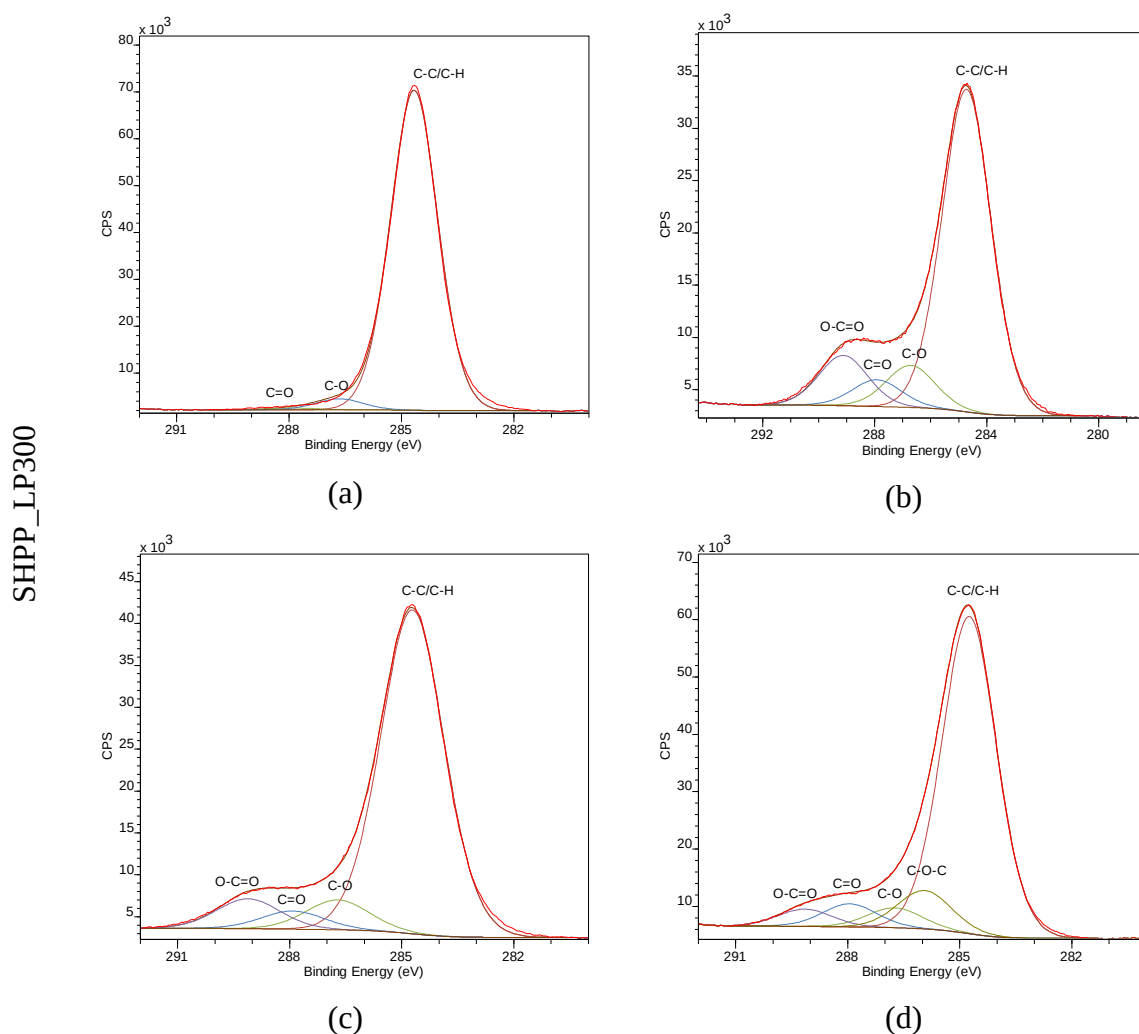


Figure 4.14 High resolution XPS spectra for PP samples, C1s spectra of SHPP_LP300 with aging time, (a) before plasma treatment, (b) immediately after plasma treatment, (c) 2 days and (d) 7 days after plasma treatment

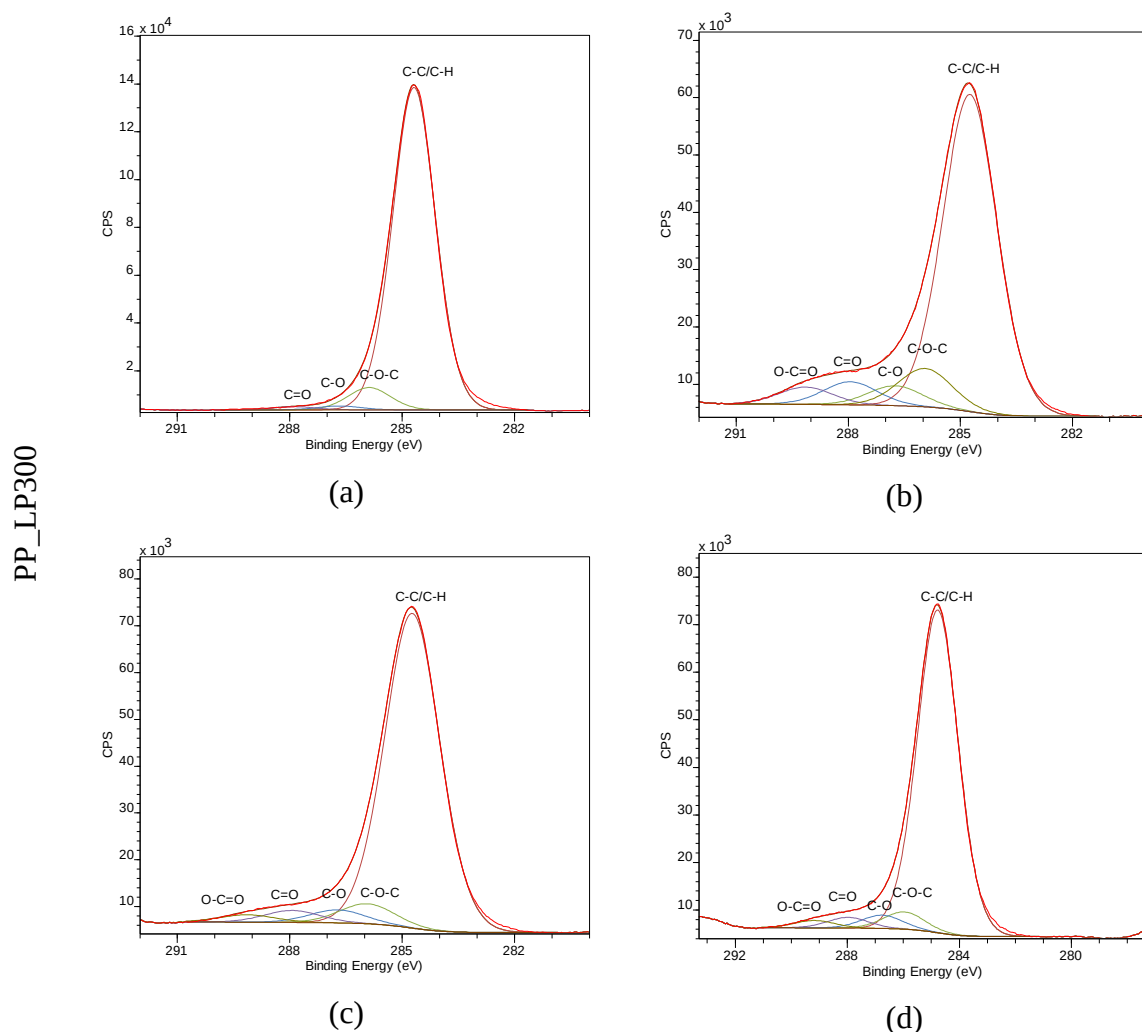


Figure 4.15 High resolution XPS spectra for PP samples, C1s spectra of PP_LP300 with aging time, (a) before plasma treatment, (b) immediately after plasma treatment, (c) 2 days and (d) 7 days after plasma treatment

As **Figure 4.14** and **Figure 4.15** show, before plasma treatment there is only a slight difference in the surface composition of SHPP_LP300 and PP_LP300 samples. In addition to the C-C/C-H peak at 284.7 eV, which presents in the highest concentration, other peaks also appeared at higher binding energies for the C-O peak at 286.7 eV, C=O peak at 287.9 eV, O-C=O peak at 289.1 eV and C-O-C peak at 286 eV. The presence of oxygen-containing groups on the surface of samples might be the result of either thermo-oxidative degradation during the extrusion of the sheets, oxidation of the samples in the air, or intrinsically contained oxygen in the samples. For both samples, due to the plasma treatment, the concentration of the polar functional groups containing oxygen increased significantly. The aging behavior of SHPP_LP300 and PP_LP300 was periodically analyzed to follow the effect of hydrophobic recovery in the first 7 days of aging. The hydrophobic recovery occurred predominantly in the first two days after the plasma treatment. After that, I did not experience significant changes in the concentration of functional groups. **Table 4.15** shows the relative

concentrations of the functional groups of SHPP_LP300 and PP_LP300 after the plasma treatment calculated from C1s high-resolution spectra.

Table 4.15 The relative concentrations of the functional groups (%) of the fitted peaks for C1s spectra of plasma-treated roughened (SHPP_LP300) and smooth (PP_LP300) samples as a function of aging time

		SHPP_LP300				PP_LP300				
		C-C/C-H	C-O	C=O	O-C=O	C-C/C-H	C-O-C	C-O	C=O	O-C=O
Before plasma treatment		96.1	3.3	0.5	-	91.6	6.4	1.1	0.9	-
Aging time (day)	0	71.8	10.1	6.2	11.9	76.3	9.2	4.8	5.6	4.1
	1	76.4	8.4	5.5	9.7	80.1	8.0	4.1	4.6	3.2
	2	78.5	7.8	5.1	8.4	83.2	6.3	3.8	4.0	2.7
	3	80.6	7.4	4.5	7.4	85.7	5.4	3.6	3.3	2.0
	7	83.9	6.4	3.7	6.0	89.9	4.6	3.6	2.9	2.0

Table 4.15 shows the decrement of the relative concentration of C–C/C–H bond from 96.1 to 71.8 % (SHPP_LP300) and from 91.6 % to 76.3 % (PP_LP300) after the plasma treatment and then a gradual increase during aging time (days). In the first two days of aging, the concentration of oxygen functional groups decreased drastically, which shows that the rate of hydrophobic recovery is faster at the beginning of aging and, after two days, slows down gradually.

Hydrophobic recovery is known to be highly dependent on aging temperature [167]. To investigate the stability of the plasma-induced hydrophilicity of SHPP_MP600 samples with roughened surfaces, I stored them at 70 and 150 °C for one hour before measuring their CA_{adv} (see Appendix **Table 8.4**). I found that heat treatment caused the surfaces to become highly hydrophobic. A one-hour heat treatment at 70 °C increased the CA_{adv} on the polymer surface to $140 \pm 1^\circ$ and a one-hour heat treatment at 150 °C caused an even greater hydrophobic recovery to $145 \pm 1^\circ$.

4.2.3. Approach to characterize droplet spreading behavior of superhydrophilic, roughened PP surfaces

As stated above, the wetting behavior of aged superhydrophilic surfaces cannot be expressed with one CA parameter, because during droplet spreading, the apparent contact angle shows a monotonous decline after water droplet deposition. The characterization of recovery requires a special approach, since wettability changes over time. To overcome this difficulty, I aimed to find functions that can be properly fitted to our measured data, and the parameters which can be used to compare these surfaces.

I investigated the kinetics of this spreading using momentary CA measurement time graphs, which I acquired by analyzing video samples frame by frame recorded

during CA measurements. I used a simple exponential function based on the study of Dabóczy et al. [168] to fit curves on the plotted graphs (Equation 4.1):

$$\theta(t) = \theta_s + (\theta_0 - \theta_s) * e^{\left(\frac{-t}{T}\right)} \quad (4.1)$$

where θ_s denotes the stationary contact angle, and θ_0 and T represent fitted initial contact angle and characteristic spreading time, the values of which I acquired by fitting the function on our graphs. A larger value of T indicates a slower rate of spreading. Since the equilibrium contact angle θ_s is 0 in our case due to near-perfect wetting, the θ_0 parameter of the fitted curves corresponds to the apparent contact angle at the start of the measurement (Equation 4.2):

$$\theta(t_0) = \theta_0 \quad (4.2)$$

Figure 4.16a depicts the decline of the apparent contact angle in time during the CA measurement on the SHPP_LP300 sample. This fitting method was used to evaluate the behavior of multiple samples prepared with various plasma treatment parameters and after various times of room temperature storage. The fitted T values are shown in **Figure 4.16b**.

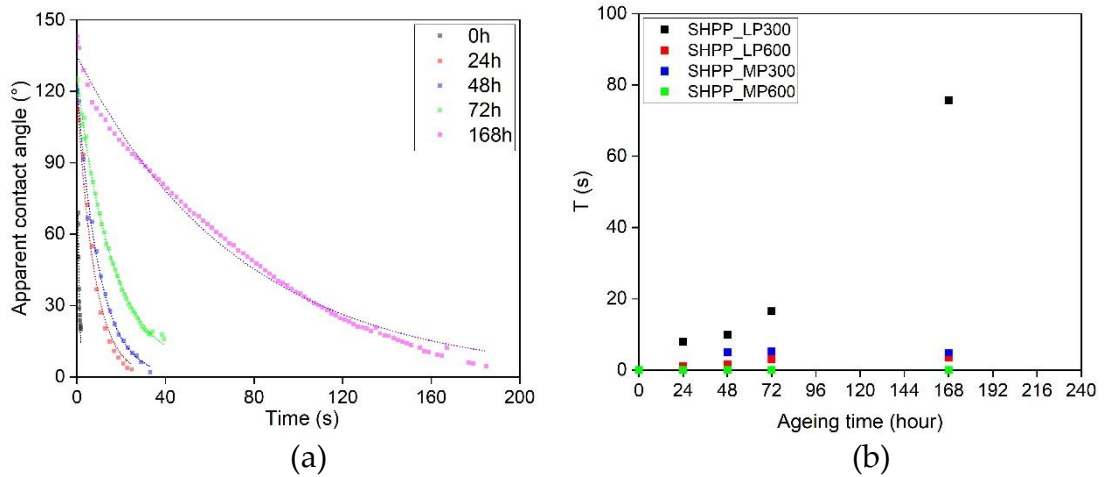


Figure 4.16 (a) Exponential function fitted to the spreading results measured on the SHPP_LP300 sample after different aging time after the plasma treatment. (b) The fitted T values plotted as a function of aging time for different plasma treatment parameters

The hydrophobic recovery of the samples prepared with different plasma parameters can be characterized with the T parameter, as shown in **Figure 4.16**. These results clearly show the effect of hydrophobic recovery on the spreading rate of samples that exhibit near-perfect wetting ($\theta(t_\infty) = 0^\circ$), SHPP_LP300, SHPP_LP600 and SHPP_MP300. The fitted T values increase monotonously with aging time in these cases. This increase represents a change in wetting kinetics, or more precisely, a decrease in the power of the exponential function. In contrast, sample SHPP_MP600 exhibited nearly instantaneous wetting even after one week of aging.

The results show the effect of the intensity of plasma treatment on the rate of hydrophobic recovery experienced on the surfaces. Plasma treatment with optimal parameters results in a sample with near-spontaneous perfect wetting and slow hydrophobic recovery. Increasing plasma power above a certain point (HP samples) leads to faster recovery due to the degradation of the sample (see Appendix **Table 8.3**).

I propose an explanation for the droplet-spreading behavior on our rough surfaces. Plasma treatment, as previously stated, greatly increases the hydrophilicity of the PP surface. This hydrophilicity, combined with the structured surface, can potentially cause water intake via the effect known as capillary wetting. Capillary wetting caused by the morphology of the samples decreases droplet volume and increases spreading speed. **Figure 4.17** shows a model for the change in wettability after plasma treatment.

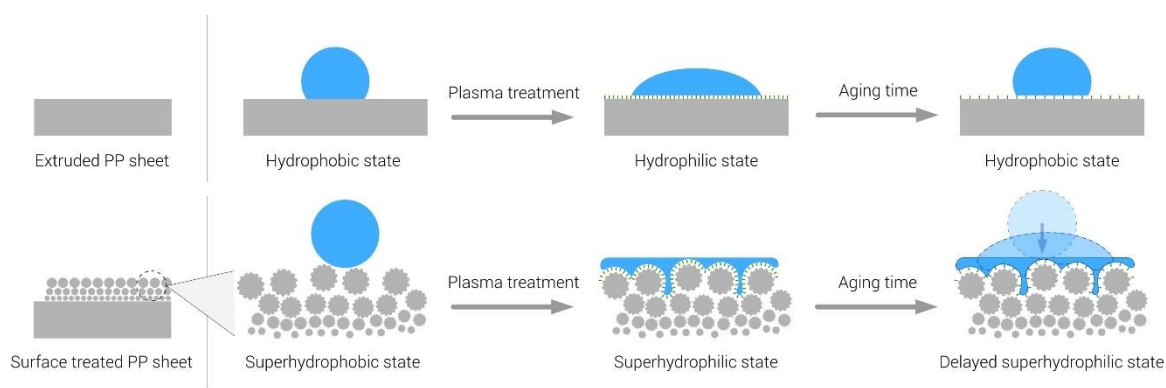


Figure 4.17 A schematic of wettability phenomena observable on plasma-treated PP surfaces depending on roughness and aging time

After solvent treatment, the originally smooth surface of extruded sheets had microscale and nanoscale roughness resulting in a porous surface structure. When the samples (smooth and roughened) were exposed to plasma treatment, hydrophilic groups formed at the outermost surface through plasma oxidation and no morphological changes were detected. The porous surface structure of the surface-modified sample and the possible capillary imbibition are demonstrated in **Figure 4.17**. Plasma exposure can mostly impact the accessible outermost surfaces of the sample and hardly diffuse into the pores. The surface of the spherulites in the deeper layers of the porous structure is subject to reduced plasma exposure because they are shielded by the spherulites on the outermost surface. However, the hydrophilic groups were also introduced in deeper layers to a lesser extent, which enabled the imbibition of water droplets into the pores below the surface and thus helping the fast spreading of water droplets and film-wetting.

Due to the porous surface structure and the high surface energy created by hydrophilic groups, the surface became superhydrophilic, leading to almost complete film wetting. Moreover, samples treated with high-power and long plasma treatment showed signs of degradation, especially at the outermost surface of the modified layer. The increased spreading time of droplets to reach complete wetting after the plasma

treatment results from the rearrangement of polar groups toward the bulk phase to reach a more stable state. The concentration of the polar groups decreased over aging time, thus more time was needed for the drop to reach its equilibrium wetting state. As **Figure 4.17** shows, presumably only the surface energy changes during recovery while the roughness remains constant, which might enable the application of PP over a wider time interval.

4.3. Solar reflectivity of porous PP samples

Environmentally friendly and efficient approaches for outdoor thermal management are equally essential [169]. High sunlight reflection is a key factor in mitigating the effect of intense solar radiation. Approaches, like solar reflective techniques, are attractive for their economic viability and suitability for buildings [2]. These methods can significantly reduce direct solar heat gain within buildings, lowering energy demand. Developing materials with reflective and thermal insulation properties across the entire solar spectrum, which can effectively help in lowering overall energy consumption is still a hot topic [170, 171].

In the previous chapters, the formation of a roughened surface layer on PP was shown to provide significant potential for modifying surface wettability, ranging from superhydrophobic to superhydrophilic properties. The primary factor influencing the wettability of solvent-treated samples is their surface structure. Additionally, the modified surface layer contributes to other functional properties of the samples. Beyond the roughened surface, the porous structure enhances solar light reflection. This chapter focuses on investigating the relationship between the morphology of the porous layer—specifically the size and size distribution of spherulites and the thickness of the porous layer—and solar light reflection. Furthermore, the synergistic effects of reflective materials (TiO_2 , BaSO_4) and the porous structure on light scattering are also analyzed in detail.

4.3.1. *The solar reflectivity of porous surface layer*

Figure 4.18 shows the total and diffuse reflectance of untreated and solvent-treated PP samples in the 200–2500 nm wavelength range. Usually, we can use TSR to evaluate the reflectivity of a material.

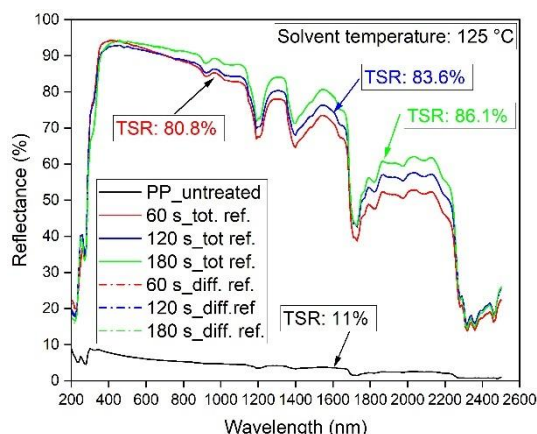


Figure 4.18 Total and diffuse reflectance of solvent-treated PP sheets after different immersion times (60 s, 120 s, 180 s) compared to the untreated sample

As expected, the untreated extruded PP sheet, which has a smooth surface, reflects less than 10 % of solar irradiance, the lowest value among all the investigated structures. In the 200–2500 nm range, the spectrophotometry results of solvent-treated samples show almost an order of magnitude stronger reflection compared to their untreated counterparts due to the porous layers created. By increasing immersion time from 60 s to 180 s, TSR increased from 11.0 % (untreated PP) to 80.3 % (60 s) and 86.1 % (180 s). However, while immersion time increased threefold, TSR only increased by 7 %.

In typical porous polymer structures, the scattering units are circular air voids of different sizes, and scattering can be described with the Mie theory. In contrast to these structures with circular air voids, the collective sunlight scattering behaviour of the porous PP samples are more complex and may be attributed to different factors, including the thickness and surface roughness of the top dense layer (skin layer) and the underlying porous layer. At the polymer–air interface in the porous PP layer, the refractive index mismatch between PP spheres ($n_{PP}=1.49$) and the surrounding air ($n_{air}=1.0$) results in intense sunlight scattering. Increasing porosity thus the specific surface area by reducing spherulite size, leads to higher scattering ability. Additionally, the anisotropic porous structure and the hierarchical spherulite structure amplify scattering across ultraviolet, visible, and near-infrared wavelengths, leading to a robust scattering effect [172, 173].

In addition to the scattering from the interfaces, the spherulite itself could also act as a birefringent scattering center [173]. However, in our case, spherulites as scattering units have a minor contribution as the light-scattering effect of spherulites decreases proportionally with the decrease in their size [173], which is the opposite of what I observed.

Above 800 nm, in the NIR range, absorption bands are superposed on the spectra, which can be attributed to methylene and methyl overtones and combination bands, characteristic of PP. The absorption bands remain practically the same (determined by the chemical structure), while the physical (scattering) processes determine the shift of the spectra. The differences in total and diffuse reflectance amount to a few percent

(Figure 4.18), because the incident light reflected from the surface is almost entirely reflected diffusely, which can be explained with the surface structure. In the following, only the total reflectance values are used. Solvent-induced recrystallization leads to a restructured, partly porous layer in PP samples (covered by a protective skin layer), resulting in an enhanced diffuse reflection in the whole solar spectrum. Incoming light will suffer multiple reflections and be reflected back diffusely as it meets the surface and near-surface porous layers.

Similarly, I investigated the importance of the skin layer on reflectance by comparing the total reflectance of solvent-treated samples with and without the skin layer (Figure 4.19).

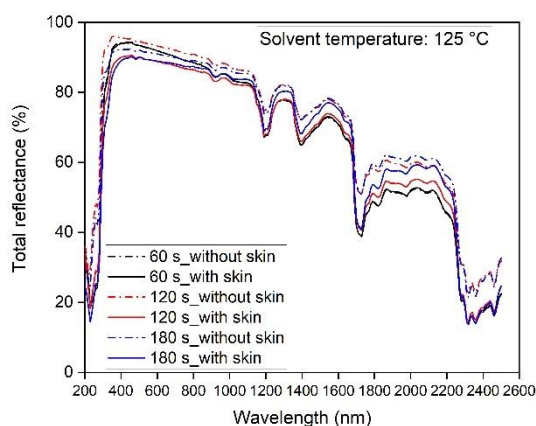


Figure 4.19 Total reflectance of solvent-treated PP sheets with and without the skin layer

I found that the reflectance of the samples without intact skin layers is slightly higher than that of the unpeeled samples solvent-treated for 60 s, 120 s, and 180 s, by 0.01 %, 1.05 %, and 1.84 %, respectively. I concluded that the skin layer is practically transparent and forms a “conformal layer” on the sample surface, therefore it does not influence scattering. Moreover, the skin layer protects the subsurface, so I retained it [116]. Leaving out the peeling process simplifies the manufacturing process and improves the mechanical stability of the product.

To achieve better solar reflection and thus more effective thermal insulation, I examined two strategies: the combination of a porous structure with reflective additives and the modification of the porous structure with a PP nucleating agent.

4.3.2. Combination of the porous structure with reflective additives

To further improve solar reflection, I distributed TiO_2 and BaSO_4 particles in the PP matrix in different amounts (5, 10 and 20 wt%) and solvent-treated the samples at 125 °C for 60, 120 and 180 s. I investigated how the porous PP composite samples composed of a three-dimensional porous structure, and internally distributed reflective additives can enhance solar reflectance.

In **Figure 4.20a** and **Figure 4.20b**, I compare the solar reflectance of PP composites (PP/TiO₂ and PP/BaSO₄) before and after the solvent treatment. Only the samples solvent-treated for 180 s are plotted in **Figure 4.20**.

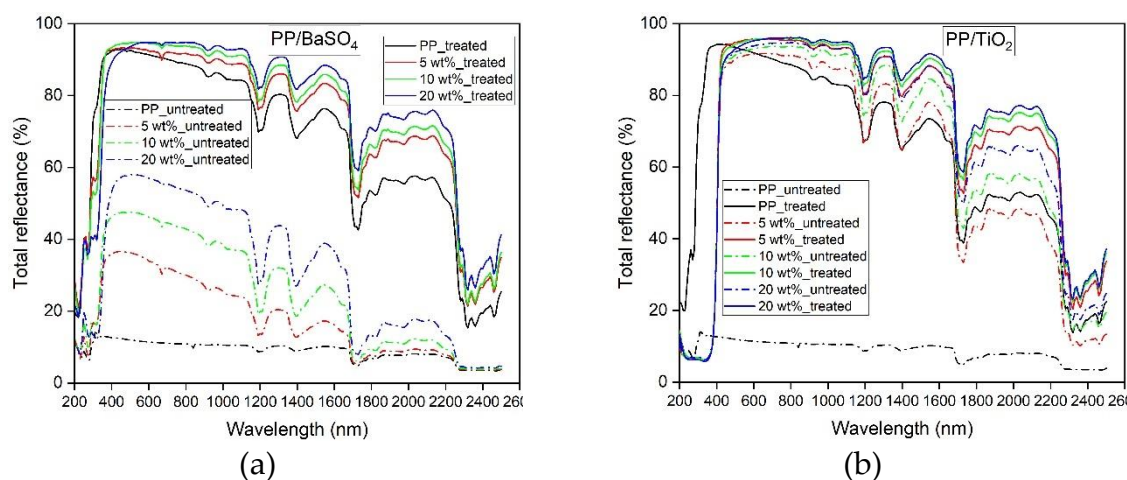


Figure 4.20 Total reflectance of (a) PP/BaSO₄ samples and (b) PP/TiO₂ samples before (noted as untreated) and after (noted as treated) the solvent treatment. The solvent treatment conditions were: 125 °C, 180 s

In **Figure 4.20a** and **Figure 4.20b**, samples labeled with dash-dot lines represent extruded PP and PP composites before the solvent treatment and the solid lines represent the solvent-treated PP and PP composites. In the case of PP composites, I experienced a different increment in solar reflectance compared to the extruded PP samples depending on the type of additives, as I increased filler content from 5 % to 20 %. The solar reflectance of PP/BaSO₄ composite samples without solvent treatment shows moderate improvement. One possible reason for the moderate reflection performance of PP/BaSO₄ composite samples might be the lower refractive index of BaSO₄ ($n \approx 1.64$) that of TiO₂ ($n \approx 2.6$). A pleasant surprise is that the solar reflectance of all the PP/TiO₂ composite samples is dramatically higher. As it is known, TiO₂ is a strong UV absorber, thus the PP/TiO₂ composite samples show a sharp drop in the UV range (200–400 nm), resulting in a UV reflectance below 10 %. However, in the VIS (400–800 nm) and NIR (700–2500 nm) wavelength range, reflectance increases significantly with increasing concentration of TiO₂. In the VIS wavelength range, the reflectance of porous PP treated for 180 s is exceeded by the reflectance of PP/TiO₂ composite samples with 5 wt% of TiO₂, which also shows the excellent VIS reflection property of the TiO₂ additive. Above 10 wt% of TiO₂ concentration, PP/TiO₂ samples show better performance in both the VIS and NIR light ranges compared to the porous PP sample treated for 180 s. **Table 4.16** shows the TSR values of extruded PP and PP composites with different reflective additive concentrations.

Table 4.16 The TSR values of PP/BaSO₄ and PP/TiO₂ composites

Weight percent of additives (wt%)	TSR (%)	
	PP/BaSO ₄	PP/TiO ₂
0	11.6	11.6
5	28.3	82.0
10	39.3	84.9
20	49.1	86.7

A comparison of the TSR of extruded PP samples with and without an additive shows that both additives increased TSR considerably but TiO₂ was far more effective. 5 wt% BaSO₄ in the PP matrix led to a 2.5 times increase in TSR, and 20 wt% produced an increase in TSR a little over four times. TiO₂ on the other hand, produced a very high TSR even in 5 wt% (over 7 times more than the neat PP sheet). However, increasing the amount of TiO₂ did not increase TSR much.

The synergistic effect of the porous structure and reflective particles results in the enhancement of solar reflection performance. I extended the solar reflection comparison of solvent-treated PP composites with different concentrations of additives and varied the immersion times (60, 120 and 180 s). The TSR results of solvent-treated PP composite samples as a function of the thickness of the porous layer can be seen in **Figure 4.21a** and **Figure 4.21b**.

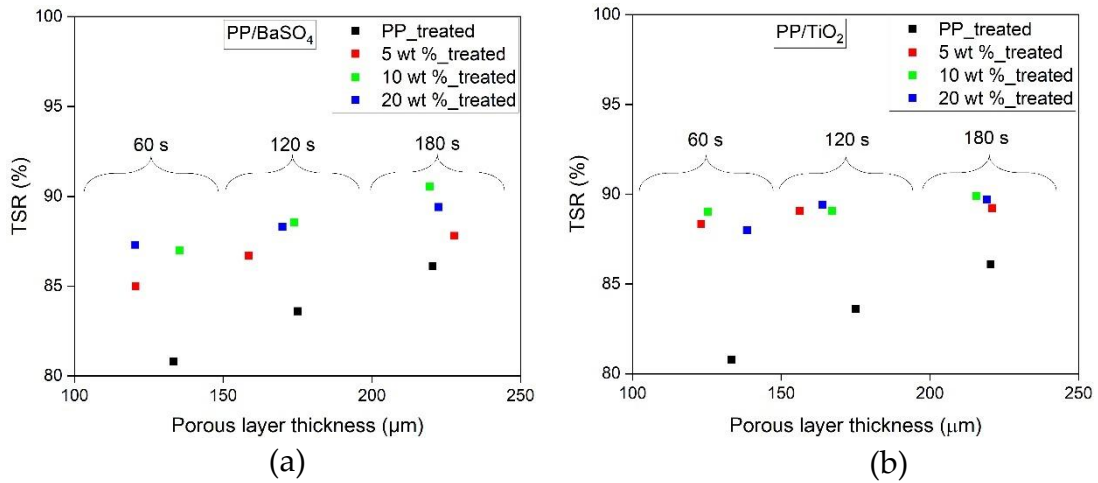


Figure 4.21 TSR result of (a) porous PP/BaSO₄ samples and (b) porous PP/TiO₂ samples as a function of porous layer thickness. The solvent treatment conditions were 125 °C, and 60, 120 and 180 s.

In **Figure 4.21**, the TSR values of porous PP samples (without any additive) with different pore thicknesses are shown with black squares. Porous PP/BaSO₄ (TSR: 85.0 %) and porous PP/TiO₂ (TSR: 88.3 %) samples, both containing a minimal amount of additives (5 wt%) and with a brief solvent treatment (60 s), exhibit comparable or even enhanced performance compared to porous PP, solvent treated for 180 s (TSR: 86.1 %). This improvement can be attributed to the effective light scattering caused by the additives, as well as the inherent internal pore structure formed by spherulites.

Through increasing additive concentration and solvent immersion time, the solvent-treated PP/BaSO₄ composite samples exhibited a gradual rise in TSR, reaching approximately 90 %. This increase was achieved by simultaneously enhancing additive concentration and prolonging immersion time, resulting in an increase in porous layer thickness. Notably, the PP/BaSO₄ composite samples containing 10 wt% and 20 wt% BaSO₄, with porous layer thicknesses of 219.5 μm and 222.5 μm , respectively, had a TSR of 90 %. These specific layer thicknesses corresponded to a solvent treatment time of 180 s.

In the case of PP/TiO₂ composite samples, increased additive content or solvent treatment time did not increase TSR significantly over the TSR of the sample with the lowest amount of the additive and solvent-treated for the shortest time (60 s). The 5 wt% PP/TiO₂ samples prepared with the shortest immersion time (60 s) had a porous layer thickness of 123.1 μm and a TSR of 88.3 %. The 10 and 20 wt% PP/TiO₂ samples surface-treated for 180 s have porous layer thicknesses of 215.5 and 219.1 μm and a TSR of 89.9 and 89.6 %, respectively. A shorter treatment time is preferable for the solvent-treated PP/TiO₂ composite samples since treatment is faster and the thickness of the porous layer of samples treated for 60 s is half that of samples solvent-treated for 180 s, resulting in samples with better mechanical properties. Furthermore, the lower content of additives also contributes to reducing the cost of production.

Comparing the solvent-treated and non-solvent-treated PP/BaSO₄ and PP/TiO₂ composite samples, I can conclude that the hierarchical porous surface structure formed on the outer layer of the porous PP/BaSO₄ composite samples contributes significantly to the further increase in solar reflectance. In the solvent-treated PP/TiO₂ composite samples, the porous layer contributed to a lesser extent to the increase in the solar reflectance of the untreated samples.

Nonetheless, recent research has highlighted potential environmental and human health concerns related to the utilization of TiO₂ [174]. Moreover, these additives have the potential to considerably increase the overall cost of the final product. Therefore, it becomes imperative to explore alternatives for reflective additive-based systems, taking into account optical performance, as well as environmental and health considerations. Additionally, considering the increasing demand for monomaterial usage to facilitate product recyclability, I investigated the solar reflectance of nucleated PP samples.

4.3.3. Adjusting the porous layer with a nucleating agent

During the solvent treatment process, the nucleating agents provide more nuclei, so during recrystallization, spherulite growth can start from several points, which results in a decrease in their size. The selected NA 21E nucleating agent was added to PP in a concentration of 500 ppm. The SEM cross-sectional morphology of solvent-treated, nucleated PP samples are shown in **Figure 4.22**.

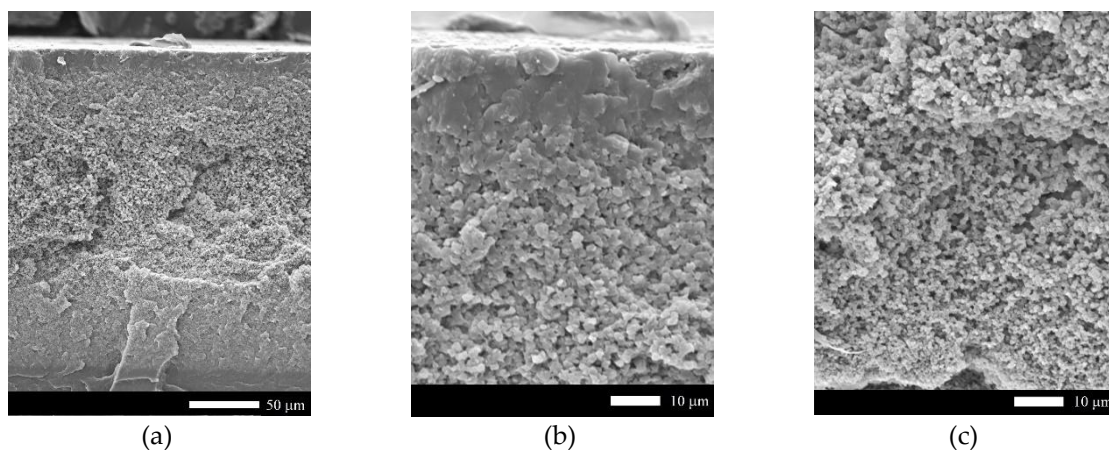


Figure 4.22 Cross-sectional SEM image (a) of a porous layer of a nucleated PP sample. SEM image of the magnification of the upper (b) and the lower (c) part of the porous layer in (a). The applied solvent treatment conditions are 125 °C and, 60 s

With the application of a nucleating agent, it is possible to effectively reduce the size of the spherulite in the cross-section, thus we can create a more structured porous layer with a larger specific surface area. The diameter of each spherulite is less than 5 μm , and the nucleated PP samples also keep their anisotropic porous structure.

Figure 4.23 shows the solar spectra of solvent-treated nucleated PP samples, treated for different immersion times, and also the TSR of the individual samples.

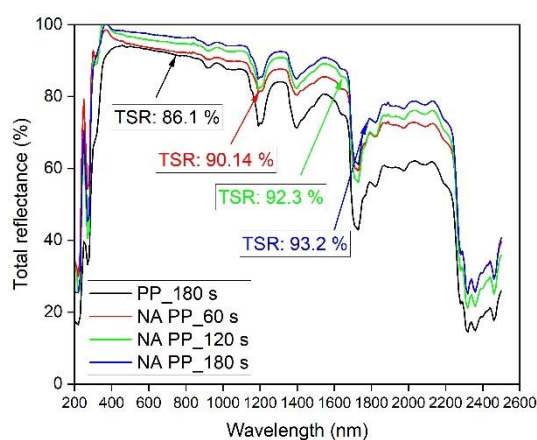


Figure 4.23 Solar reflectance of porous nucleated PP samples solvent-treated for 60 s, 120 s and 180 s. The TRS values for the spectra are marked with the same color

The solvent-treated nucleated PP samples treated for 60 s reached a TSR of 90.1 %, compared to the 86.1 % of porous PP, treated for 180 s. More notably, increasing the immersion time of nucleated PP samples increased TSR to 90.1 % (60 s), 92.3 % (120 s) and 93.2 % (180 s).

Porous nucleated PP samples showed better solar reflectance than porous PP composite samples (without nucleation) regardless of solvent treatment time, presumably due to their increased porosity and specific surface area.

I compared the solar reflectance performance of the porous nucleated PP sample treated for 180 s with that of heat-reflective commercial paints [106, 175] and recently developed light-scattering porous polymer materials [176-179] (**Table 4.17**).

Table 4.17 Comparison of reflectance of various reflective additive-based composites and porous polymers

Reference	Fabrication method	Reflectance (%)
[106]	TiO ₂ /acrylonitrile-styrene-acrylate copolymer hybrid material (volume concentration = 5 %)	59.4
[179]	Acrylonitrile-styrene-acrylate copolymer /barium titanate (volume concentration = 1 %)	70.0
[178]	Polytetrafluoroethylene porous coatings	95.0
[176]	PMMA/chlorinated poly- ethylene blending material with porous surface (porosity = 30 %)	93.7
[177]	UV irradiated HDPE/ TiO ₂ composite	71.2
This work	Porous nucleated PP	93.2

As **Table 4.17** shows, the solar reflectance of the porous surface layer in nucleated PP samples (as prepared in this study) surpasses that of additive-based composites. Also, the solar reflectance of the porous nucleated PP samples is nearly identical to that of alternative porous polymer materials.

The TSR values are the results of the theoretical calculation of the reflected solar energy, which should also be verified with a reflective insulation test. For this purpose, I performed laboratory-scale simplified sun tests by measuring the temperature inside coated boxes covered with solvent-treated samples and inside a reference box under a sunlight lamp in laboratory conditions. In **Figure 4.24**, I plotted ΔT as a function of time. The samples with the best TSR were selected for the reflective insulation test. The higher the ΔT value, the better thermal insulation is.

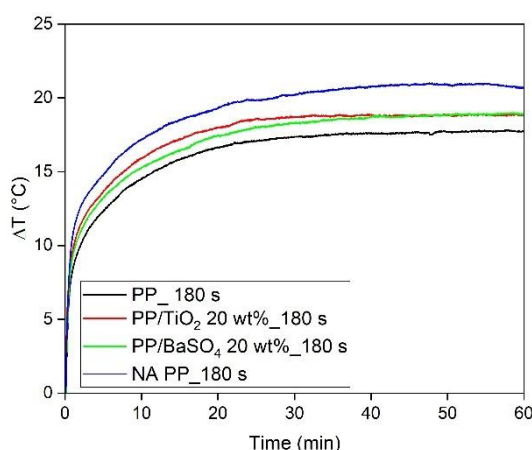


Figure 4.24 The temperature difference of porous PP, porous PP composites (with 20 wt% of PP/TiO₂ and PP/BaSO₄), and porous nucleated PP samples compared to the extruded PP sheet. The applied solvent treatment time was 180 s

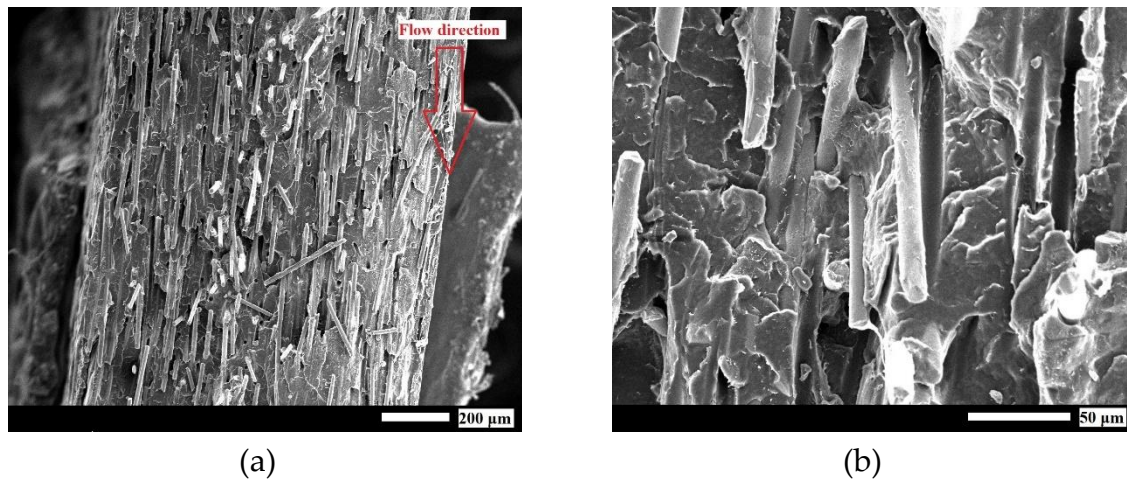
Based on the trend exhibited by these curves, ΔT curves can be separated into three distinct sections. In the initial 10 minutes, there is a notable and relatively rapid increase in temperature. Subsequently, between 10 and 40 min, the rise in ΔT follows a progressively diminishing rate. Eventually, in the last 20 min, ΔT stabilizes, reaching an equilibrium phase. The minimum ΔT value belongs to porous PP, solvent treated for 180 s, which can reduce the temperature by around 17 °C compared to the reference sample. 20 wt% porous PP/TiO₂ and 20 wt% porous PP/BaSO₄, both solvent-treated for 180 s, can reduce the temperature by ~19 °C. Nucleated PP solvent treated for 180 s reduced inner surface temperature the most, by ~23 °C.

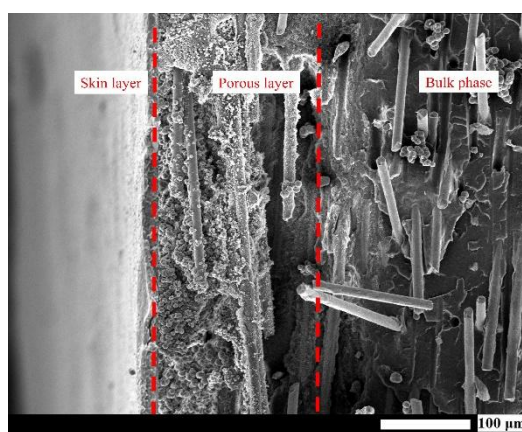
4.4. Polypropylene/glass fiber composites with a porous surface layer

4.4.1. Cross-sectional morphology of solvent-treated PP composites

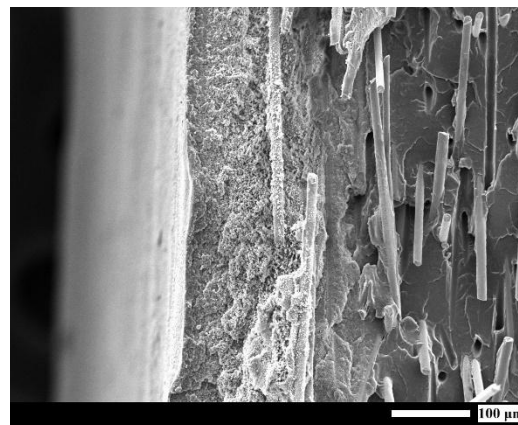
For the PP composites I chose chopped GF as a reinforcing material, because it has a high refractive index ($n = 1.55\text{--}1.58$) which results in a notable increase of the refractive index contrast between the matrix and air voids, also leading to strong scattering ability at solar wavelength.

I conducted a cross-sectional morphology study using SEM to determine the impact of solvent treatment on the morphology of composites. **Figure 4.25** shows the cross-sectional SEM micrographs of the extruded GFPP 30 both before and after the solvent treatment.

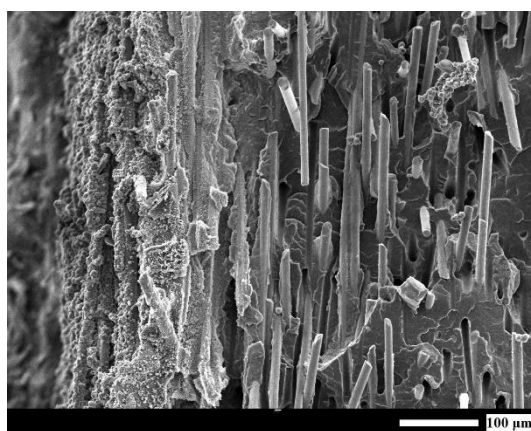




(c)



(d)



(e)

Figure 4.25 SEM images showing the cross-section of an extruded GFPP 30 sheet in (a) and (b). The (c) GFPP 20, (d) GFPP 30, and (e) GFPP 40 sheets after solvent treatment at 125 °C for 60 s

Figure 4.25a gives a representative image of the fiber orientation in the cross-section of the composite sheets produced. As expected, most fibers were oriented parallel to the flow direction, indicated with red arrows (**Figure 4.25a,c**). In **Figure 4.25b**, there were short pulled-out fibers on the fractured cross-sectional surfaces, leaving grooves at the fiber-matrix inter-face. However, it was also visible that most fibers exhibited good matrix adhesion, as shown by the retained PP matrix on the fiber surfaces. This good interfacial adhesion between the PP and GF can be attributed to MAPP—good interfacial adhesion is critical for the high tensile strength of the composites [180]. In the Appendix, **Figure 8.1** shows the weak interactions between the PP and GF without MAPP, resulting in longer pulled-out fibers and more pronounced grooves.

As was previously concluded, [181, 182] and as can be seen in Figure 4.24c–e, the solvent penetration depth into the bulk phase was a crucial parameter influencing the thickness and structure of the porous layer, the reduction of the bulk phase, and consequently, the optical and mechanical properties of the samples [182, 183].

Therefore, I investigated the penetration depth of the solvent into the bulk extruded samples, as the remaining non-porous bulk layer, unaffected by the solvent,

was expected to be responsible for the major load-bearing section. I calculated solvent penetration depth by measuring the difference between the initial thickness of the sheet (1 mm) and the remaining bulk phase after solvent treatment. The thicknesses of both the bulk phase and the porous layer were determined through the analysis of cross-sectional SEM images.

Figure 4.26 presents the correlation between solvent penetration depth and porous layer thickness in samples treated for varying durations, with the results corresponding to one side of the samples.

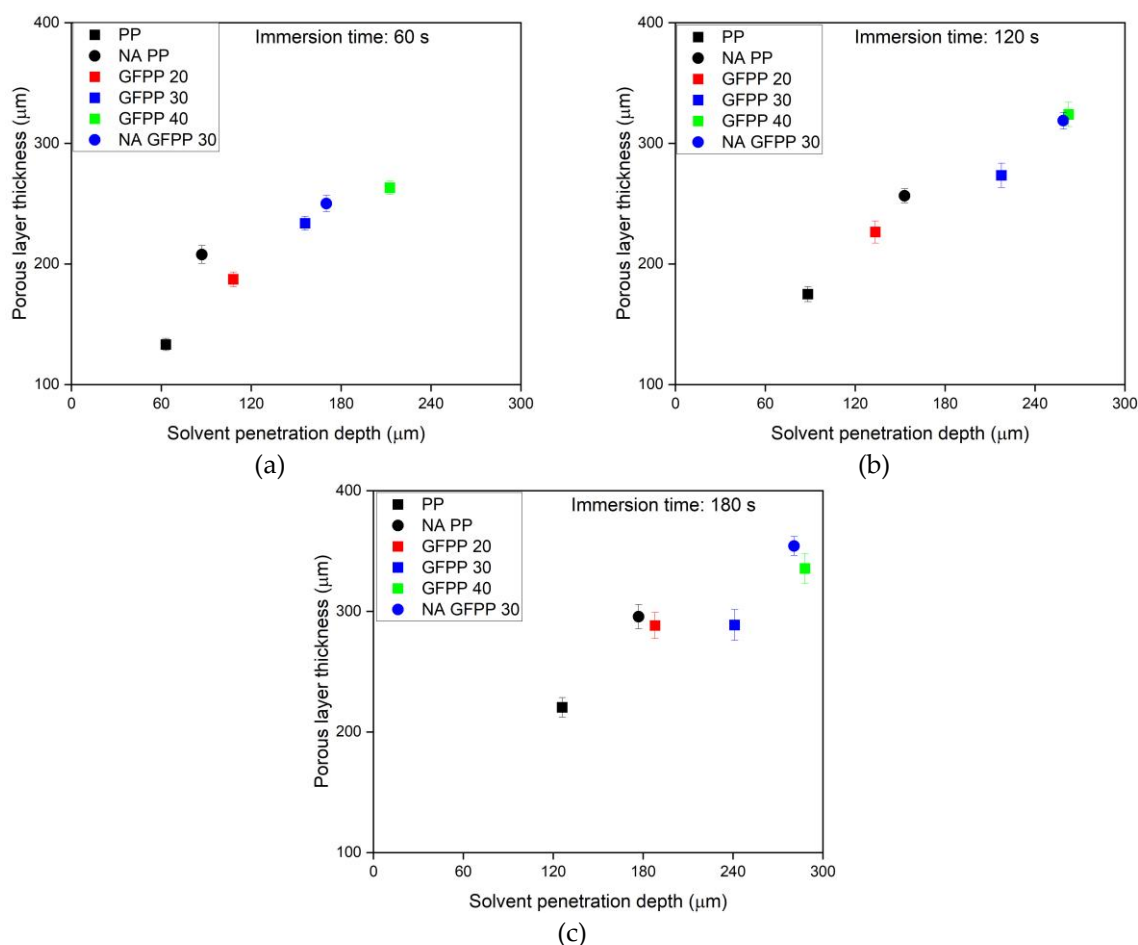


Figure 4.26 The thickness of the porous layers of the solvent-treated neat PP, GFPP, and NA GFPP 30 samples as a function of the solvent penetration depth. The samples were solvent-treated at 125 °C for (a) 60 s, (b) 120 s, and (c) 180 s.

Solvent penetration depth increased with immersion time, similar to the previously studied PP samples [183]. The composition of the PP sheets significantly influenced the diffusion rate of the solvent into the extruded sheets. With a constant immersion time, the solvent diffusion rate into the bulk phase increased with a higher GF (and MAPP) content in the samples compared to the neat PP.

The individual effects of GF and MAPP content on solvent diffusion were examined on solvent-treated samples containing only GF (PP/GF 20, PP/GF 30, PP/GF 40) or only MAPP (PP/MAPP 2, PP/MAPP 3, PP/MAPP 4) (Appendix, **Figure 8.2**). In PP/GF composites without MAPP, the solvent likely diffused more

rapidly along the fiber surfaces into the bulk phase due to insufficient interfacial adhesion between the PP matrix and GF, compared to neat PP. In PP/MAPP samples, MAPP may act as a nucleating agent in the PP phase and facilitate faster solvent diffusion [184]. Analyzing solvent penetration depth in the PP/GF, PP/MAPP, and GFPP composite systems separately demonstrated that both GF and MAPP enhanced the rate of solvent diffusion. Appendix, **Table 8.5** summarizes the results related to the crystallinity of the samples.

Figure 4.26 also shows that the presence of the NA accelerated solvent diffusion. The distinct super-morphological structures in the cross-section of the NA PP samples may facilitate faster solvent diffusion than neat PP. The POM images (Appendix, **Figure 8.3**) indicated that the extruded PP sheets exhibited the expected core-shell morphology in the cross-section. The core layer is comprised of spherulites, while the outer shell layers consist of a highly oriented structure. In contrast, the POM images of the NA PP samples showed a micro-spherulitic structure throughout the cross-section without the core-shell morphology. The thinner oriented-shell layer formed in the nucleated sheets may have led to deeper solvent penetration at the same immersion times. The combination of the NA-induced micro-spherulitic structure with GF and MAPP in the NA GFPP 30 samples resulted in the fastest solvent penetration, which can be beneficial as it enabled increased productivity during the manufacturing of the sheets.

The thickness of the porous layer increased with a longer immersion time for a given composition, with a general correlation between the thicker porous layers and greater solvent penetration depth, irrespective of the composition of the sample. The size and size distribution of the spherulites within the porous layer were influenced by the composition of the PP composites, as illustrated in **Figure 8.4** in the Appendix. In solvent-treated neat PP, the size distribution of the spherulite was wide, with sizes ranging from approximately 23 μm to 0.5 μm . However, in the solvent-treated GFPP composite samples, the distribution narrowed and spherulite sizes ranged between 10 μm and 0.5 μm . With increasing GF content, the spherulite size shifted towards smaller sizes. One may speculate that this was because during extrusion, the orientation of the GF can promote shear-induced nucleation and the formation of transcrystalline layers from the fiber surfaces [185]. These transcrystalline fronts may have disintegrated to some extent during solvent treatment. Still, the remaining order parts could act as self-nuclei during crystallization, resulting in the formation of smaller spherulites with a narrower size distribution.

In the solvent-treated NA PP and NA GFPP 30 samples, the effectiveness of the nucleating agent was clearly demonstrated by the fact that the size of the spherulites reduced drastically to 0.5–2.0 μm and was not affected by the GF content or immersion time. In these samples, the added heterogeneous nucleating agent governed the crystalline structure, promoting the formation of more crystal nuclei and thus smaller spherulites.

4.4.2. Solar reflectance

The TSR results of the samples treated for different immersion times are summarized in **Table 4.18** while **Figure 8.5** presents their corresponding reflectance spectra.

Table 4.18 The TSR results of solvent-treated samples

Immersion time (s)	TSR (%)					
	PP	NA PP	GFPP 20	GFPP 30	GFPP 40	NA GFPP 30
0	11.6 ± 0.6	12.4 ± 0.9	18.8 ± 1.4	22.7 ± 1.8	32.7 ± 2.2	28.7 ± 0.8
60	80.8 ± 1.2	90.1 ± 0.3	83.5 ± 1.6	89.9 ± 0.3	88.5 ± 0.8	93.0 ± 0.6
120	83.6 ± 0.8	92.3 ± 0.8	86.1 ± 1.3	90.7 ± 1.2	90.1 ± 0.8	93.4 ± 0.8
180	86.1 ± 1.4	93.2 ± 0.8	90.4 ± 0.6	92.4 ± 0.3	91.9 ± 1.5	92.3 ± 1.4

Table 4.18 highlights the beneficial impact of solvent treatment and the introduction of GF and NA on increasing sunlight reflectivity. In non-treated samples, the addition of GF alone enhanced TSR, achieving approximately 33 % for GFPP 40 compared to the ~12 % TSR of neat PP. Presumably, both the refractive index mismatch between PP and GF and the small diameter (10 μm compared to the generally used 20 μm) of the GF resulted in increased sunlight scattering. While the introduction of NA alone did not significantly improve TSR, its combination with GF resulted in increased reflectivity, as demonstrated by the comparison between the GFPP 30 and NA GFPP 30 samples. In solvent-treated neat PP samples, TSR values ranged between 80 % and 86 %. Increasing the GF content to 30 % raised TSR to approximately 90–92 %, regardless of the immersion time. However, further increasing the GF content to 40 % did not yield additional improvements. Beyond GF content, immersion time positively influenced the TSR of the GFPP samples. In contrast, solvent-treated NA PP and NA GFPP 30 samples exhibited TSR values exceeding 90 % and 93 %, respectively, even with the shortest immersion times. Notably, the TSR of NA GFPP 30 remained unchanged with longer immersion durations. These findings further reinforced that when NA is introduced, it plays a dominant role in governing the resulting structure. The enhanced TSR observed in solvent-treated samples could be attributed to the previously presented spherulite size reduction in the upper layer.

In typical porous polymer structures described in the literature, the scattering units are circular air voids of varying sizes, and the scattering can be described with the Mie theory. In contrast to these structures, which either feature a porous surface or exhibit porosity throughout the material, the collective solar light scattering behavior of our porous PP samples is more complex. First, our porous structure is buried beneath a skin layer, unlike the exposed porous surfaces or fully porous materials described in the literature. As such, scattering may be attributed to various factors, including the characteristics of the skin layer, such as its thickness and surface roughness, as well as the properties of the underlying porous layer.

In our previous study, I investigated the effect of the skin layer on optical properties and found that it only slightly affected reflectivity [186]. In our case, at the

polymer–air interface in the porous PP layer, the refractive index mismatch between PP spheres ($n_{PP} = 1.49$) and the surrounding air ($n_{air} = 1.0$) resulted in intense sunlight scattering. Increasing the porosity of the specific surface area by reducing the spherulite size led to higher scattering ability. The hierarchical spherulite structure also amplified scattering across the ultraviolet, visible, and near-infrared wavelengths, leading to a robust scattering effect [187, 188]. The introduction of the GF contributed to the reduction in spherulite size in the porous layer and created additional fiber/air and fiber/polymer scattering interfaces within the highly porous layer, which enhanced the total sunlight reflection and scattering.

In addition to the scattering from the interfaces, the spherulites themselves could also have acted as a birefringent scattering center [173]. However, in our case, spherulites as scattering units could only have had a minor contribution as the light-scattering effect of the spherulites decreases proportionally with a decreases in their size [173], which is the opposite of what I observed. **Figure 4.27** illustrates the relationship between the TSR values and the thickness of the porous layers.

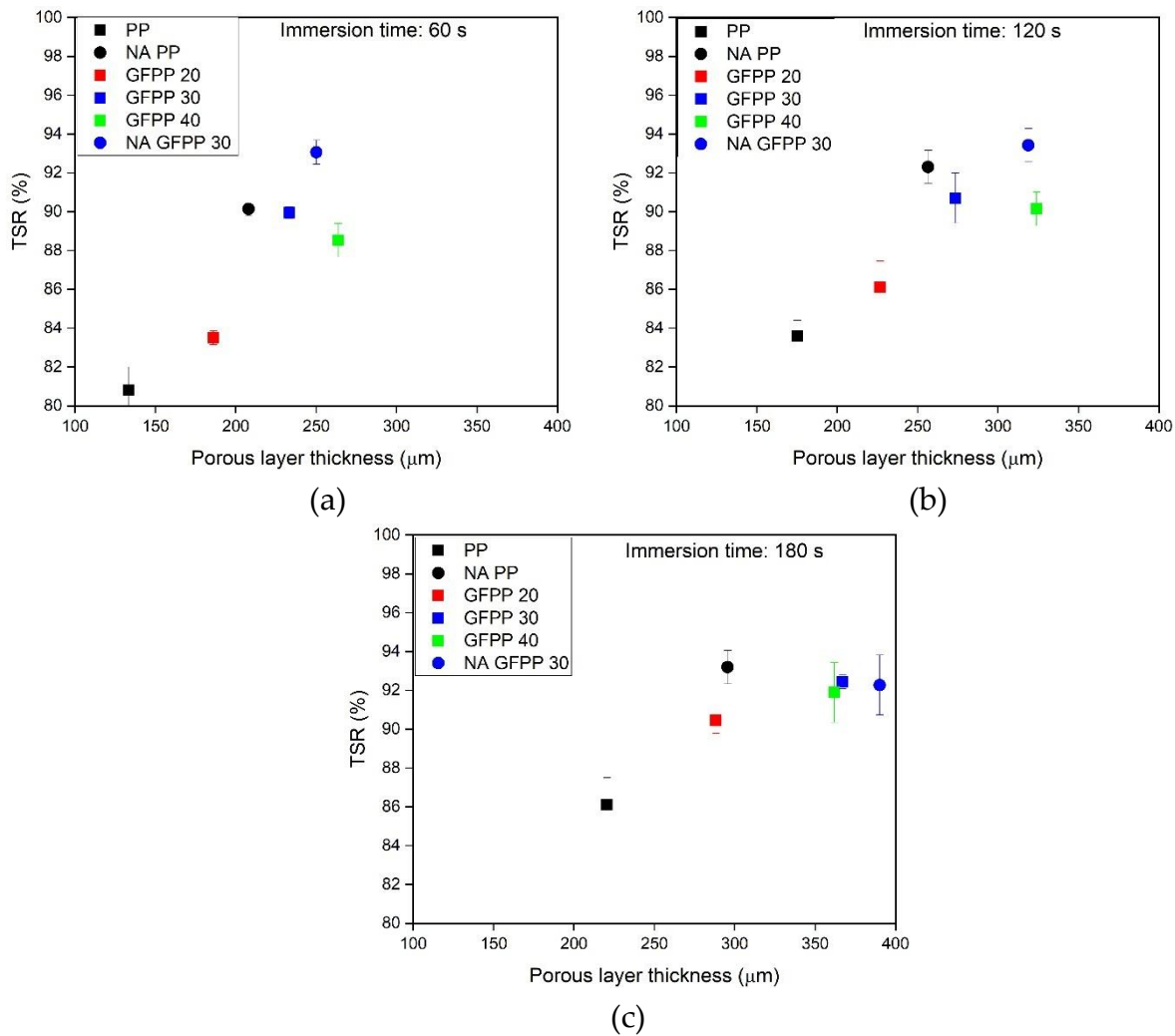


Figure 4.27 TSR of solvent-treated samples as a function of porous layer thickness. The samples were solvent treated at 125 °C for (a) 60 s, (b) 120 s and (c) 180 s

The TSR results of the solvent-treated samples increased significantly with increasing porous layer thickness, up to 250–300 μm . Beyond this point, further increases in thick-ness did not enhance TSR, which tended to plateau around 93 %. This suggested that light cannot be efficiently reflected back from deeper than 250–300 μm , indicating that creating a thicker porous layer would not further improve optical properties.

The highest TSR values (93 %) were observed in the NA GFPP 30 samples with an immersion time of 60 s (Figure 4.26a). With longer immersion times, no significant changes were observed (Figure 4.26b,c). The surface-treated NA GFPP 30 samples also showed that a porous layer thicker than 250 μm did not increase reflectivity.

4.4.3. Reflective thermal insulation ability

Considering the TSR results, the reflective thermal insulation properties of the solvent-treated samples with the shortest immersion time (60 s) as the most promising candidates will be discussed in the following section. First, the indoor test of the samples was evaluated using a laboratory-based sun test. I recorded the changes in inner temperature for the two boxes covered with an extruded PP sheet and a solvent-treated GFPP composite sample. The registered inner temperatures of the boxes are shown in **Figure 4.28a**, and the final inner temperature after 1 h is shown in **Figure 4.28b**.

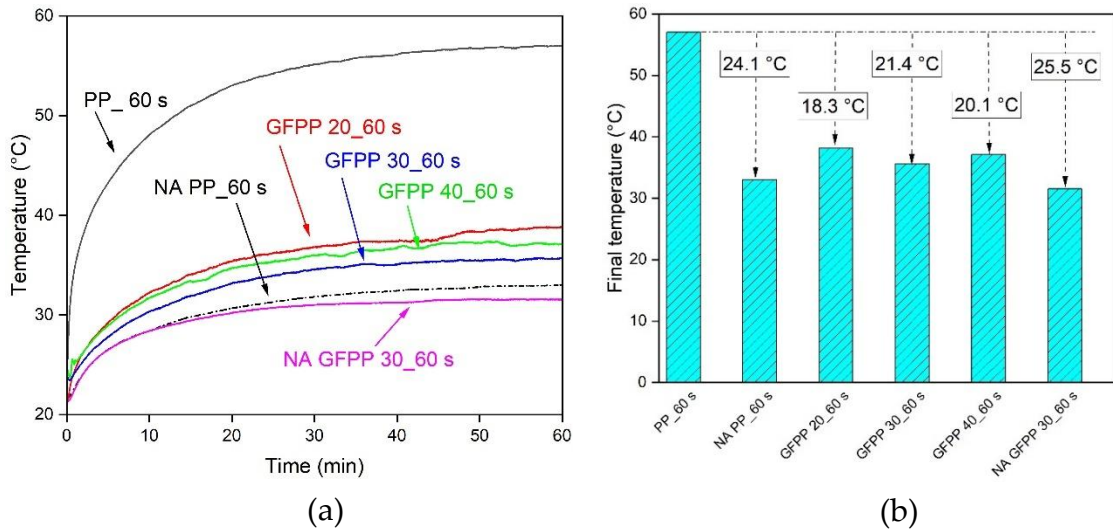


Figure 4.28 (a) Temperature tracking during the laboratory-based sun test and (b) final temperatures inside the boxes

The final temperature in the device covered with an extruded neat PP sheet was the highest, around 57 °C. The final temperatures of the solvent-treated GFPP composite samples, treated for 60 s, showed excellent alignment with the previously shown TSR results. The registered inner temperature was lower by 24.1 °C for NA PP_60 s, 18.3 °C for GFPP 20_60 s, 21.4 °C for GFPP 30_60 s, 20.1 °C for GFPP 40_60 s, and 25.5 °C for NA GFPP 30_60s.

To measure the reflective thermal insulation effect of the solvent-treated NA GFPP 30_60 s composite under more realistic conditions, I put the testing setup on the roof of a building in Budapest, where it was directly exposed to sunlight. The temperature data were recorded from 10:30 to 14:00 on 18 July 2024. The results are shown in **Figure 4.29**.

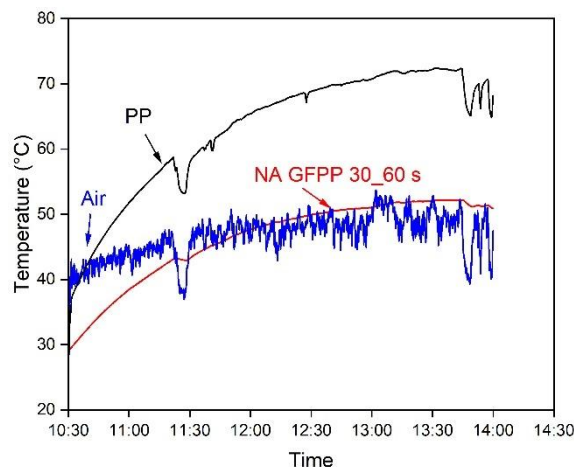


Figure 4.29 Recorded temperature of NA GFPP 30_60 s, ambient air and neat PP

The average inside temperature of the chambers covered with NA GFPP 30_60 s was 46 °C, compared with the temperature of ambient air (~48 °C) on average between 11:00 and 14:00, after exposure to direct sunlight. In contrast, the temperature of the box covered with neat PP extruded sheet increased significantly, reaching above 67 °C under peak solar irradiation. The selected sample prevented the temperature rise far more than the reference extruded PP sheet under sunlight. The sample received significant solar irradiance, which was a clear sign of reflective thermal insulation ability.

4.4.4. Mechanical properties

As the solvent treatment induced morphological changes and reduced the thickness of the bulk cross-section, it was crucial to evaluate its effect on mechanical properties.

The actual GF content and the average fiber length of the GFPP sheet samples are presented in **Table 4.19**. As expected, the average fiber length decreased with increasing GF content.

Table 4.19 Fiber characteristics before and after processing

Designation	Nominal fiber diameter (μm)	Nominal fiber length (μm)		Actual GF content (wt%)
		Before extrusion	After extrusion	
GFPP 20	10	3000	1008 ± 278	21.1 ± 0.79
GFPP 30			800 ± 234	31.1 ± 1.10
GFPP 40			746 ± 199	40.7 ± 0.51
NA GFPP 30			812 ± 178	30.5 ± 0.31

The impact of the solvent treatment on the mechanical properties of the GFPP composite samples was investigated with varying GF content and immersion times. Typical stress–strain curves of the samples treated for different immersion times are shown in **Figure 4.30**.

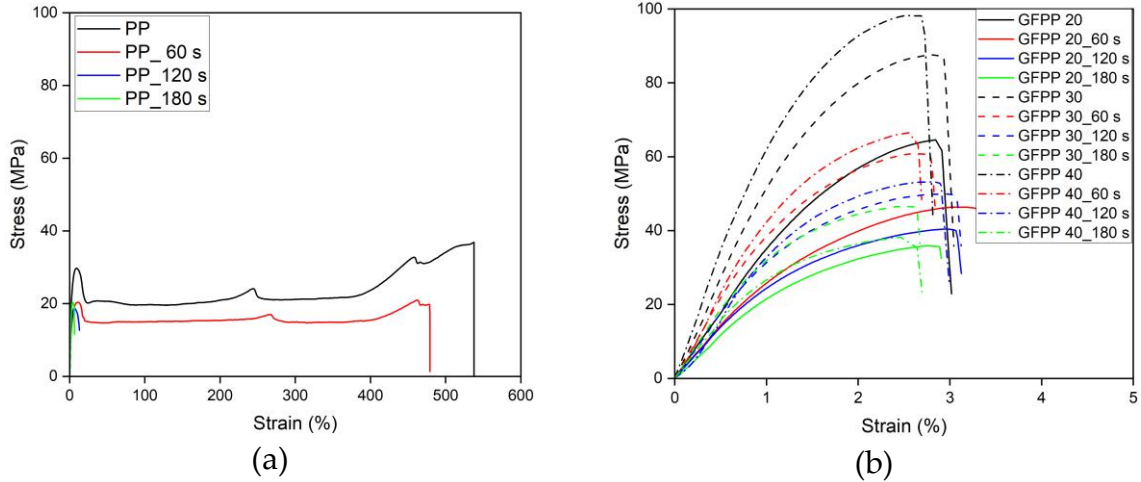


Figure 4.30. Typical stress–strain curves of solvent-treated (a) PP and (b) GFPP samples.

An increase in immersion time during solvent treatment primarily caused significant changes in the failure mode of PP samples, as shown in **Figure 4.30a**. A ductile-to-brittle transition was observed with increasing immersion time, accompanied by a dramatic decrease in elongation at break. The stress–strain curves for the PP and GFPP samples differed significantly due to the presence of GF. GFPP composite samples exhibited brittle fracture behavior (**Figure 4.30b**), which is commonly attributed in the literature to stress concentrations at the fiber ends. These stress concentrations induce matrix cracking and, ultimately, matrix failure [187].

The results of the tensile tests are shown in **Figure 4.31**. The non-treated samples of 0 s immersion time represent the reference samples.

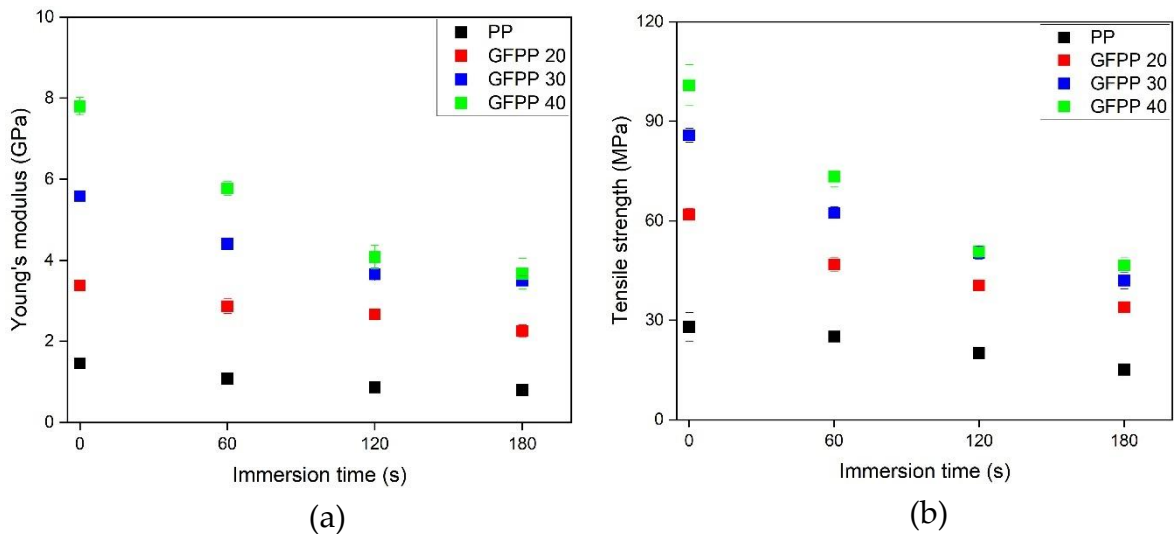


Figure 4.31 Effect of fiber content and immersion time on (a) the stiffness, (b) the tensile strength of solvent-treated samples

As shown in **Figure 4.31a,b**, adding GF met the expectations and significantly increases stiffness (Young's modulus) and tensile strength. Specifically, Young's modulus increased from 1.5 GPa to approximately 8.0 GPa, while tensile strength improved from ~30 MPa to 100 MPa within the tested composition range. These findings aligned well with the literature data. Pukánszky et al. [188] investigated injection-molded GF-reinforced polypropylene (GFPP) with varying GF content from 0 to 50 wt%. Their study reported that increasing GF content to 30 and 40 wt% resulted in Young's modulus values of approximately 6 GPa and 8 GPa, respectively. Furthermore, they emphasized the effect of MAPP as a coupling agent on tensile strength. GFPP samples containing 30 wt% GF exhibited ~40 MPa tensile strength without MAPP, which increased to ~70 MPa when MAPP was used.

As expected, the solvent treatment with longer immersion times resulted in decreased Young's modulus and tensile strength. This reduction was attributed to the reduced thickness of the bulk phase, as the solvent diffused deeper in the treated samples (**Figure 4.26**), thereby reducing the effective load-bearing cross-section and affecting the mechanical properties negatively. Despite this trend, the GFPP 30 and GFPP 40 samples, even after the longest immersion time of 180 s, still exhibited significantly higher Young's modulus and tensile strength values than the untreated neat PP.

The deformability of the composites is crucial for practical applications. However, it is well known that while reinforcing fibers increase stiffness, they reduce deformability and toughness. Appendix, **Table 8.6** summarizes the elongation at break of the solvent-treated composites as a function of immersion time. The highest deformation was observed in non-solvent-treated samples, although increasing GF content negatively impacted deformability. For the GFPP composites, both solvent-treated and non-treated samples exhibited low (~1 %) elongation at break because of matrix fracture.

The effect of applied heterogeneous NA on the mechanical properties of the solvent-treated NA PP and NA GFPP 30 samples is shown in **Figure 4.32**.

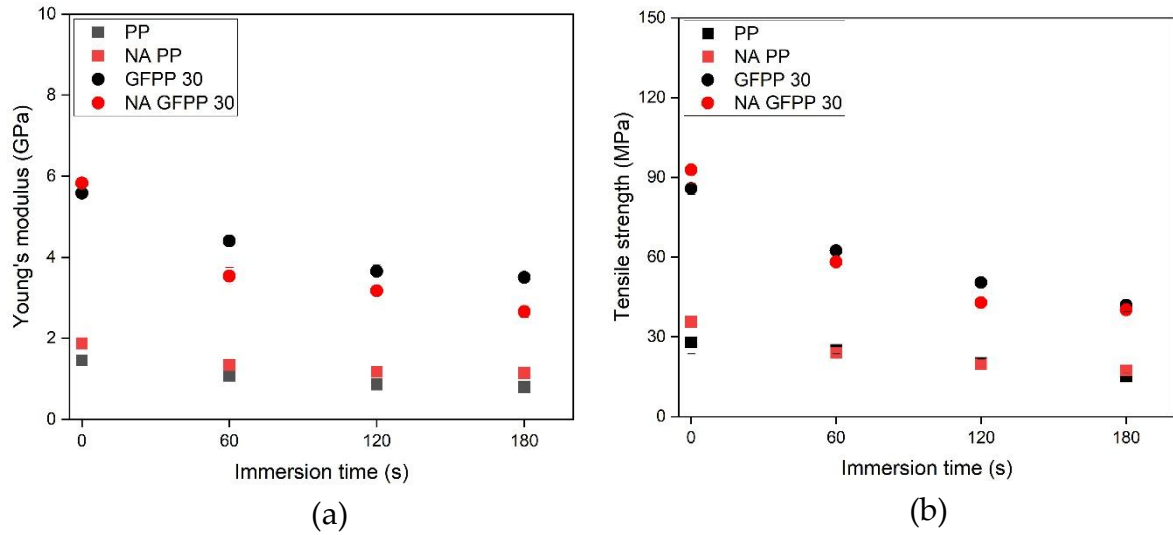


Figure 4.32 Effect of fiber content, NA and immersion time on (a) the stiffness, (b) the tensile strength and (c) the elongation at break of solvent-treated samples

In **Figure 4.32a**, the non-treated NA PP resulted in a modulus increase of about 20 % compared to the non-treated PP. A comparison of the non-treated GFPP 30 samples and the non-treated NA GFPP 30 samples showed that the nucleated PP composites exhibited a slightly higher Young's modulus (**Figure 4.32a**) and tensile strength (**Figure 4.32b**), as expected. Unexpectedly, the solvent treatment decreased Young's modulus and tensile strength to a greater extent in the case of the NA GFPP 30 samples than for the GFPP 30 samples. This reduction was likely due to the combined effects of NA, GF, and MAPP, which accelerate solvent diffusion into the bulk phase, leading to a thinner load-bearing bulk layer. The elongation at break results (Appendix, **Table 8.6**) indicated that the presence of NA reduced sample deformability. Interestingly, while NA decreased elongation in samples without GF, it appeared to enhance elongation at break in GF-reinforced composites.

The composition and immersion time dependence of E_{perf} is presented in **Figure 4.33**.

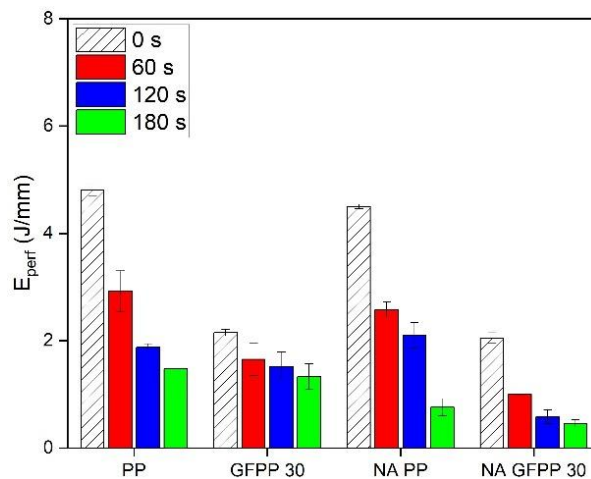


Figure 4.33 Perforation energy of solvent treated composites sample

For non-treated samples, E_{perf} is showed a slight decrease in samples containing NA, while the addition of 30 wt% GF resulted in a significant reduction in perforation energy. In general, E_{perf} is decreased gradually with increasing immersion time. The E_{perf} is of NA GFPP 30 deteriorated the most with immersion time.

4.4.5. Surface wettability

Following the solvent treatment, the surface morphology of the composite samples was analyzed using SEM images. **Figure 4.34** presents the SEM images of the surfaces of the treated samples.

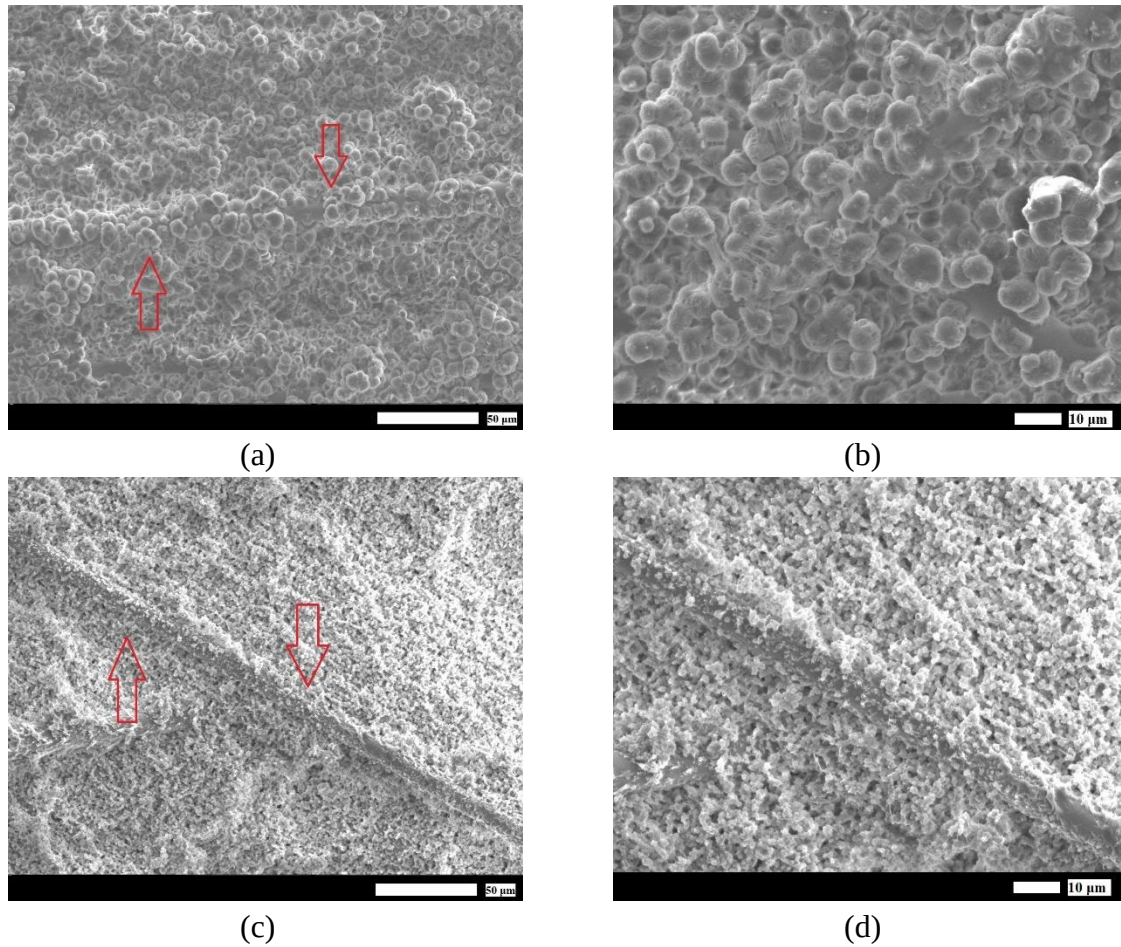


Figure 4.34 SEM images of the surfaces of solvent-treated, peeled samples: (a) and (b) GFPP 30, and (c) and (d) NA GFPP 30. Images (a) and (c) were captured at 500x magnification, while images (b) and (d) were captured at 1000x magnification

The spherulite sizes on the surfaces of GFPP 30 and NA GFPP 30 samples exhibit a reduction compared to non-nucleated PP samples. However, the decrease in spherulite size is less pronounced in GFPP 30 samples compared to NA GFPP 30 samples. The reinforcing fibers are visible on the surfaces of both samples (marked by red arrows). Despite this, the majority of the surface area in both cases is covered by spherulites, as shown in **Figures 4.34b** and **Figures 4.34d**.

Figure 4.35 presents the spherulite size distribution on the composite surfaces and Table 4.20 shows the average spherulite sizes and surface wettability of the treated samples.

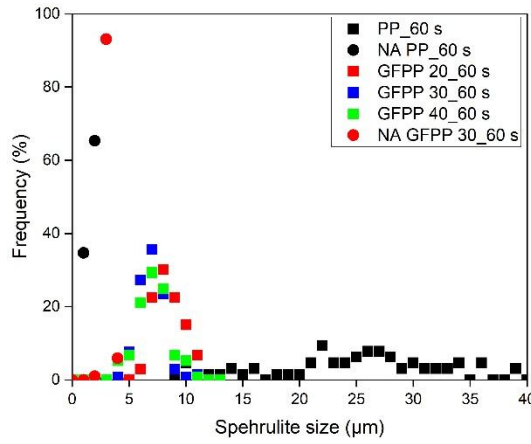


Figure 4.35 Dependence of spherulite size distribution on sample composition. Samples were solvent- treated at 125 °C for 60 s

Table 4.20 Average spherulite sizes and CA_{adv} values measured on the surfaces of treated samples

Designation	Spherulite size (μm)	CA_{adv} ($^\circ$)
PP_60 s	23.20 ± 5.50	154.9 ± 1.2
NA PP_60 s	1.55 ± 0.21	163.1 ± 1.2
GFPP 20_60 s	7.96 ± 1.18	155.8 ± 2.2
GFPP 30_60 s	6.46 ± 1.17	156.4 ± 2.1
GFPP 40_60 s	6.59 ± 1.38	155.9 ± 3.6
NA GFPP 30_60 s	1.52 ± 0.25	159.8 ± 1.9

In the case of the treated PP_60 s sample, the spherulite size distribution is relatively broad, ranging between approximately 10 and 40 μm . In contrast, for solvent-treated composite PP samples and nucleated PP samples (NA PP_60 s and NA GFPP 30_60 s), the distribution becomes narrower and shifts towards smaller size ranges. The average spherulite size in nucleated samples decreases by an order of magnitude compared to non-nucleated PP. Consistent with previously observed trends, as spherulite size decreases, a more finely structured surface layer forms, resulting in an increase in CA_{adv} . However, the reduction in spherulite size from $\sim 23 \mu\text{m}$ to $\sim 6 \mu\text{m}$ in the GFPP samples did not lead to a significant change in surface wettability; the CA_{adv} showed an improvement of only about $\sim 2^\circ$ compared to the treated PP_60 s sample. In contrast, the porous structures formed by $\sim 1.5 \mu\text{m}$ spherulites in the NA GFPP 30_60 s and the NA PP_60 s samples resulted in a significant increase in CA_{adv} , reaching 159° and 163° , respectively. The CAH values measured on the surfaces of the treated samples did not exceed 5° . Based on wettability measurements, the GFPP composite samples were able to maintain their superhydrophobic and water-repellent properties.

5. SUMMARY

Methods based on solvent-induced recrystallization for creating a roughened surface have been known for decades. These methods create a polymer solution that can be used as a coating layer on a selected product. I investigated new solvent-induced recrystallization-based method for creating roughened surfaces. Unlike methods based on the above method, I did not apply a polymer solution on the surface of the selected product to produce a roughened surface coating. Instead, I created the modified structure by gelling the upper layer of the sample. The first stage of my research started with experiments exploring the possible processing conditions that are suitable for creating the modified structure and describing the structure formed. I found that the resulting surface structure differs from those previously presented in the literature. The modified layer consists of a hierarchically structured, anisotropic porous structure composed of spherulites of varying sizes and a skin layer on the surface of the porous region. Depending on the application requirements, the skin layer can be removed from the surface.

Simultaneously, I investigated the effect of treatment parameters and the addition of a heterogeneous nucleating agent on the formation of the structure. I found that in non-nucleated PP samples, the temperature of solvent treatment had a significant influence on surface texture, while solvent treatment time affects only the thickness of the skin and the porous layer. In addition, the temperature of drying also influenced the structure. A higher drying temperature makes the skin layer more difficult to remove—above 70 °C I could not remove it. By incorporating a nucleating agent, the surface morphology can be precisely tailored through the controlled adjustment of spherulite sizes. I also investigated the correlation between the cross-sectional morphology of the extruded sheets and the created morphology after the solvent treatment. I found that after the solvent treatment, in the case of a core-shell-like structure of the extruded sheet, I received a thicker skin layer and thinner porous layer compared to the extruded sheets with a full spherulite structure in the cross-section, where the formation of a thinner skin layer and a thicker porous layer was more likely.

Removing the top layer, we can see that the porous surface layer has superhydrophobic characteristics. After optimizing treatment parameters, I achieved a CA_{adv} higher than 155° and CAH lower than 5°. With the addition of a nucleating agent, the method makes it possible to control surface morphology and to prepare superhydrophobic surfaces with an outstanding CA_{adv} (>160°) and designable spherulite sizes.

The manipulation of surface wettability on roughened PP surfaces, ranging from superhydrophobic to superhydrophilic, is known. However, the achievable wettability range can vary depending on the selected methods. The investigated solvent treatment enabled us to achieve a CA_{adv} greater than 150°. Hierarchically structured PP surfaces with superhydrophobic characteristics were produced to transition into a superhydrophilic wetting regime via air plasma treatment. With the

optimal treatment parameters, I developed a superhydrophilic surface that exhibited no measurable hydrophobic recovery effect even after one week of aging in air. I found that, using XPS and surface wettability measurements, there is a strong correlation between surface composition on the plasma-treated roughened PP surfaces and wetting kinetics, specifically in the context of droplet spreading speed over different aging periods. The roughness of the surface, influenced by capillary effects and the reorientation of oxygen-containing functional groups over time, played a key role in controlling the reduced speed of droplet spreading. I introduced a simple model for the characterization of droplet spreading behavior on superhydrophilic, roughened PP surfaces, which I used to describe the spreading time of droplets as a function of aging time. The proposed model, which shows good agreement with experimental data, offers a novel method for comparing the droplet spreading behavior on plasma-treated roughened surfaces.

Reflective thermal insulation materials have gained widespread use in everyday life and various industries by utilizing sunlight reflection to reduce the inside temperature without requiring additional energy input. As it is known, porous polymer materials possess the unique ability to provide both thermal insulation and solar reflection due to their inherent multiphase structure. The created porous surface structure resulted in another beneficial property for PP products, namely, solar light reflection. Based on the spectrophotometry investigations, I found that solar reflectance significantly increased from that of the extruded PP (11 %) to 86 %. I found that the morphological properties of the porous layer, including its thickness, spherulite size, and size distribution, primarily determine the solar reflectivity. To achieve better solar reflection and thermal insulation ability, I examined two strategies: (i) the combination of a porous structure and reflective additives (BaSO_4 , TiO_2) and (ii) the modification of the porous structure with the addition of a PP nucleating agent. I found that the solvent-treated nucleated PP samples outperformed the solvent-treated PP/ TiO_2 and PP/ BaSO_4 samples in solar light reflection. With the shortest immersion time (to make the process more cost-effective), the PP/ TiO_2 and PP/ BaSO_4 samples with 10 wt% of additives exhibit a TSR of 88 % and 87 %, while the TSR of nucleated PP samples reached a TSR of 90 %. Reflective insulation performance tests showed that porous PP, porous PP composite (with 10 wt% of PP/ TiO_2 and PP/ BaSO_4), and porous nucleated PP samples reduced the temperature by 17, 19, and 23 °C, respectively, compared to extruded PP sheets. I found that based on the reflective insulation tests, the porous nucleated PP samples showed the best insulation performance.

From a practical application perspective, the effective use of the product with the created surface layer as Reflective thermal insulation materials may be limited by mechanical stresses induced by environmental factors. The stiffness and strength of porous PP were not sufficient for such applications. I aimed to address this gap by investigating GF-reinforced PP composite sheets. With the combined effect of 30 wt% GF and the highly porous surface layer featuring spherulite sizes of 0.5–2.0 μm , I achieved an average solar reflectance of up to 93 %. Outdoor reflective insulation

performance measurements demonstrate a temperature reduction of up to 21 °C compared to standard extruded PP sheets, particularly during midday hours. The solvent-treated GFPP composite samples exhibit enhanced mechanical strength with the optimal incorporation of GF. This highlights the possibility of using mechanically reinforced and porous PP composites as reflective thermal insulation materials.

6. THESIS STATEMENTS AND PUBLICATIONS

6.1. Thesis statements

Thesis 1

I developed a xylene solvent treatment that enables the formation of a two-layer structure near the surface of 1 mm thick extruded isotactic polypropylene homopolymer sheets. As a result of recrystallization during the treatment process, an anisotropic internal porous layer is created, which consists of spherulites of varying sizes. Additionally, a continuous, dense outer skin layer is formed, covering the surface. This outer skin layer can be easily peeled off following the application of appropriate treatment parameters (xylene solvent temperature of 125 °C, immersion times ranging from 30 to 300 seconds, and drying temperature of 30 °C; xylene solvent temperature between 120 and 130 °C, immersion time of 60 s, and drying temperature of 30 °C; xylene solvent temperature of 125 °C, immersion time of 60 s, and drying temperatures of 30 °C and 50 °C). The porous structure beneath the removed skin layer exhibits a hierarchical micro- and nanoscale structure. My findings regarding the formation of this novel structure have been validated through scanning electron microscopy and calorimetric analyses.

1. Tézis

Kidolgoztam egy olyan xilol oldószeres kezelést, amellyel 1 mm vastag, extrudált izotaktikus polipropilén homopolimer lemezek felszínének közelében kialakítható egy két eltérő rétegből álló szerkezet. A kezelés során átkristályosodás eredményeként létrejön egy anizotróp szerkezetű belső pórusos réteg, amely különböző méretű szferolitokból épül fel, valamint egy felszínt alkotó külső, összefüggő, tömör bőrréteg, amely a megfelelő kezelési paraméterek (125 °C-os xilol oldószer hőmérséklet, 30-300 másodperces bemerítési idő és 30 °C-os szárítási hőmérséklet továbbá 120 és 130 °C közötti xilol oldószer hőmérséklet 60 másodperces bemerítési idő és 30 °C-os szárítási hőmérséklet, valamint 125 °C-os xilol oldószer hőmérséklet, 60 másodperces bemerítési idő és 30 és 50 °C-os szárítási hőmérséklet) alkalmazása esetében a kezelést követően könnyedén eltávolítható. Az eltávolított bőrréteg alatti pórusos szerkezet mikro- és nanoléptékű hierarchikus struktúráltságot mutat. A kialakított új szerkezetre vonatkozó állításaimat, pásztázó elektronmikroszkópos és kalorimetriás mérésekkel igazoltam.

Related publications

Vámos Cs., Rácz I, Marosfői B., Kállay-Menyhárd A., Bárány T.: Novel, solvent-based method for the production of polymer sheets with superhydrophobic surface. *Polymer Engineering and Science*. 63(4), 1289-1302 (2023) IF: 3.2

Vámos Cs., Honti Sz., Marosfői B., Rácz I., Bárány T.: Extrudált lemez morfológiájának hatása az oldószeres kezeléssel kialakított szuperhidrofób felületi struktúrára. Polimerek. 8, 227-232 (2022)

Rácz I., Marosfői B., **Vámos Cs.**, Nakai Y., Mimura Y.: Method for producing molded body with ultra-water-repellent surface and molded body with ultra-water-repellent surface WO/2021/201156; EP4130115A1 published, pending; CN115335441B (2024.07.23.) granted.

Thesis 2

I have demonstrated that by modifying the technological parameters of the solvent treatment process and adjusting the initial cross-sectional morphology of the extruded sheet, the morphology of the porous layer formed during surface treatment, as well as the thickness of the skin layer, can be controlled within a range of 10 to 100 μm . During xylene surface treatment, an increase in immersion time from 30 to 180 s at a solvent temperature of 125 °C, or an increase in solvent temperature from 120 °C to 130 °C (with a 60 s immersion time), leads to a simultaneous increase in the thickness of both the porous and skin layers. Furthermore, increasing the immersion time gradually enhances the porosity of the samples. Conversely, raising the drying temperature from 30 °C to 120 °C results in a reduction in the thickness of both the porous layer and the skin layer.

2. Tézis

Kimutattam, hogy az oldószeres kezelés technológiai paramétereinek változtatásával, valamint az extrudált lemez kezdeti keresztmetszeti morfológiájának módosításával a felületkezelés során kialakuló pórusos réteg morfológiája és a bőrréteg vastagsága, 10 és 100 μm között szabályozható.

Xilolos felületkezelés során a bemelegítési idő 125 °C-os oldószerhőmérsékleten 30 s-ról 180 s-ig történő növelésével, vagy az oldószer hőmérsékletének 120 °C-ról 130 °C-ra történő emelésével (60 s bemelegítési idő mellett) a pórusos réteg és a bőrréteg vastagsága egyaránt növekszik. A bemelegítési idő növelésével a minták porozitása is fokozatosan növekszik. Ugyanakkor a szárítási hőmérséklet emelése 30 °C-ról 120 °C-ra mind a pórusos réteg, mind a bőrréteg vastagságának csökkenését eredményezi.

Related publications

Vámos Cs., Rácz I., Marosfői B., Kállay-Menyhárd A., Bárány T.: Novel, solvent-based method for the production of polymer sheets with superhydrophobic surface. Polymer Engineering and Science. 63(4), 1289-1302 (2023) IF: 3.2

Vámos Cs., Varga L. J., Marosfői B., Bárány T.: Role of extruded sheet morphology in phase separation and final morphology of superhydrophobic polypropylene. Periodica Polytechnica Mechanical Engineering. 66, 260-271 (2022)

Vámos Cs., Honti Sz., Marosfői B., Rácz I., Bárány T.: Extrudált lemez morfológiájának hatása az oldószeres kezeléssel kialakított szuperhidrofób felületi struktúrára. Polimerek. 8, 227-232 (2022)

Thesis 3

Through scanning electron microscopy and water contact angle measurements, I have demonstrated that solvent surface treatment of nucleated extruded polypropylene sheets at 125 °C for 60 s enables the achievement of an advancing contact angle exceeding 160° and a contact angle hysteresis below 5°. This phenomenon is attributed to the hierarchical (micro- and nanoscale) structure of the porous surface layer formed by spherulites after solvent treatment and drying, as well as the reduced spherulite size (1–2 µm) resulting from the application of a heterogeneous nucleating agent. My findings have been proven by the analysis of polypropylene samples treated with xylene solvent, containing 500 and 1000 ppm of the NA-21 E heterogeneous nucleating agent.

3. Tézis

Pásztázó elektronmikroszkópos és vízperemszög-vizsgálatokkal igazoltam, hogy a heterogén gócképzett polipropilén extrudált lemezek 125 °C-on, 60 másodperces oldószeres felületkezelése lehetővé teszi a 160°-nál nagyobb haladó peremszög és az 5°-nál kisebb peremszög-hiszterézis értékek elérését. Ennek oka az oldószeres kezelést és szárítást követően kialakuló, szferolitokból álló pórusos felületi réteg hierarchikus (mikro- és nanoléptékű) strukturáltsága, valamint a heterogén gócképző alkalmazásával elért kisebb szferolitméret (1-2 µm). Állításomat, az NA-21 E heterogén gócképzőt 500 és 1000 ppm mennyiségben tartalmazó, xilol oldószerrel kezelt polipropilén minták vizsgálatával igazoltam.

Related publications

Vámos Cs., Rácz I., Marosfői B., Kállay-Menyhárd A., Bárány T.: Novel, solvent-based method for the production of polymer sheets with superhydrophobic surface. Polymer Engineering and Science. 63(4), 1289-1302 (2023) IF: 3.2

Vámos Cs., Varga L. J., Marosfői B., Bárány T.: Role of extruded sheet morphology in phase separation and final morphology of superhydrophobic polypropylene. Periodica Polytechnica Mechanical Engineering. 66, 260-271 (2022)

Thesis 4

Through X-ray photoelectron spectroscopy and water contact angle measurements, I have demonstrated that the hierarchically structured, porous surface layer of solvent-treated extruded polypropylene sheets can be transformed into a superhydrophilic surface (advancing contact angle: 0°) through plasma treatment. Due

to its hierarchical structure, the porous surface layer can retain the wettability state induced by plasma treatment for an extended period (at least one week) at room temperature, in contrast to smooth-surfaced extruded polypropylene samples. My findings have been validated by characterizing extruded polypropylene sheets, which were solvent-treated at 125 °C for 60 s and subsequently plasma-treated at 0.15 mbar pressure with power of 24, 40, and 64 W for 300 and 600 s.

4. Tézis

Röntgen-fotoelektron spektroszkópiás és vízperemszög-mérésekkel kimutattam, hogy az oldószeres felületkezelésen átesett extrudált polipropilén lemezek hierarchikusan strukturált, pórusos felületi rétege plazmakezeléssel szuperhidrofil felületté (haladó peremszög: 0°) alakítható. A pórusos felületi réteg, strukturáltsága következtében, a plazmakezeléssel kialakított nedvesítési állapotát, hosszabb ideig (legalább egy hét) képes szobahőmérsékleten fenntartani, összehasonlítva a sima felületű extrudált polipropilén mintákkal. Állításomat, 125 °C-on, 60 másodpercig oldószeresen felületkezelt, majd 0,15 mbar nyomásom, 24, 40, 64 W teljesítménnyel, 300 és 600 másodpercig plazmakezelt extrudált polipropilén lemezek jellemzésével igazoltam.

Related publications

Vámos Cs., Rácz I, Marosfői B., Kállay-Menyhárd A., Bárány T.: Novel, solvent-based method for the production of polymer sheets with superhydrophobic surface. *Polymer Engineering and Science*. 63(4), 1289-1302 (2023) IF: 3.2

Vámos Cs., Füredi M., Hórvölgyi Z., Krafcsik O., Kiss G., Bárány T, Marosfői B. B.; Analysis of time-dependent hydrophobic recovery on plasma-treated superhydrophobic polypropylene using XPS and wettability measurements. *Scientific Reports*. 14(1), 23715 (2024) IF: 3.8

Thesis 5

I have demonstrated that the reflectance of polypropylene sheets with a porous surface layer formed through solvent surface treatment is significantly enhanced compared to extruded polypropylene sheets. Based on spectrophotometric measurements, I have shown that solvent-treated samples at 125 °C for 60, 120, and 180 s increase the total solar reflection within the 200–2500 nm wavelength range to 81%, 83%, and 86%, respectively, compared to untreated extruded polypropylene sheets, which exhibit only 11%. I demonstrated that for samples treated with solvent for 60 s at 125 °C, the reflectance remained unchanged after the removal of the skin layer, while for samples treated for 120 and 180 s, the reflectance increased slightly after the skin layer was removed by 1.05% and 1.84%, respectively.

5. Tézis

Igazoltam, hogy az oldószeres felületkezeléssel kialakított porózus felületi réteggel rendelkező polipropilén lemezek fényvisszaverő képessége jelentősen megnő az extrudált polipropilén lemezekhez képest. Spektrofotometriás mérések alapján kimutattam, hogy a 125 °C-on 60, 120 és 180 másodpercig oldószeresen felületkezelt minták a 200–2500 nm hullámhossz tartományban a visszavert sugárzás energiáját a kezeletlen extrudált polipropilén lemezekhez képest (11 %) jelentősen, 81, 83 és 86 %-ra növelik. Igazoltam továbbá, hogy a 125 °C-on 60 s ideig oldószerrel kezelt minták reflexiója a bőrréteg eltávolítását követően nem változott, míg a 120 és 180 s ideig oldószerrel kezelt minták reflexiója a bőrréteg eltávolítását követően kis mértékben nő: 1,05 %-kal, illetve 1,84 %-kal.

Related publications

Vámos Cs., Bárány T., Marosfői B.: Porous polypropylene produced by phase separation for high solar reflectivity and passive cooling. *Journal of Thermal Analysis and Calorimetry*. 149, 10851-10863 (2024) IF: 3.0

Vámos Cs., Bárány T., Glass fiber reinforced polypropylene composites with high solar reflectance for thermal insulation Applications. *Polymers*. 17(3), 274 (2025) IF: 4.7

Thesis 6

Through spectrophotometric analysis and outdoor measurements assessing reflective thermal insulation performance, I have demonstrated that 1 mm thick polypropylene composite samples containing glass fiber, maleic anhydride-grafted polypropylene, and a heterogeneous nucleating agent exhibit enhanced solar reflectance following xylene solvent treatment. This improvement is attributed to the synergistic effect of the heterogeneous nucleating agent and glass fiber content, which results in a more effective reflective thermal insulation capability compared to xylene-treated polypropylene samples without these additives. My findings have been verified through spectrophotometric measurements and outdoor thermal insulation performance assessments of xylene-treated nucleated polypropylene composite samples containing 30 wt% of glass fiber.

6. Tézis

Spektrofotometriás vizsgálatokkal, valamint kültéren végzett reflektív hőszigetelő képességet jellemző méréssel igazoltam, hogy a xilolos oldószeres kezelését követően az 1 mm vastagságú, üvegszálat, maleinsav-anhidriddel ojtott polipropilént, valamint heterogén gócképzőt tartalmazó polipropilén kompozit minták, a heterogén gócképző és az üvegszál tartalom szinergisztikus hatásának

következtében, a visszavert napsugárzás energiáját növelik, ezáltal hatékonyabb reflektív hőszigetelő tulajdonsággal bírnak, mint az azonos körülmények mellett xilol oldószeresen kezelt polipropilén minták. Az állításomat, xilol oldószerrel felületkezelt, 30 m% üvegszál tartalmú, göcképzett polipropilén kompozit minták spektrofotometriás és a kültéren végzett hőszigetelő képességet jellemző mérésekkel igazoltam.

Related publications

Vámos Cs., Bárány T., Glass fiber reinforced polypropylene composites with high solar reflectance for thermal insulation Applications. *Polymers*. 17(3), 274 (2025) IF: 4.7

Vámos Cs., Bárány T., Az üvegszál erősítés hatása az oldószeresen kezelt polipropilén kompozit felület szuperhidrofobicitására, *Polimerek*, 1, 25-31 (2025)

Vámos Cs., Bárány T., Marosfői B.: Porous polypropylene produced by phase separation for high solar reflectivity and passive cooling. *Journal of Thermal Analysis and Calorimetry*. 149, 10851-10863 (2024) IF: 3.0

6.2. Publications supporting the thesis statements

6.2.1. Publications

1. **Vámos Cs.,** Rácz I, Marosfői B., Kállay-Menyhárd A., Bárány T.: Novel, solvent-based method for the production of polymer sheets with superhydrophobic surface. *Polymer Engineering and Science*. 63(4), 1289-1302 (2023) IF: 3.2

2. **Vámos Cs.,** Varga L. J., Marosfői B., Bárány T.: Role of extruded sheet morphology in phase separation and final morphology of superhydrophobic polypropylene. *Periodica Polytechnica Mechanical Engineering*. 66, 260-271 (2022)

3. **Vámos Cs.,** Honti Sz., Marosfői B., Rácz I., Bárány T.: Extrudált lemez morfológiájának hatása az oldószeres kezeléssel kialakított szuperhidrofób felületi struktúrára. *Polimerek*. 8, 227-232 (2022)

4. **Vámos Cs.,** Bárány T., Marosfői B.: Porous polypropylene produced by phase separation for high solar reflectivity and passive cooling. *Journal of Thermal Analysis and Calorimetry*. 149, 10851-10863 (2024) IF: 3.0

5. **Vámos Cs.,** Füredi M., Hórvölgyi Z., Krafcsik O., Kiss G., Bárány T, Marosfői B. B.; Plasma treated, superhydrophobic Polypropylene surfaces: a detailed study of hydrophobic recovery by XPS and wettability measurements. *Scientific Reports*. 14(1), 23715 (2024) IF: 3.8

6. **Vámos Cs.,** Bárány T., Az üvegszál erősítés hatása az oldószeresen kezelt polipropilén kompozit felület szuperhidrofobicitására, *Polimerek*, 1, 25-31 (2025)

7. **Vámos Cs.,** Bárány T., Glass fiber reinforced polypropylene composites with high solar reflectance for thermal insulation Applications. *Polymers*. 17(3), 274 (2025) IF: 4.7

6.2.2. *Patents*

1. RÁCZ I., MAROSFŐI B., **Vámos Cs.**, NAKAI Y., MIMURA Y.: Method for producing molded body with ultra-water-repellent surface and molded body with ultra-water-repellent surface WO/2021/201156; EP4130115A1 publicated, pending; CN115335441B (2024.07.23.) granted.
2. MAROSFŐI B., **Vámos Cs.**, RÁCZ I., NAKAI Y., MIMURA Y: Articles that diffusely reflect ultraviolet, visible and/or infrared radiation and methods of making the same JP2021-161976; PCT/JP2022/036457 publicated, pending.

6.3. Applicability

Two international patents related to the technology and the properties resulting from the developed structure have previously been submitted and published.

The patent for "Method for producing molded body with ultra-water-repellent surface and molded body with ultra-water-repellent surface" presents the solvent-based surface treatment technology described in the dissertation and the superhydrophobic, water-repellent surface it can create.

Meanwhile, the patent "Articles that diffusely reflect ultraviolet, visible and/or infrared radiation and methods of making the same" demonstrates the optical properties of the structured surface across different wavelength ranges.

For the experimental work, I conducted solvent-based surface treatments on extruded PP samples using a laboratory-scale, batch process. This approach allowed for the separate examination of the effects of individual technological steps (solvent treatment, drying, and removal of the skin layer) on the structure-property relationship of the samples and was sufficient for the small quantities of samples. As a further development of the technology and for scaling up, we designed and built a pilot-scale surface treatment device capable of a continuous process with "synchronizing" the individual process steps. This pilot-scale equipment (**Figure 6.1**) enables the treatment of other semicrystalline polymers, in addition to PP, and can process larger quantities of material.

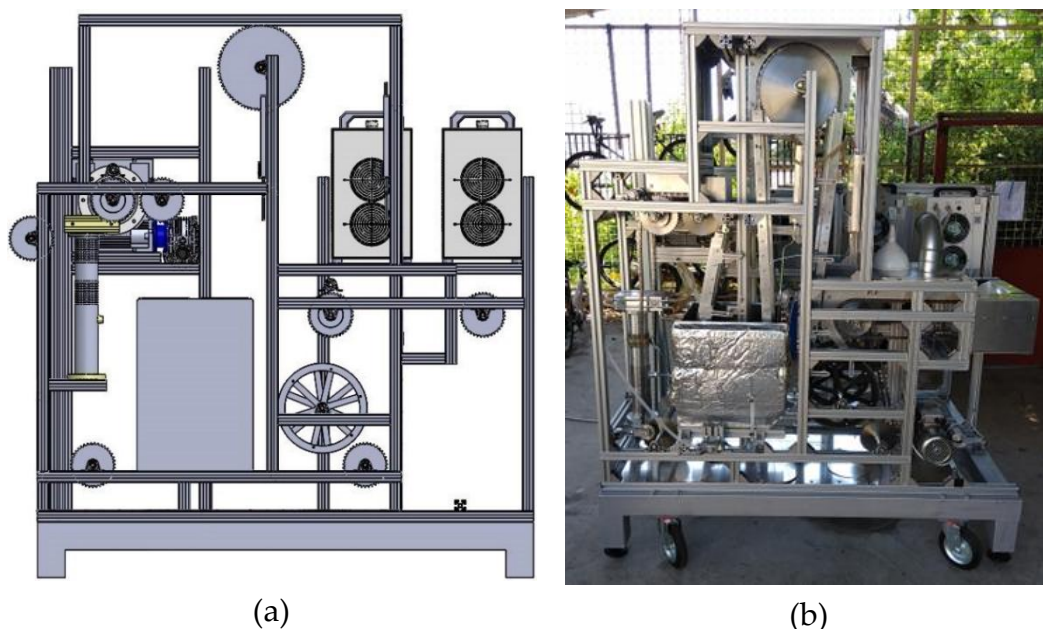


Figure 6.1 Schematic diagram of the continuous solvent-treatment machine (a) and the current state of the actual machine (b)

Regarding applicability, both newly developed properties (water repellency and solar light reflectivity) open up further opportunities for products made from PP. By utilizing the treated sheet's superhydrophobic and self-cleaning nature, as well as its light-scattering properties, it can be used in lighting technology or as reflective thermal

insulation elements, such as coverings for outdoor components (e.g., electrical enclosures).

6.4. Future challenges

The following investigations are planned regarding the solvent-based surface treatment technology:

- Examination of solvent surface treatment on extruded sheets of varying thicknesses
- Investigation of the surface treatment of samples produced by different processing methods, such as injection molding and compression molding
- Solvent-treatment of different polymer types in addition to PP.

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8. APPENDIX

Table 8.1 Effect of plasma pressure on CA

Sample	Pressure (Pa)	Power (W)	Treatment time (s)	CA (°)
SHPP001	15	24	60	40 ± 3
SHPP002	30	24	60	131 ± 2

Table 8.2 The values of θ_s , θ_0 and T acquired by fitting for various samples with an exponential function. A zero in parentheses '(0)' indicates near-spontaneous wetting where video capture and curve fitting was not possible.

Designation	$\theta(t) = \theta_s + \theta_0 * e^{\left(\frac{-t}{T}\right)}$ θ_s [-] θ_0 [-] and T [s] acquired by fitting									
	Immediately after treatment		1 day after treatment		2 days after treatment		3 days after treatment		7 days after treatment	
	θ_0	T	θ_0	T	θ_0	T	θ_0	T	θ_0	T
SHPP_LP300	130	0.8	120	7.9	123	10.0	123	16.0	14	73.3
SHPP_LP600	-	(~0)	0.8	1.1	0.9	1.7	2.2	3.0	1.8	3.6
SHPP_MP300	-	(~0)	-	(~0)	3.6	5.1	3.2	5.3	4.0	4.9
SHPP_MP600	-	(0)	-	(0)	-	(0)	-	(0)	-	(0)

Table 8.3 Effect of degradation due to high power

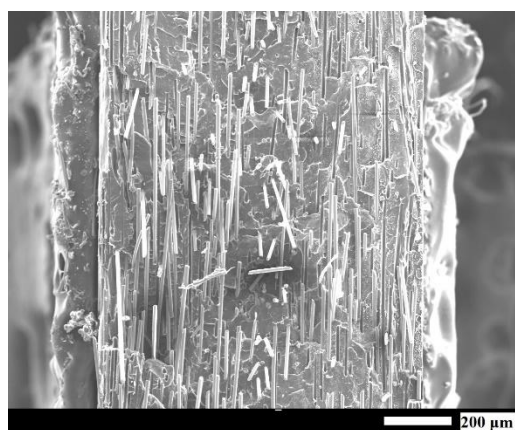
Designation	$\theta(t) = \theta_s + \theta_0 * e^{\left(\frac{-t}{T}\right)}$ θ_s [-] θ_0 [-] and T [s] acquired by fitting														
	Immediately after treatment			1 day after treatment			2 days after treatment			3 days after treatment			7 days after treatment		
	θ_s	θ_0	T	θ_s	θ_0	T	θ_s	θ_0	T	θ_s	θ_0	T	θ_s	θ_0	T
SHPP_MP300							0	3.6	5.1	0	3.2	5.3	0	4.0	4.9
SHPP_MP600	(0)	-	(0)	(0)	-	(0)	(0)	-	(0)	(0)	-	(0)	(0)	-	(0)
SHPP_HP300	(0)	-	(0)	(0)	-	(0)	0	2.3	2.5				0	1.6	5.3
SHPP_HP600	(0)	-	(0)	(0)	-	(0)							0	1.0	4.1

Table 8.4 CAs of SHPP_MP600 samples after heat treatment at 70 °C and 150 °C for one hour

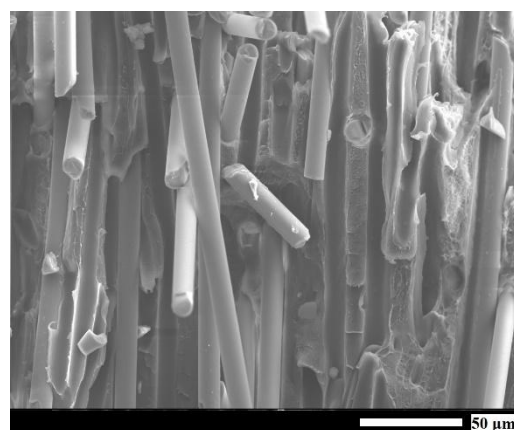
Samples	Aging temperature (°C)	CA _{adv} (°)
Before plasma treatment	25	152 ± 1
	25	≈ 0
After plasma treatment	70	140 ± 1
	150	145 ± 1

Table 8.5 Crystalline parameters of the extruded PP, GFPP, NA PP and NA GFPP samples

Designation	T _m (°C)	T _c (°C)	ΔH _m (J/g)	X _c (%)*
PP	167	113	81	39
PP_ 60 s	166.	114	86	41
PP_ 120 s	164.	114	90	43
PP_ 180 s	166	114	93	45
NA PP	165	129	85	41
NA PP_ 60 s	167	129	89	43
NA PP_ 120 s	167	129	91	44
NA PP_ 180 s	167	128	93	45
GFPP 30	166	119	48	38
GFPP 30_ 60 s	166	119	66	45
NA GFPP 30	166	130	61	42
NA GFPP 30_ 60 s	166	130	65	45



(a)



(b)

Figure 8.1 SEM images showing the cross-section of an extruded PP/GF 30 sheet at (a) 100x magnification and at (b) 500x magnification

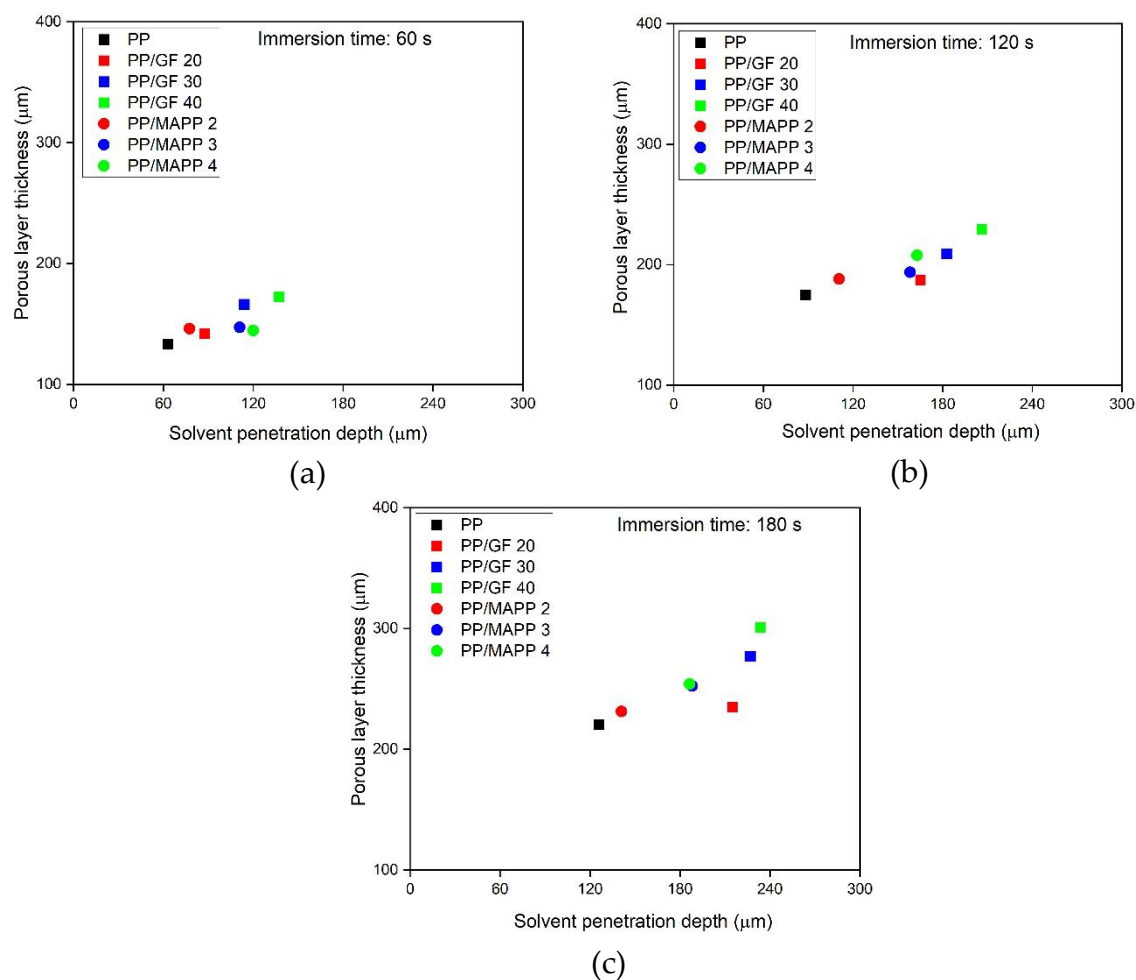


Figure 8.2 The thickness of the porous layer of solvent-treated neat PP, PP/GF samples, and PP/MAPP samples as a function of solvent penetration depth. The samples were solvent-treated at 125 °C for (a) 60 s, (b) 120 s and (c) 180 s

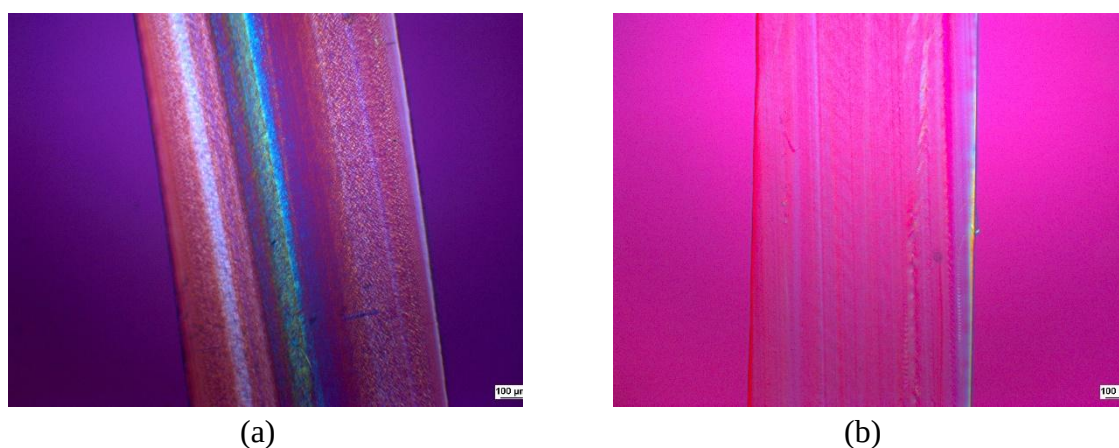
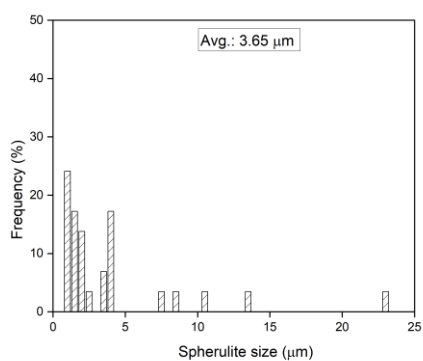


Figure 8.3 POM images of (a) extruded PP sheet and (b) extruded NA PP sheet at 50x magnification

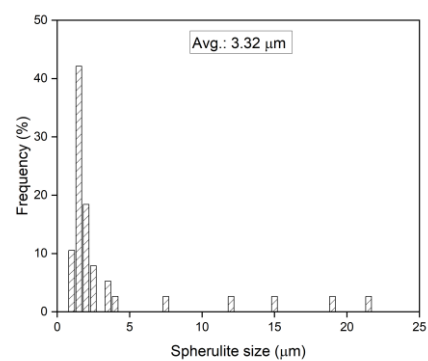
Immersion time (s)

60s

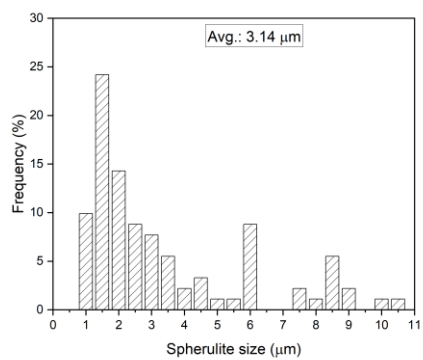
180s



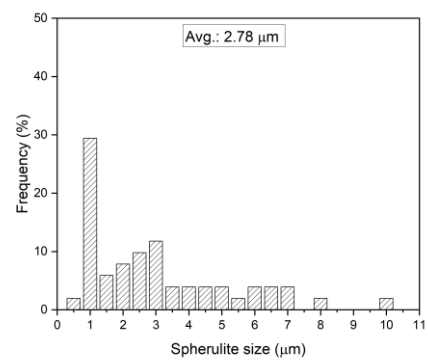
(a)



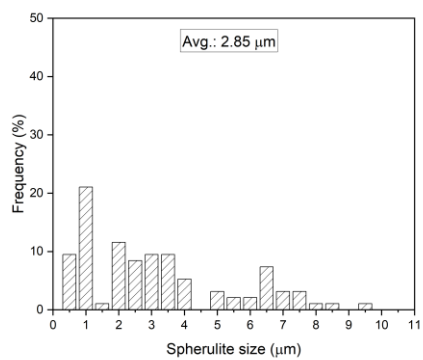
(b)



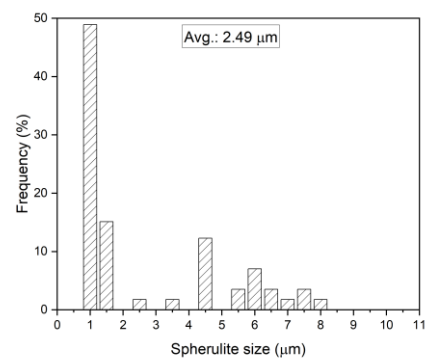
(c)



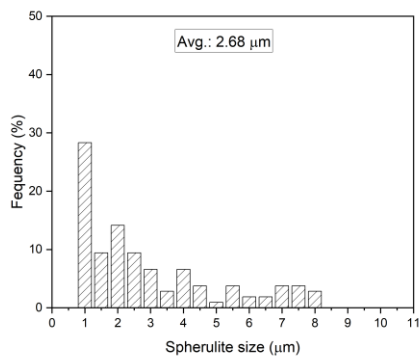
(d)



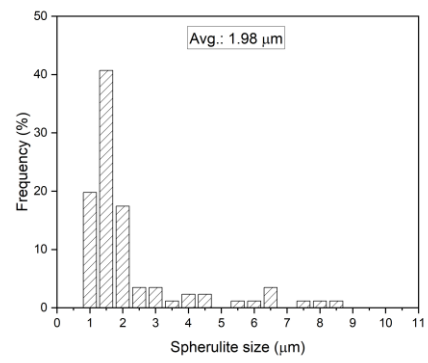
(e)



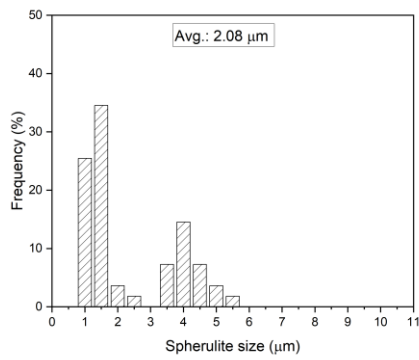
(f)



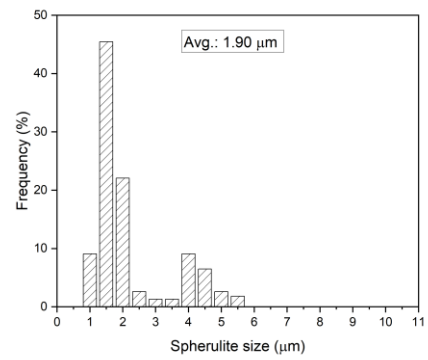
(g)



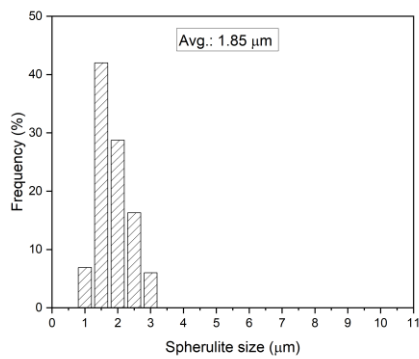
(h)



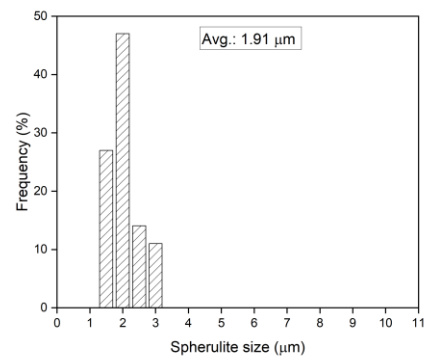
(i)



(j)



(k)



(l)

Figure 8.4 Spherulite size distribution along the porous layer of (a) PP, (c) GFPP 20 %, (e) GFPP 30 %, (g) GFPP 40 %, (i) NA PP, (k) NA GFPP 30 % solvent treated for 60 s and (b) PP, (d) GFPP 20 %, (f) GFPP 30 %, (h) GFPP 40 %, (j) NA PP, (l) NA GFPP 30 %, solvent treated for 180 s

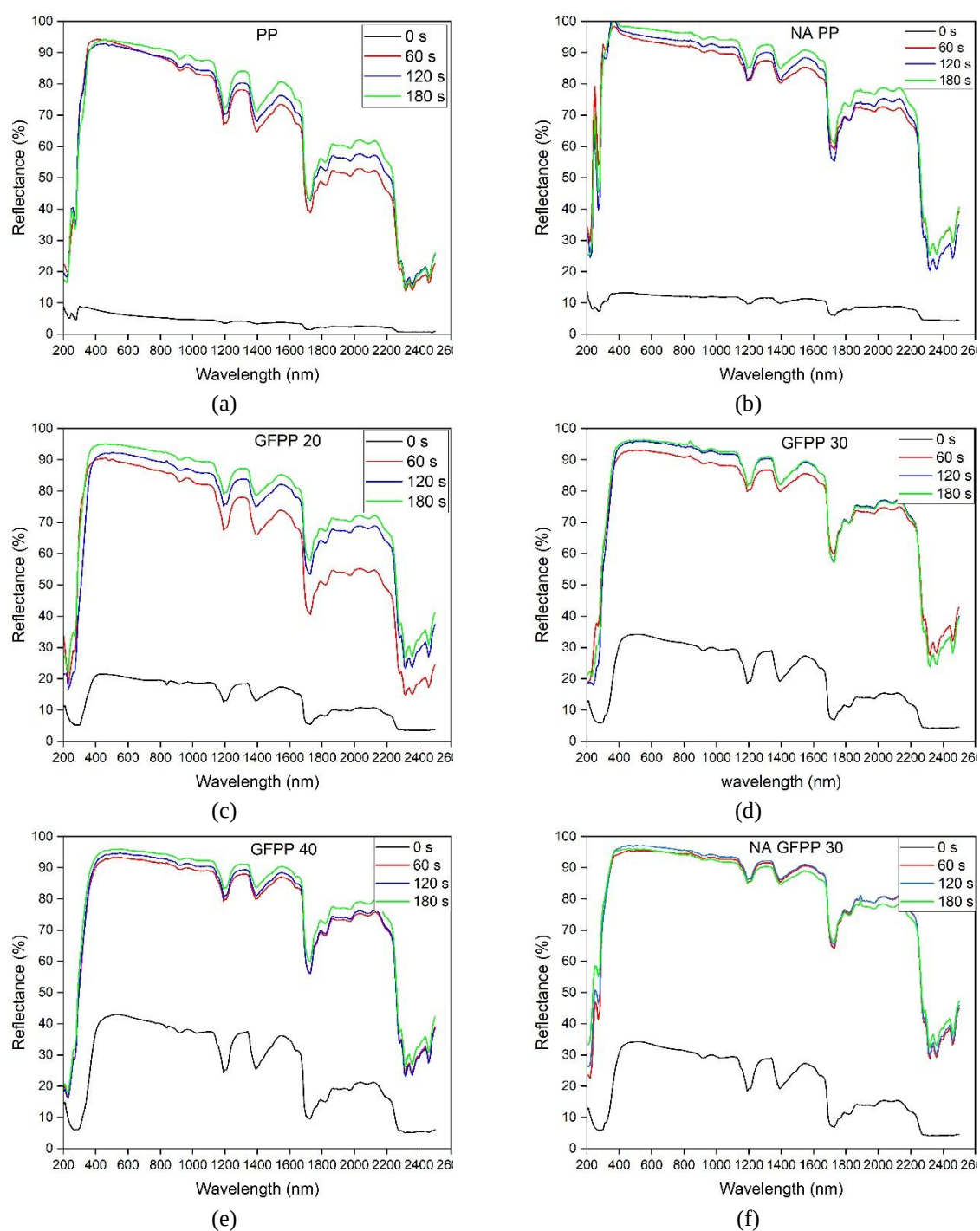


Figure 8.5 Reflectance spectra of (a) PP, (b) NA PP, (c) GFPP 20, (d) GFPP 30, (e) GFPP 40, (f) NA GFPP 30, treated with different immersion times

Table 8.6 The effect of fiber content and immersion time on the elongation at break of solvent-treated samples

Immersion time (s)	Elongation at break (%)					
	PP	NA PP	GFPP 20	GFPP 30	GFPP 40	NA GFPP 30
0	464.6 ± 32.4	23.7 ± 5.1	1.0 ± 0.1	0.8 ± 0.1	0.8 ± 0.1	2.6 ± 0.1
60	425.8 ± 12.5	20.7 ± 1.9	1.0 ± 0.1	1.0 ± 0.2	1.0 ± 0.1	2.8 ± 0.1
120	287.3 ± 15.0	19.2 ± 0.6	1.1 ± 0.2	0.8 ± 0.1	0.9 ± 0.1	2.5 ± 0.1
180	25.6 ± 4.0	7.2 ± 0.5	1.1 ± 0.1	1.0 ± 0.1	0.9 ± 0.1	2.6 ± 0.1